

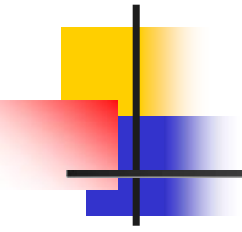
Quartus II 軟體使用練習

練習三種輸入方法：

(1). Bottom-Up

(2). Top-Down

(3). One-Chip



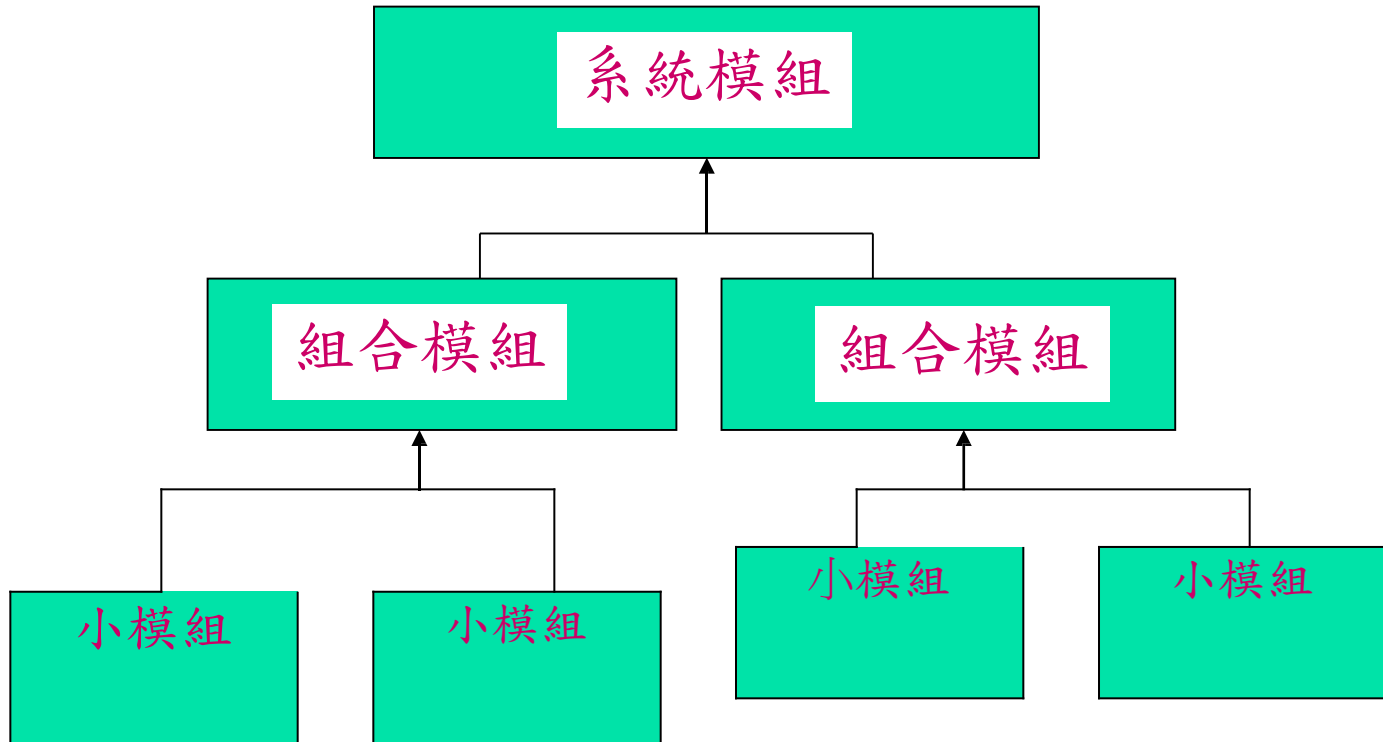
Bottom-Up Design

Bottom-Up

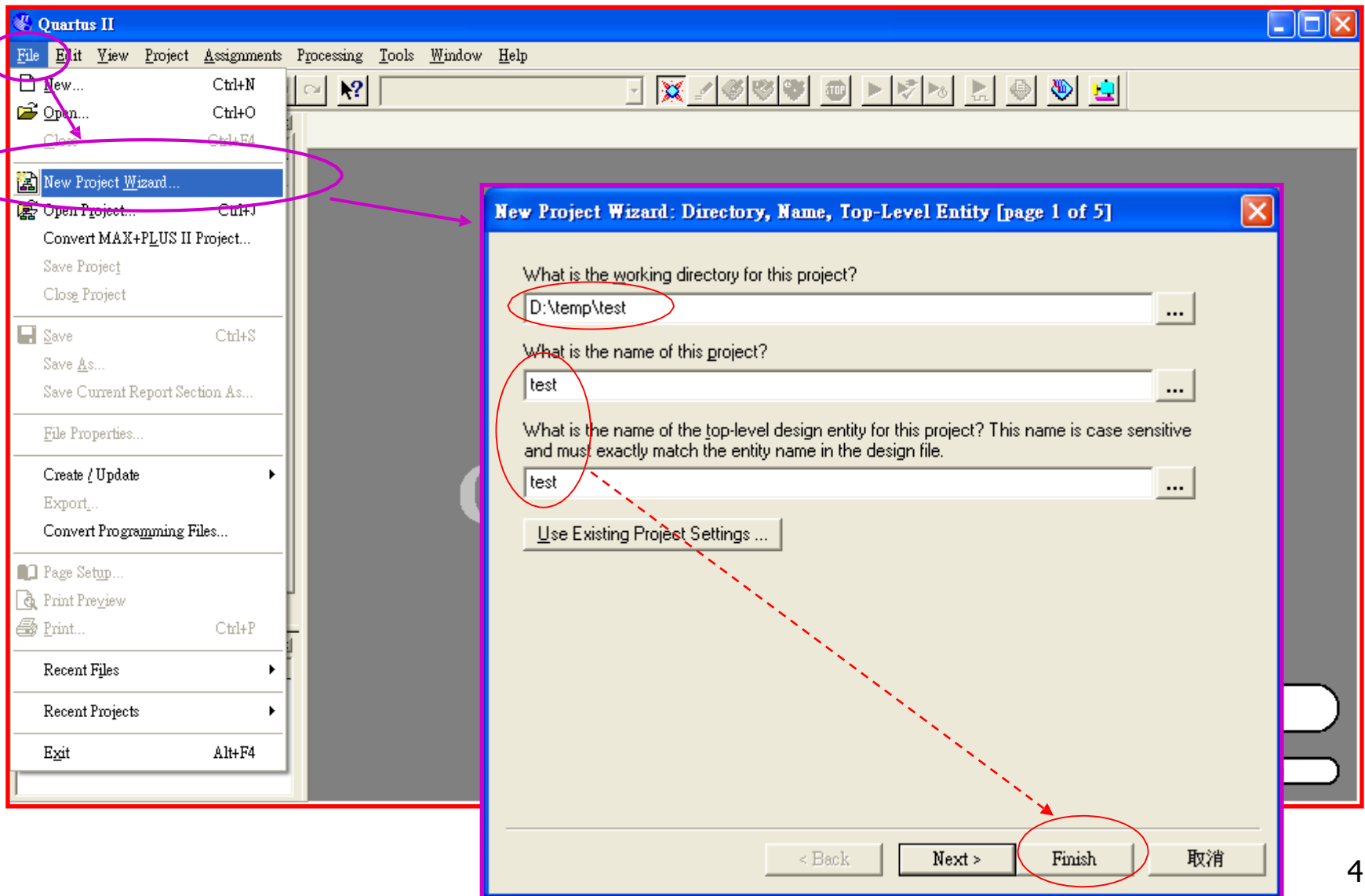
Design Flow

Bottom-Up Design

設計的順序



Bottom-Up Design (1)



Bottom-Up Design (2)

按右鍵選擇加入其他的檔案

Additional Function

“Files” → 指定 “Files” → 按右鍵 → 選定
: “Add/Remove Files in project”.
→ Select “103_tcfst_FPGA_example\CH04_Design
three methods\d_ff_main_code\d_ff_main_code.v”
→ ADD

OK Cancel

Bottom-Up Design (3)

Quartus II - D:/temp/test/test - test - [test.v]

File Edit View Project Assignments Processing Tools Window Help

test

Project Navigator

- Files
 - Device Design Files
 - test.v
 - Software Files
 - Other Files

test.v

```
1 module test (clk_in, div_2);
2
3 input    clk_in;
4         div_2;
5
6
7
8 always @ (posedge clk_in)
9 begin
10     div_2 = div_2 ^ 1;
11 end
12
13 endmodule
14
```

點二下叫出程式內容

Additional Function => Compiler => wrong ???
=> Model name "d_ff_main_code" ≠ Project name "test".

Bottom-Up Design (4)

The screenshot displays the Quartus II interface during a compilation process. The main window is titled "Quartus II - D:/temp/test/test - test - [Compilation Report - Flow Summary]". The "Project Navigator" on the left shows the project structure, including "Files", "Device Design Files", "Software Files", and "Other Files". The "Compilation Report - Flow Summary" window is open, showing a successful compilation status for "test.v". A dialog box titled "Quartus II" is overlaid on the report, indicating that the "Create Symbol File" operation was successful. The dialog box has a blue header bar with the Quartus II logo and a close button. The main text area says "Create Symbol File was successful" and features a button labeled "確定" (OK) in Japanese. A red arrow points from the "File→Creat/Update => Create Symbol Files for current File." text box to the "確定" button. The status bar at the bottom left shows the progress of the "Create Symbol File" task as 100%.

Quartus II - D:/temp/test/test - test - [Compilation Report - Flow Summary]

File Edit View Project Assignments Processing Tools Window Help

Project Navigator

- Files
- Device Design Files
 - test.v
- Software Files
- Other Files

test.v

- Compilation Report
 - Legal Notice
 - Flow Summary
 - Flow Settings
 - Flow Elapsed Time
 - Flow Log
- Analysis & Synthesis
- Fitter
- Assembler
- Timing Analyzer

Compilation Report - Flow Summary

Flow Summary

Flow Status	Successful - Mon Aug 28 00:25:34 2006
Quartus II Version	5.1 Build 216 03/06/2006 SP 2 SJ Full Version
Revision Name	test
Top-level Entity Name	test
Family	Cyclone
Device	EP1C6Q240C8
Final	
ments	Yes
1 / 5,980 (< 1 %)	
2 / 185 (1 %)	
0	
0 / 92,160 (0 %)	
0 / 2 (0 %)	

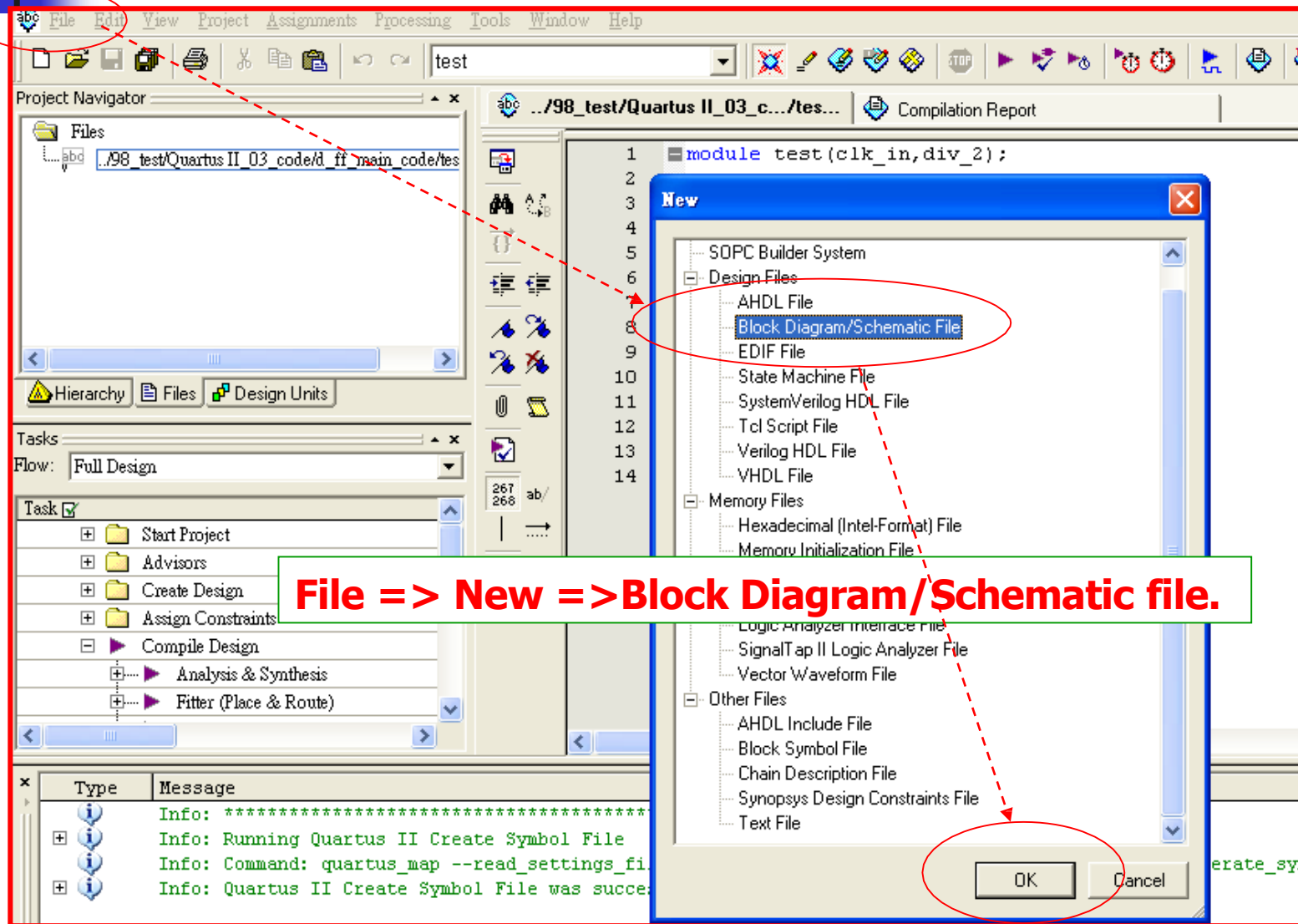
Quartus II

Create Symbol File was successful

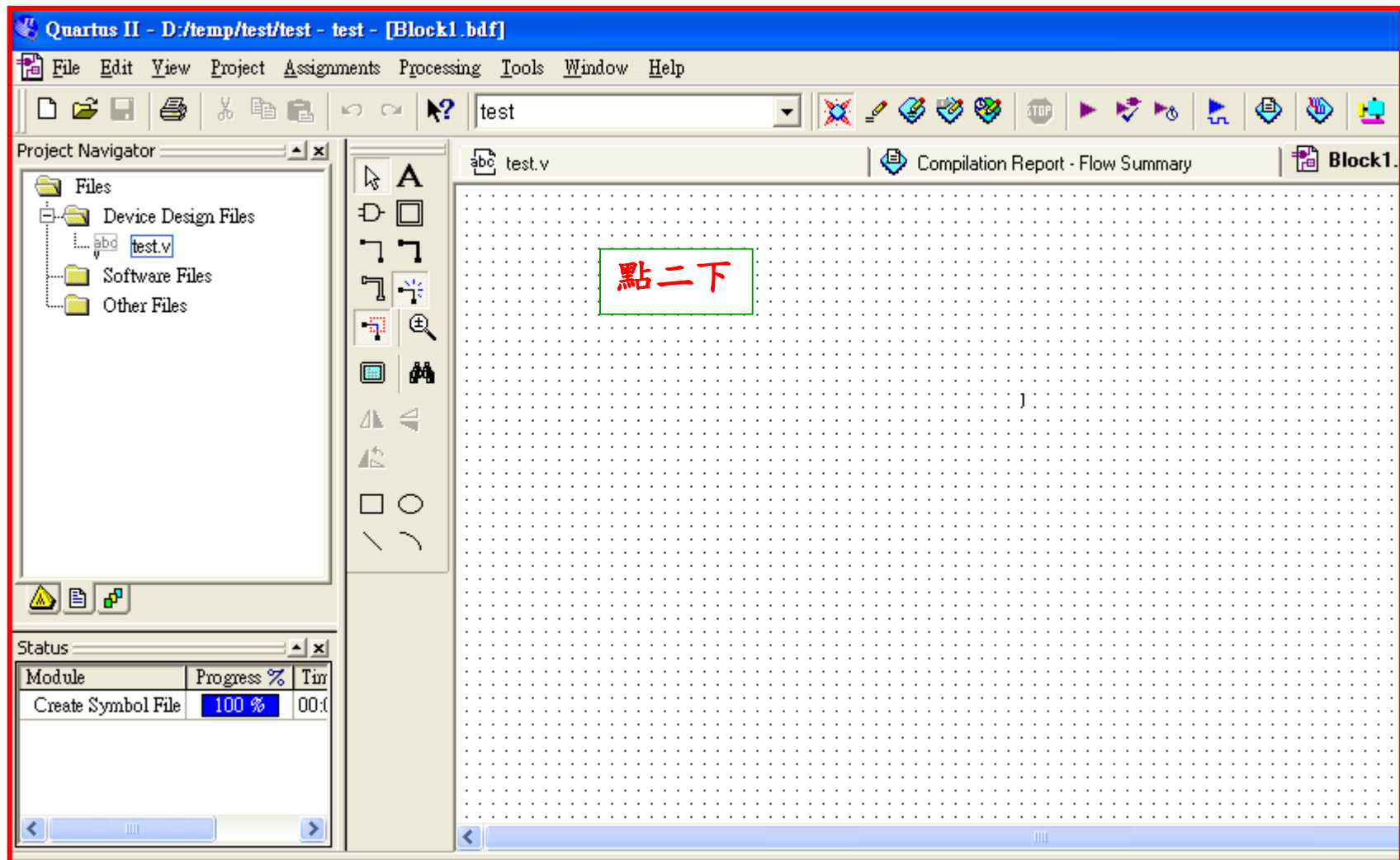
確定

File→Creat/Update => Create Symbol Files for current File.

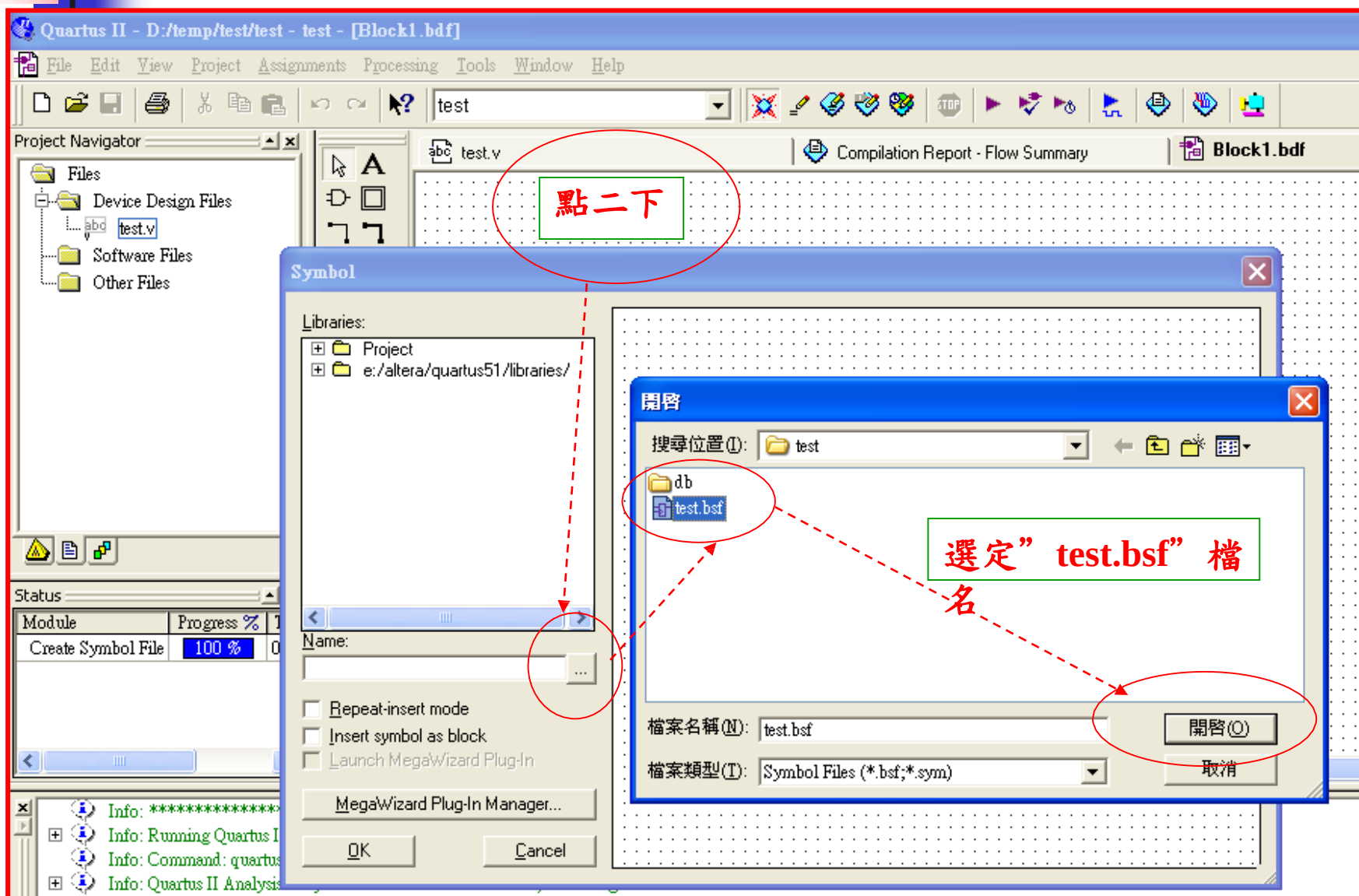
Bottom-Up Design (5)



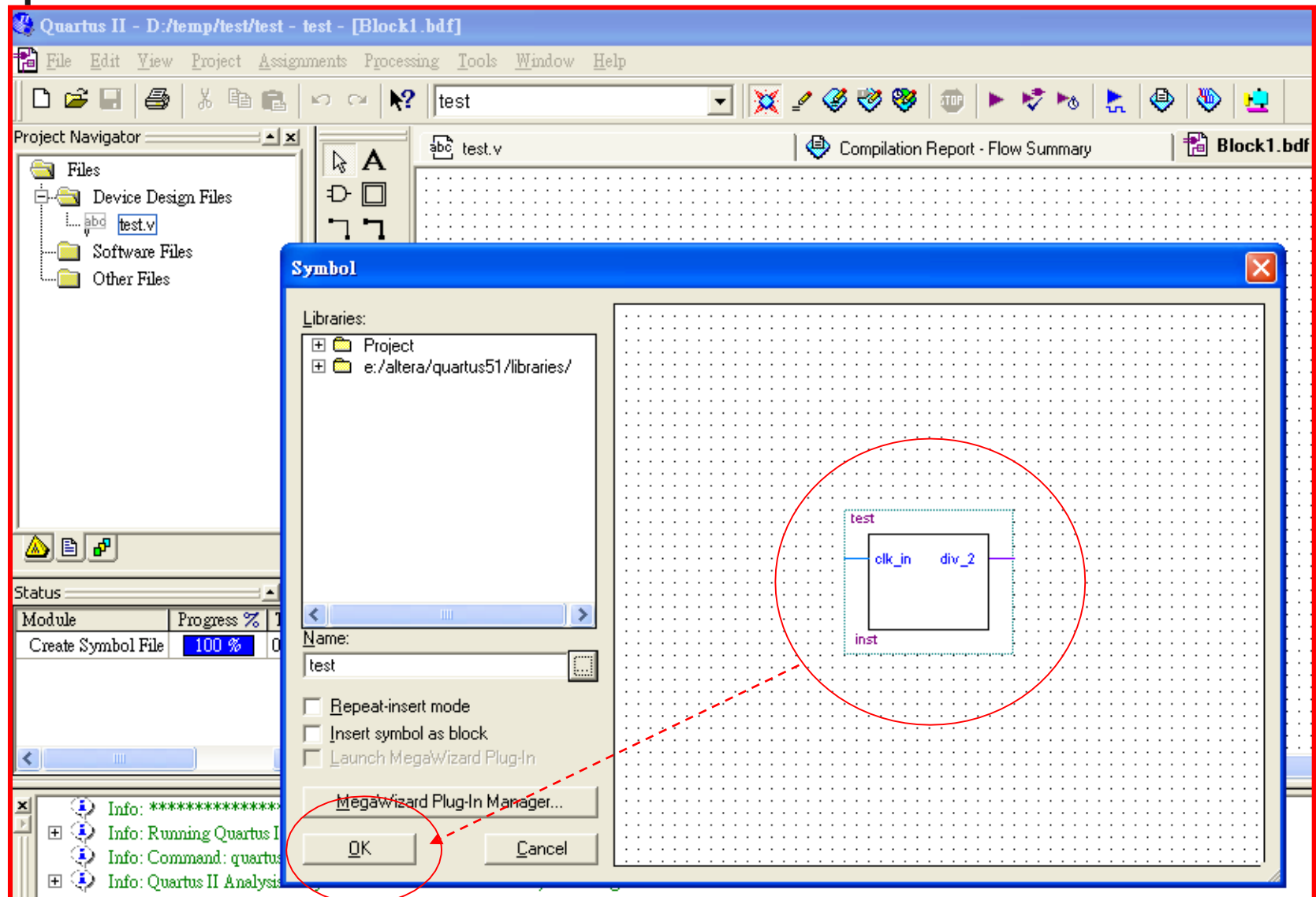
Bottom-Up Design (6)



Bottom-Up Design (7)



Bottom-Up Design (8)



Bottom-Up Design (9)

Quartus II - D:/temp/test/test - test - [Block1.bdf*] 在 Block 環境中呼叫 "symbol"

File Edit View Project Assignments Processing Tools Window Help

test

Project Navigator

- Files
 - Device Design Files
 - test.v
 - Software Files
 - Other Files

test.v

Compilation Report - Flow

test

clk_in div_2

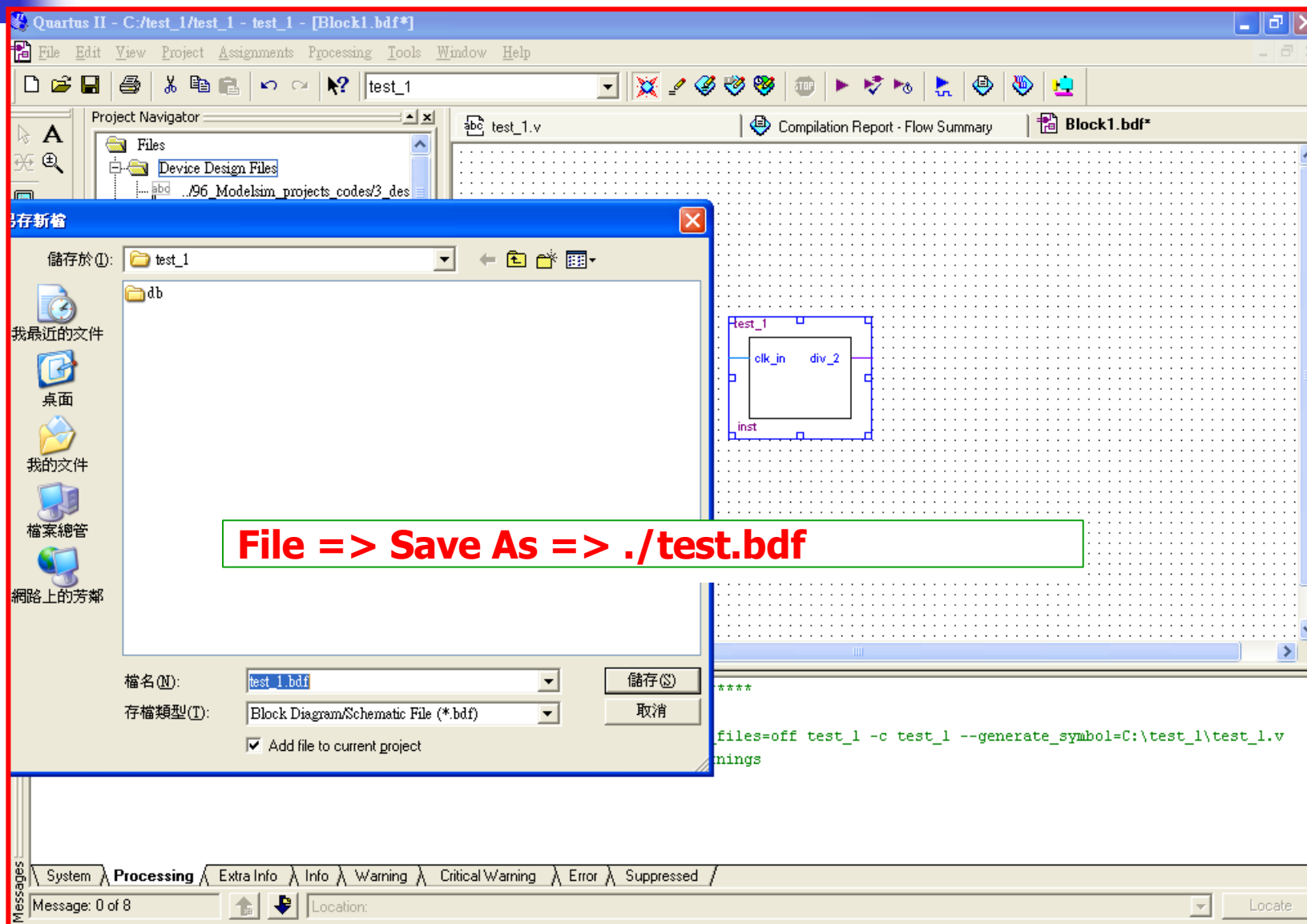
inst

CTRL+Z = 復原

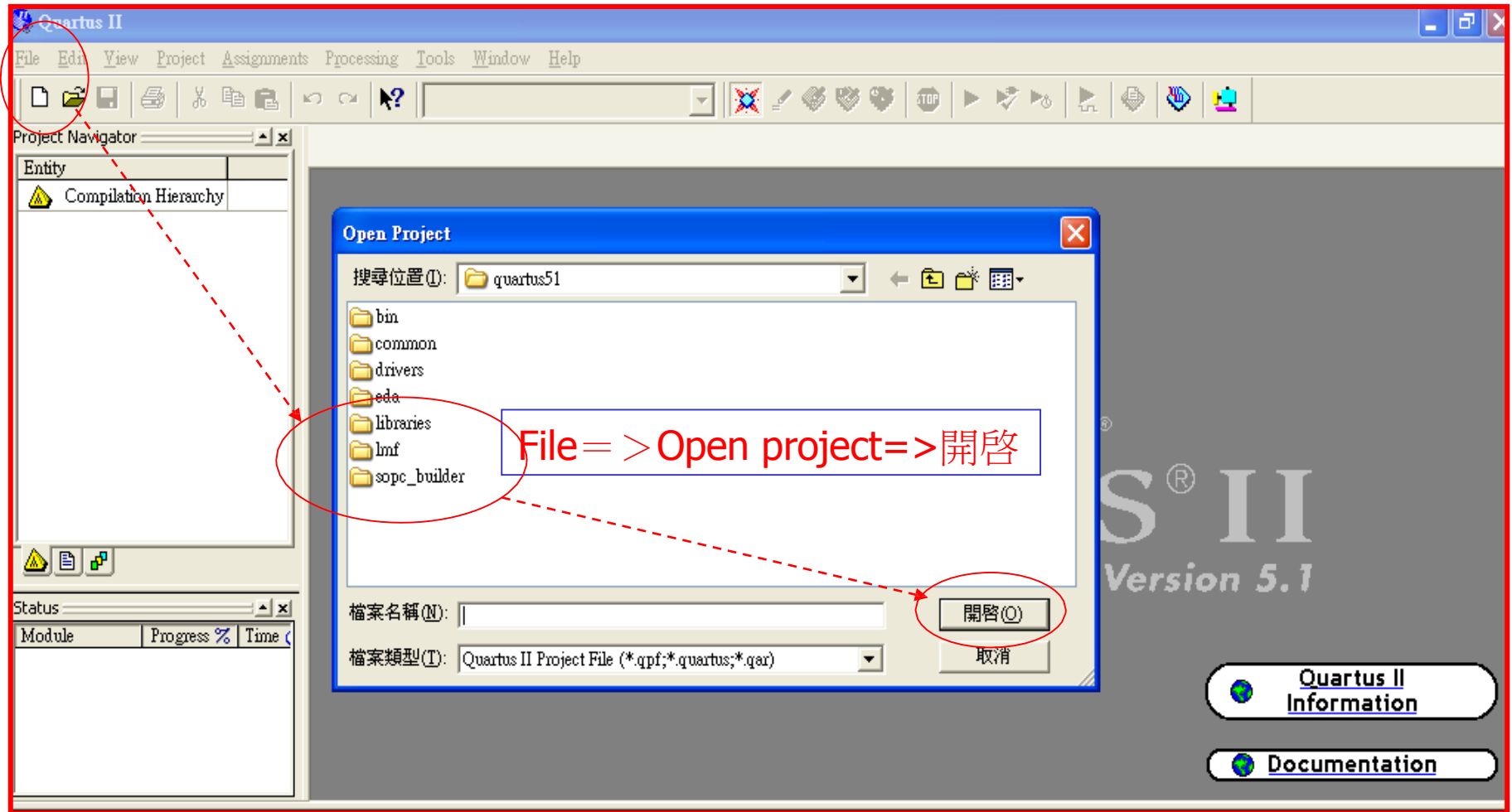
Status

Module	Progress %	Time
Create Symbol File	100 %	00:00

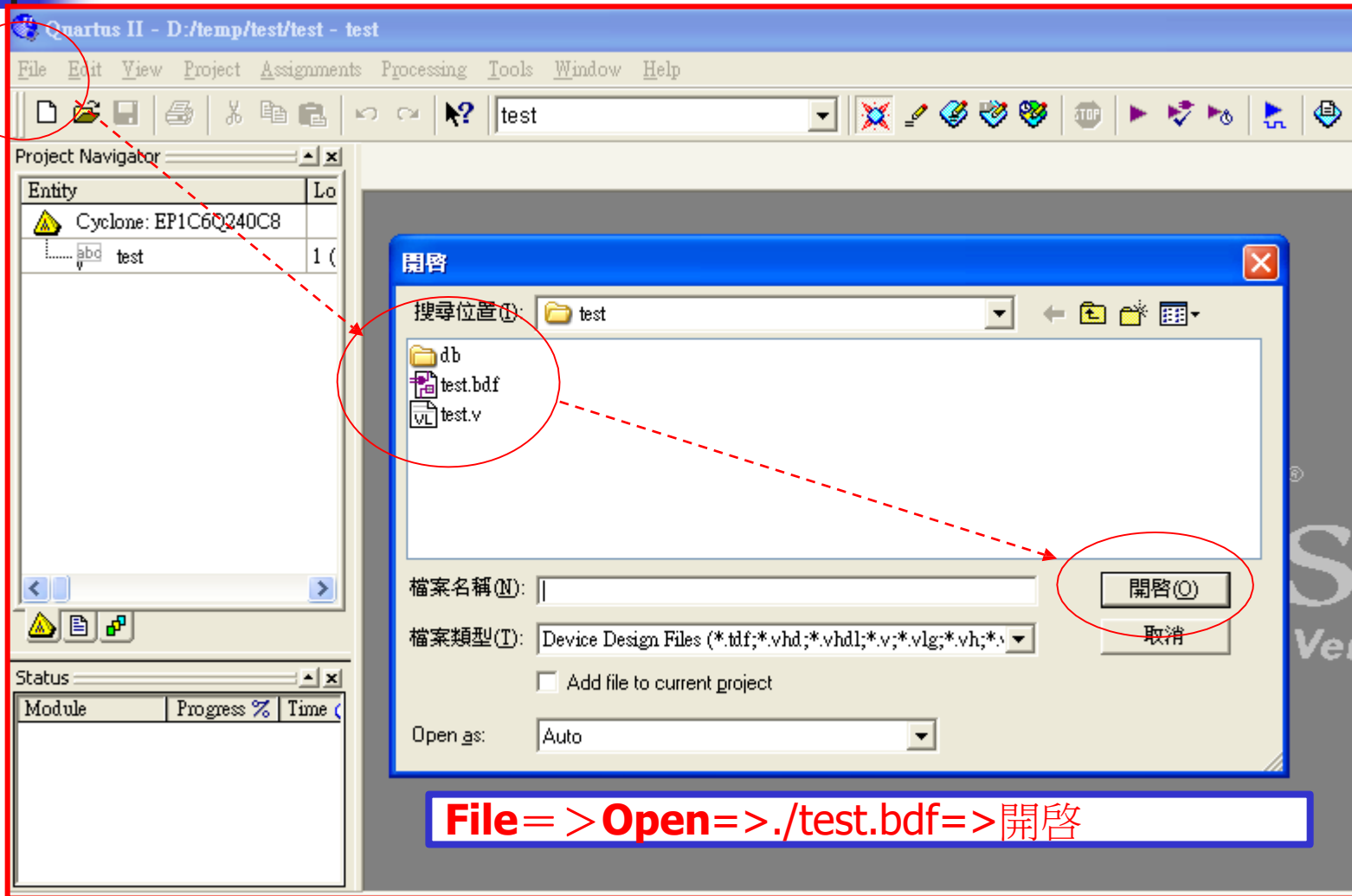
Bottom-Up Design (9)



Bottom-Up Design (10)



Bottom-Up Design (11)



Bottom-Up Design (12)

The screenshot displays the Quartus II software interface for a project named "test" located at "D:/temp/test/test". The interface includes a menu bar (File, Edit, View, Project, Assignments, Processing, Tools, Window, Help) and a toolbar with various icons for file operations and simulation.

The **Project Navigator** on the left shows the project hierarchy. The **Entity** list is expanded, showing the "test" entity selected. A red circle highlights the "test" entity in the list.

The **Entity List** on the left shows the "test" entity selected. A red circle highlights the "test" entity in the list.

The **HDL Editor** on the right shows the "test.bdf" file open. The code in the editor is:

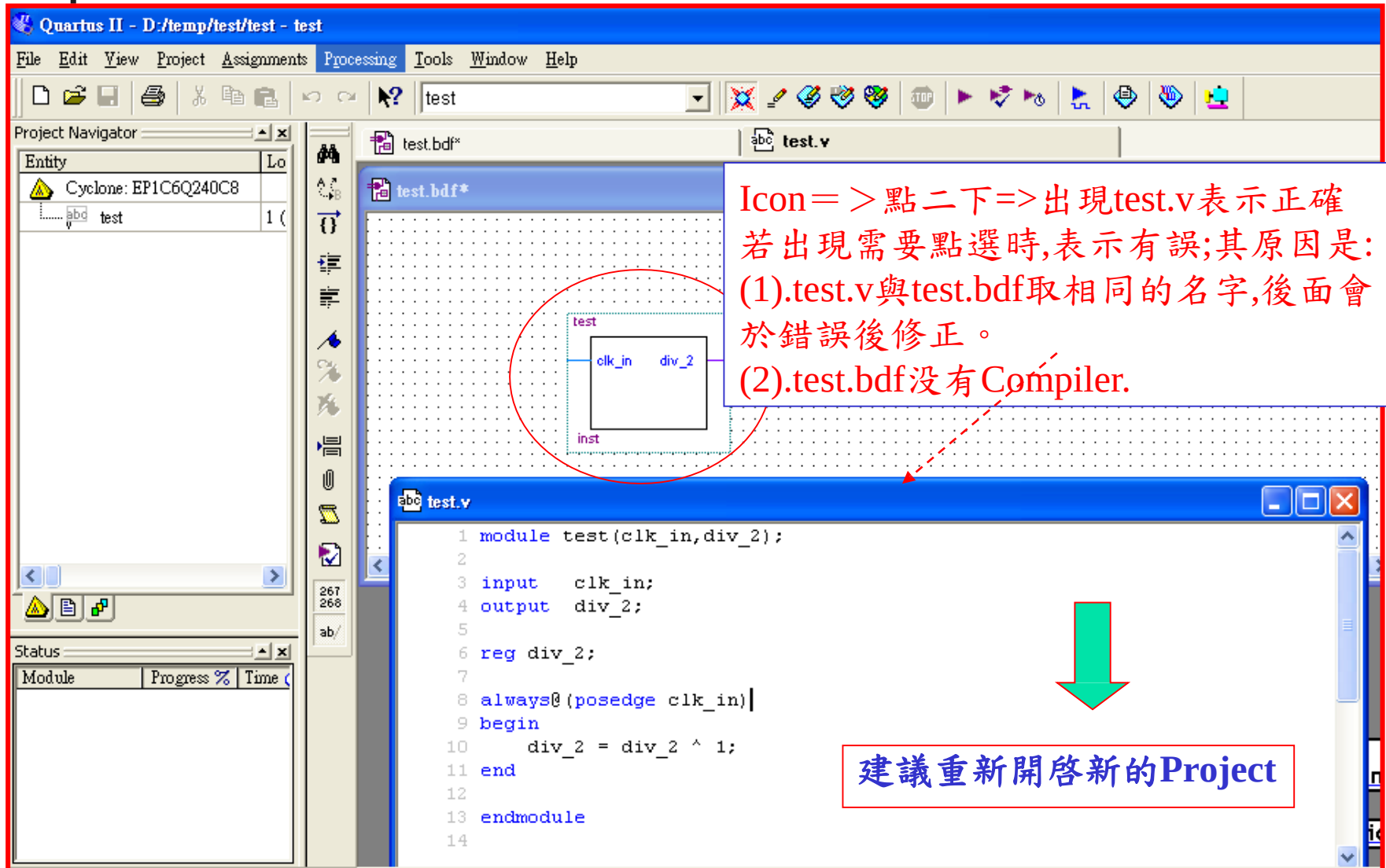
```
module test(clk_in, div_2);  
    // ...  
endmodule
```

A red circle highlights the "test.bdf" file in the editor's title bar.

The **Status** window at the bottom left shows the "Module" tab, with columns for "Progress %" and "Time".

A red box at the bottom right contains the text: **File => Open => 開啓**

Bottom-Up Design (13)



Icon => 點二下 => 出現 test.v 表示正確
若出現需要點選時, 表示有誤; 其原因是:
(1). test.v 與 test.bdf 取相同的名字, 後面會於錯誤後修正。
(2). test.bdf 沒有 Compiler.

```
1 module test(clk_in, div_2);  
2  
3 input  clk_in;  
4 output div_2;  
5  
6 reg div_2;  
7  
8 always@ (posedge clk_in) |  
9 begin  
10     div_2 = div_2 ^ 1;  
11 end  
12  
13 endmodule  
14
```

建議重新開啓新的Project

Bottom-Up Design (14)

Quartus II - D:/temp/test/test - test - [test.bdf*]

File Edit View Project Assignments Processing Tools Window Help

test

Project Navigator

Entity	Location
Cyclone: EP1C6Q240C8	
abd test	1 (

test.bdf*

test

clk_in div_2

inst

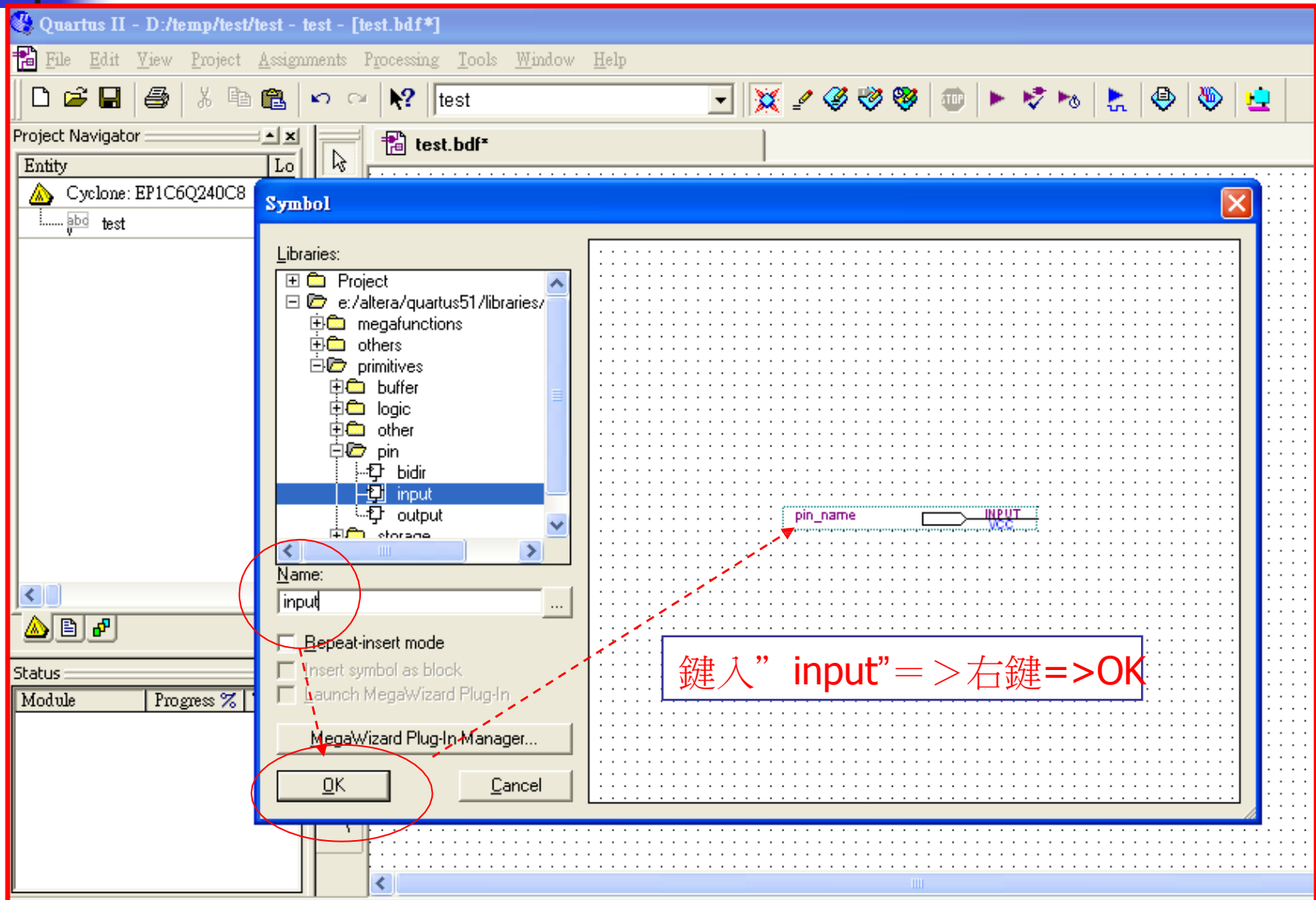
inst1

inst2

選定icon => 右鍵=>Copy=>Paste

點二下

Bottom-Up Design (15)



Bottom-Up Design (16)

Quartus II - D:/temp/test/test - test - [test.bdf*]

File Edit View Project Assignments Processing Tools Window Help

Project Navigator

Entity	Location
Cyclone: EP1C6Q240C8	
test	1 (

test.bdf*

test

inst

inst3

inst4

clk_in

div_2

pin_name

INPUT

VCC

選定適當位置=>按右鍵擺放” input Pin” =>” Esc” 離開

點二下

Status

Module	Progress %	Time (
--------	------------	--------

Bottom-Up Design (17)

Quartus II - D:/temp/test/test - test - [test.bdf*]

File Edit View Project Assignments Processing Tools Window Help

test

Project Navigator

Entity

Cyclone: EP1C6Q240C8

test

test.bdf*

Symbol

Libraries:

- Project
- e:/altera/quartus51/libraries/
- megafuncions
- others
- primitives
 - buffer
 - logic
 - other
 - pin
 - bidir
 - input
 - output
 - storage

Name: output

☒ Repeat-insert mode

☐ Insert symbol as block

☐ Launch MegaWizard Plug-In

MegaWizard Plug-In Manager...

OK Cancel

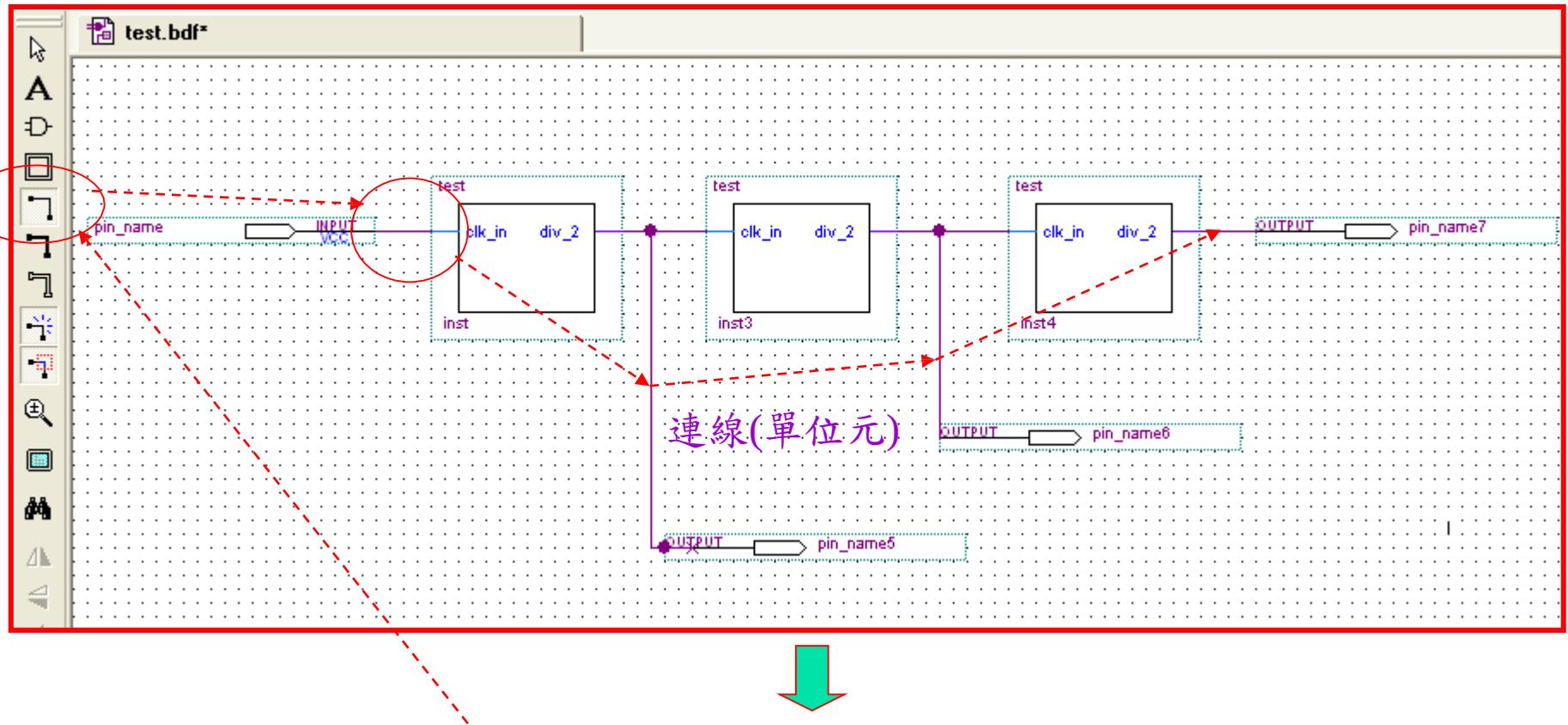
div_2

OUTPUT pin_name

鍵入" output" => 勾選" Repeat icon mode" => OK

 拉線

Bottom-Up Design (19)



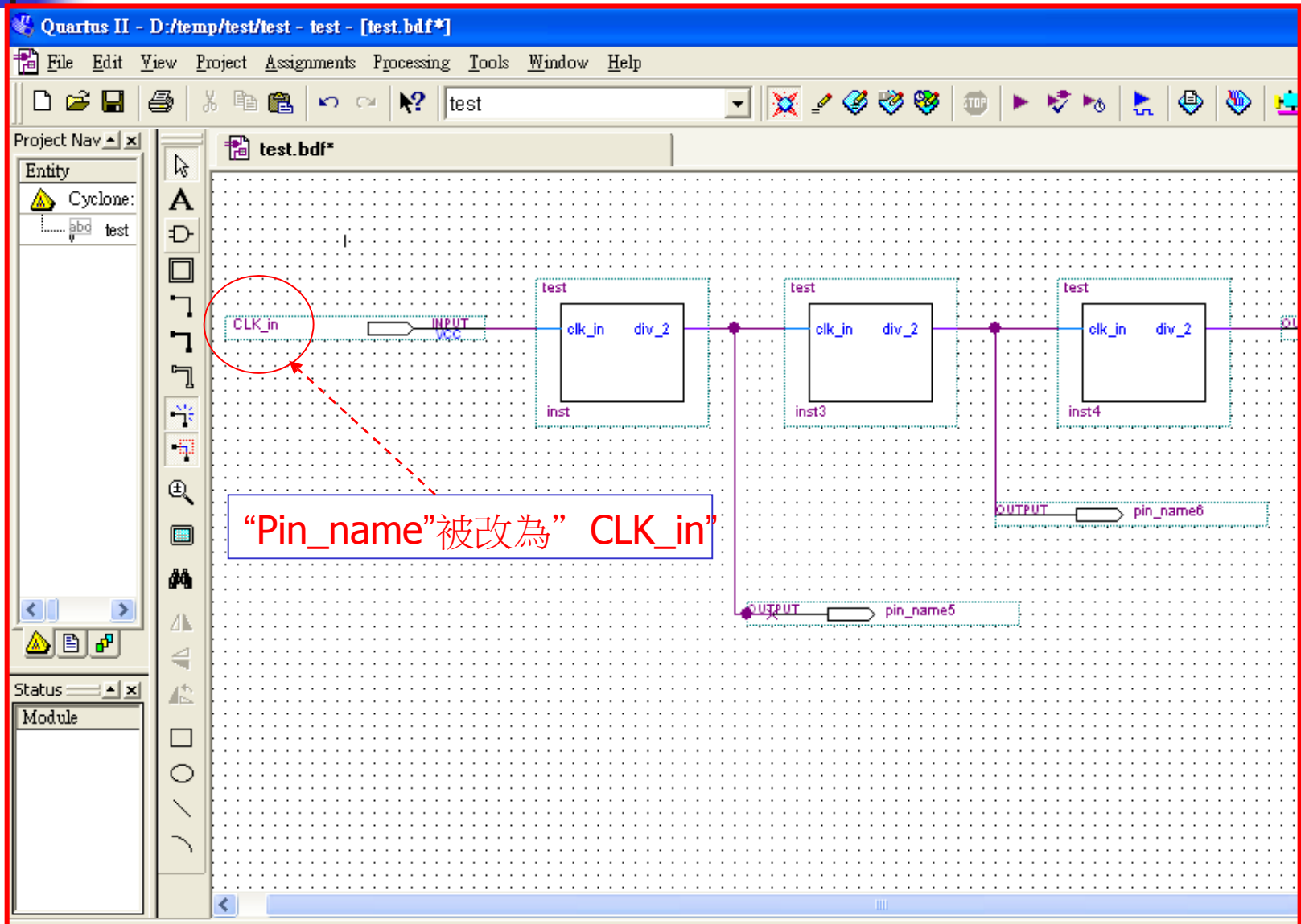
選定” 連線icon”=> 連線(單位元) =>更改輸出入接腳的名稱

Bottom-Up Design (20)

The screenshot shows the Quartus II interface with the Pin Properties dialog box open. The dialog box has two tabs: General and Format. The General tab is selected. The Pin name(s) field contains "CLK_in" and the Default value dropdown is set to "VCC". The dialog box is titled "Pin Properties".

選定input接腳 => 右鍵 => Properties
(Last icon) => "CLK_in" => 確定或
"點右鍵二下"

Bottom-Up Design (21)





Bottom-Up Design (23)

Project Navigator

Files

..98_test/Quartus II_03_code/d_ff_main_code/test.bdf

test.bdf

Compilation Report - Flow Summary

Flow Summary

Flow Status	Flow Failed - Thu Apr 30 08:45:09 2009
Quartus II Version	8.0 Build 215 05/29/2008 SJ Full Version
Revision Name	test
Top-level Entity Name	test
Family	Stratix II
Met timing requirements	N/A
Logic utilization	N/A until Partition Merge
Combinational ALUTs	N/A until Partition Merge
Total DLLs	N/A until Partition Merge

Quartus II

Full Compilation was NOT successful (5 errors)

確定

Error: Can't compile duplicate declarations of entity "test" into library "work"

Info: Found 1 design units, including 1 entities, in source file test.bdf

Error: Quartus II Analysis & Synthesis was unsuccessful. 3 errors, 0 warnings

Error: Quartus II Full Compilation was unsuccessful. 5 errors, 0 warnings

Task

Flow: Full Design

Task

Start Project

Advisors

Create Design

Assign Constraints

Compile Design

Analysis & Synthesis

Fitter (Place & Route)

Bottom-Up Design (24)

The screenshot displays the Quartus II IDE interface. The Project Navigator on the left shows the file hierarchy, with `test_1.bdf` highlighted. The main window shows the 'Flow Summary' for the compilation of `test_1.bdf`. A 'Quartus II' dialog box is overlaid, indicating that the full compilation was successful with 7 warnings. A green box at the bottom contains red text explaining the design process.

Flow Summary

Flow Status	Successful - Thu Apr 30 08:55:59 2009
Quartus II Version	8.0 Build 215 05/29/2008 SJ Full Version
Revision Name	test
Top-level Entity Name	test_1
Family	Stratix II
Met timing requirements	Yes
Logic utilization	< 1 %
Combinational ALUTs	3 / 12,480 (< 1 %)
Dedicated logic registers	3 / 12,480 (< 1 %)
Global registers	3
Global pins	4 / 343 (1 %)
Global virtual pins	0
Global block memory bits	0 / 419,328 (0 %)
Global block 9-bit elements	0 / 96 (0 %)
Total PLLs	0 / 6 (0 %)
Total PLLs	0 / 6 (0 %)
Device	F484C3

Quartus II
Full Compilation was successful (7 warnings)
確定

修改” bdf”而非修改” test.v”，因為test.v已經建立為” Symbols”，修正程序相當繁複。
=> 修改為” test_1.dbf” => 並刪除” test.dbf”及設定” test_1.dbf”為” Set as Top Entity” => Compiler.



Bottom-Up Design (26)

The screenshot shows the Quartus II software interface. The main window displays a waveform editor for a test bench. The waveform is titled 'test.bdf' and shows a signal at 12.95 ns. A red box highlights a point on the waveform, and a red arrow points from this box to the 'Node Finder...' button in the 'Insert Node or Bus' dialog box.

Project Navigator:

- Files
- Device Design Files
 - test.v
- Software Files
- Other Files

Master Time Bar: 12.95 ns | Pointer: 250 ps | Interval: -12.7 ns | Start: 0 ps

Waveform Editor:

Name	Value at 12.95 ns

Insert Node or Bus Dialog Box:

- Name: []
- Type: INPUT
- Value type: 9-Level
- Radix: Binary
- Bus width: 1
- Start index: 0
- ☐ Display gray code count as binary count
- Buttons: OK, Cancel, Node Finder...

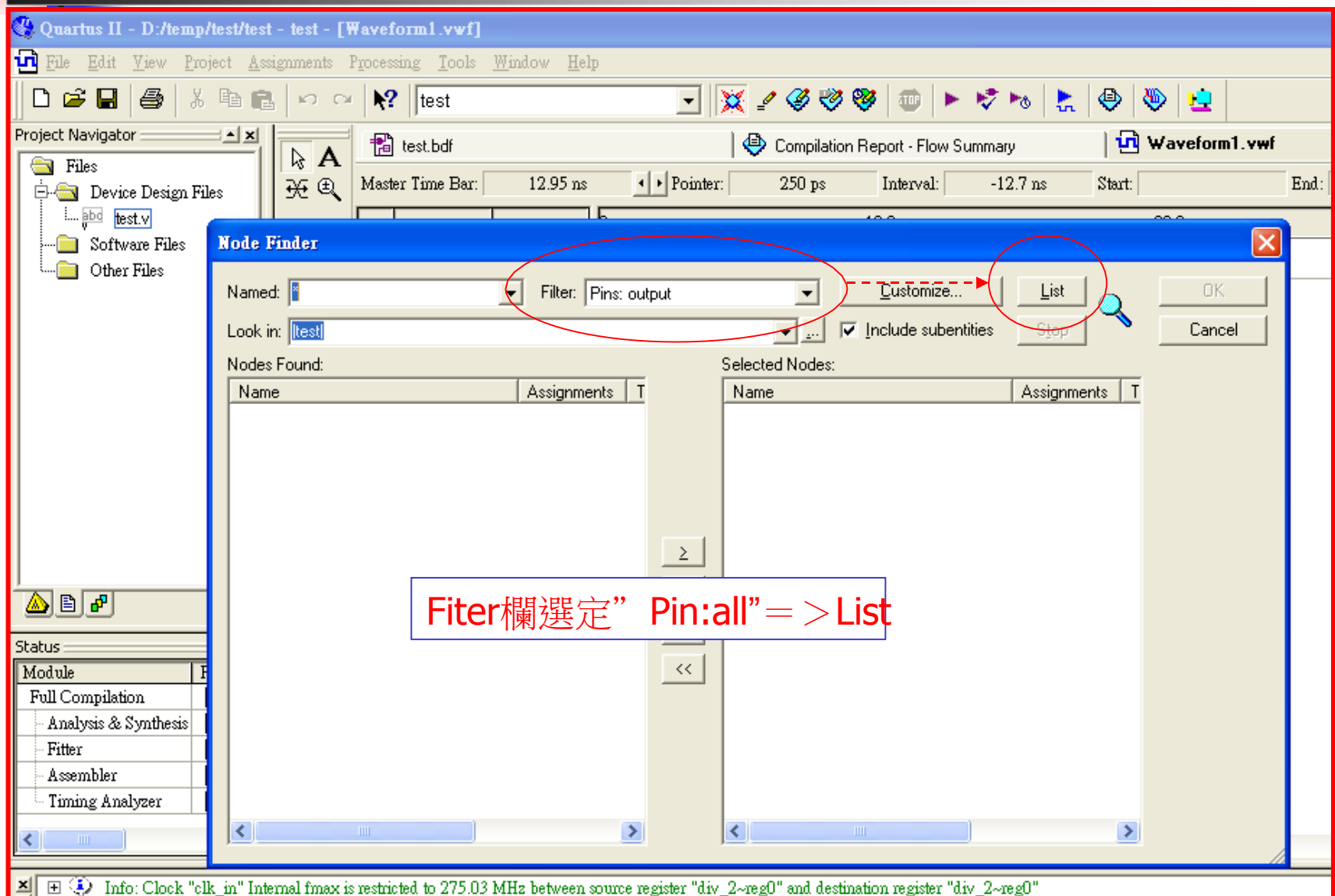
Status Window:

Module	Progress
Full Compilation	100 %
Analysis & Synthesis	100 %
Fitter	100 %
Assembler	100 %
Timing Analyzer	100 %

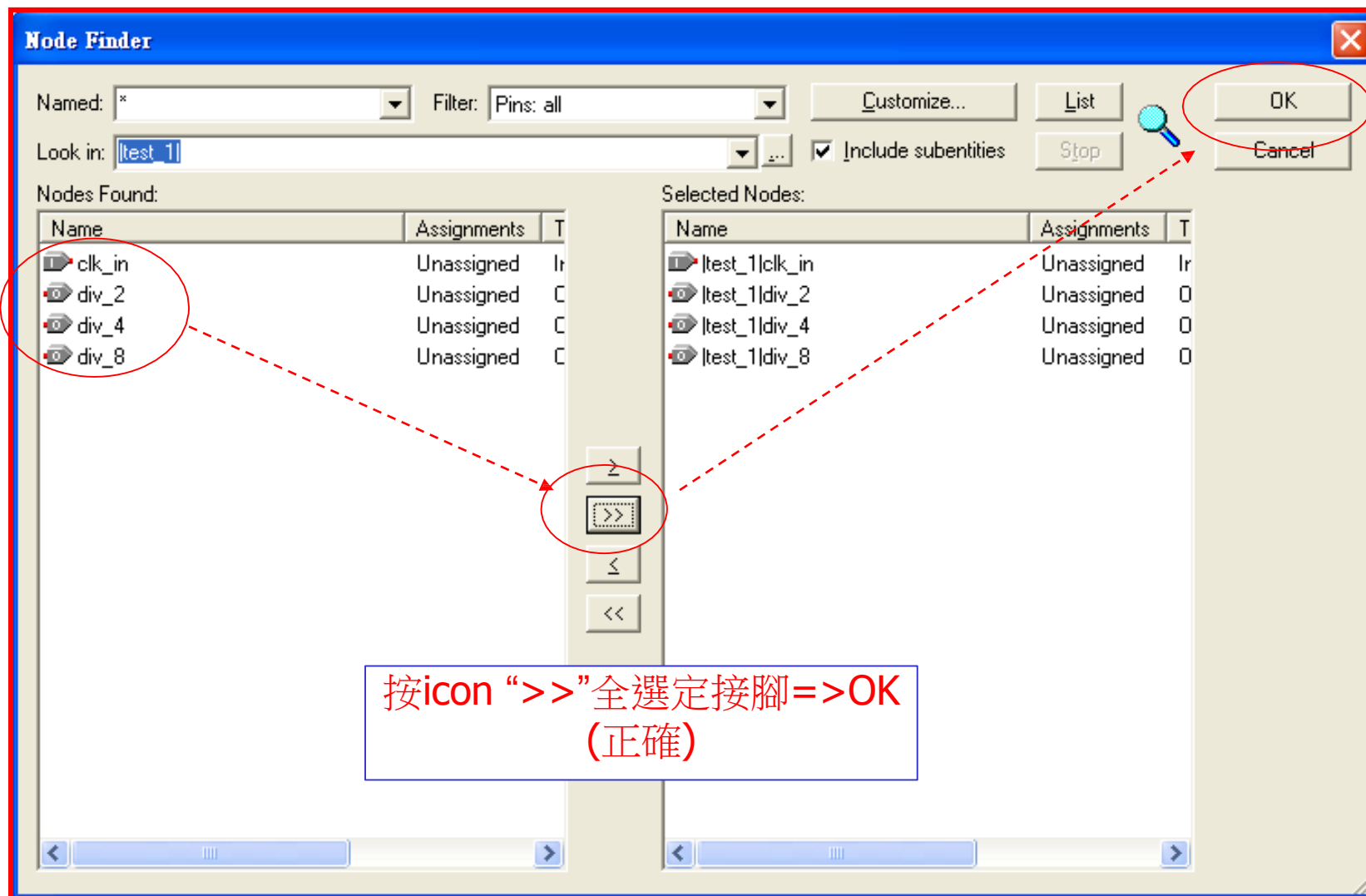
Annotations:

- 點二下 (Click twice)
- 空白處點二下 => Node Finder (Click twice in blank area => Node Finder)

Bottom-Up Design (27)



Bottom-Up Design (28)



Bottom-Up Design (29)

The screenshot displays the Quartus II software interface. The main window shows a waveform viewer with a time axis from 0 ps to 20.0 ns. A node is inserted at 12.95 ns. An 'Insert Node or Bus' dialog box is open, with the 'OK' button highlighted by a red circle. The dialog box contains the following fields:

- Name:
- Type:
- Value type:
- Radix:
- Bus width:
- Start index:
- ☐ Display gray code count as binary count

The status bar at the bottom indicates: Info: Clock "CLK_in" Internal fmax is restricted to 275.03 MHz between source register "testinstdiv_2" and destination register "testinstdiv_2".

Bottom-Up Design (30)

The screenshot displays the Quartus II software interface. The Project Navigator on the left shows the project structure, including the file `test_1.bdf`. The main window shows a timing diagram with a clock signal `clk_in` and a data signal `div_2`. The Master Time Bar indicates a value of 8.625 ns. A dialog box titled "Clock" is open, showing the "Time range" (Start time: 0 ps, End time: 1.0 us) and "Base waveform on" (Time period: 10.0 ns, Offset: 0.0 ns, Duty Cycle (%): 50). The "Duty Cycle (%)" field is circled in red, and a red dashed line points to it from the Project Navigator. The bottom status bar shows the command: `=off test -c test --timing`.

Project Navigator: Files, test_1.bdf

Master Time Bar: 8.625 ns

Pointer: 150 ps

Interval: -8.48 ns

Start: 0 ps

Value: 8.63

10.0 ns

8.625 ns

clk_in

div_2

0 ps

1.0 us

Time range

Start time: 0 ps

End time: 1.0 us

Base waveform on

Time period:

Period: 10.0 ns

Offset: 0.0 ns

Duty Cycle (%): 50

OK

Cancel

Type: Message

Info: *****

Info: Running Quartus II Classic Timing

Info: Command: quartus_tan --read_settin

=off test -c test --timing

Bottom-Up Design (31)

Quartus II

No simulation input file assignment specified on Simulator page of the Settings dialog box

選定" save" => OK => Simulation

若出現” 讀不到檔案,表示其VWF(test_1.vwf)檔名與test_1.bdf相同,改為不同即可。

Bottom-Up Design (32)

The screenshot displays the Quartus II software interface during a simulation. The Project Navigator on the left shows the project files: test_1.bdf, test_1.vwf, and test.vwf. The Tasks pane indicates the 'Full Design' flow, with 'Compile Design', 'Analysis & Synthesis', and 'Fitter (Place & Route)' completed. The Simulation Waveforms window shows a timing diagram for 'clk_in' and 'div_2'. A 'Quartus II' dialog box in the center confirms 'Simulator was successful' with a '確定' (OK) button.

File Edit View Project Assignments Processing Tools Window Help

test

Project Navigator

Files

- ..98_test/Quartus II_03_code/d_ff_main_code/test_1.bdf
- test_1.vwf
- test.vwf

Hierarchy Files Design Units

Tasks

Flow: Full Design

Task

- Start Project
- Advisors
- Create Design
- Assign Constraints
- Compile Design
- Analysis & Synthesis
- Fitter (Place & Route)

Simulation Rep

Legal Notice

Flow Summ

Flow Setting

Simulator

Summar

Settings

Simulati

Simulati

INI Usa

Message

Simulation Waveforms

Simulation mode: Timing

Master Time Bar: 8.625 ns

Pointer: Interval: Start: End:

Name Value

0 ps 10.0 ns 20.0 ns 30.0 ns 40.0 ns 50.0 ns

8.625 ns

0 clk_in B

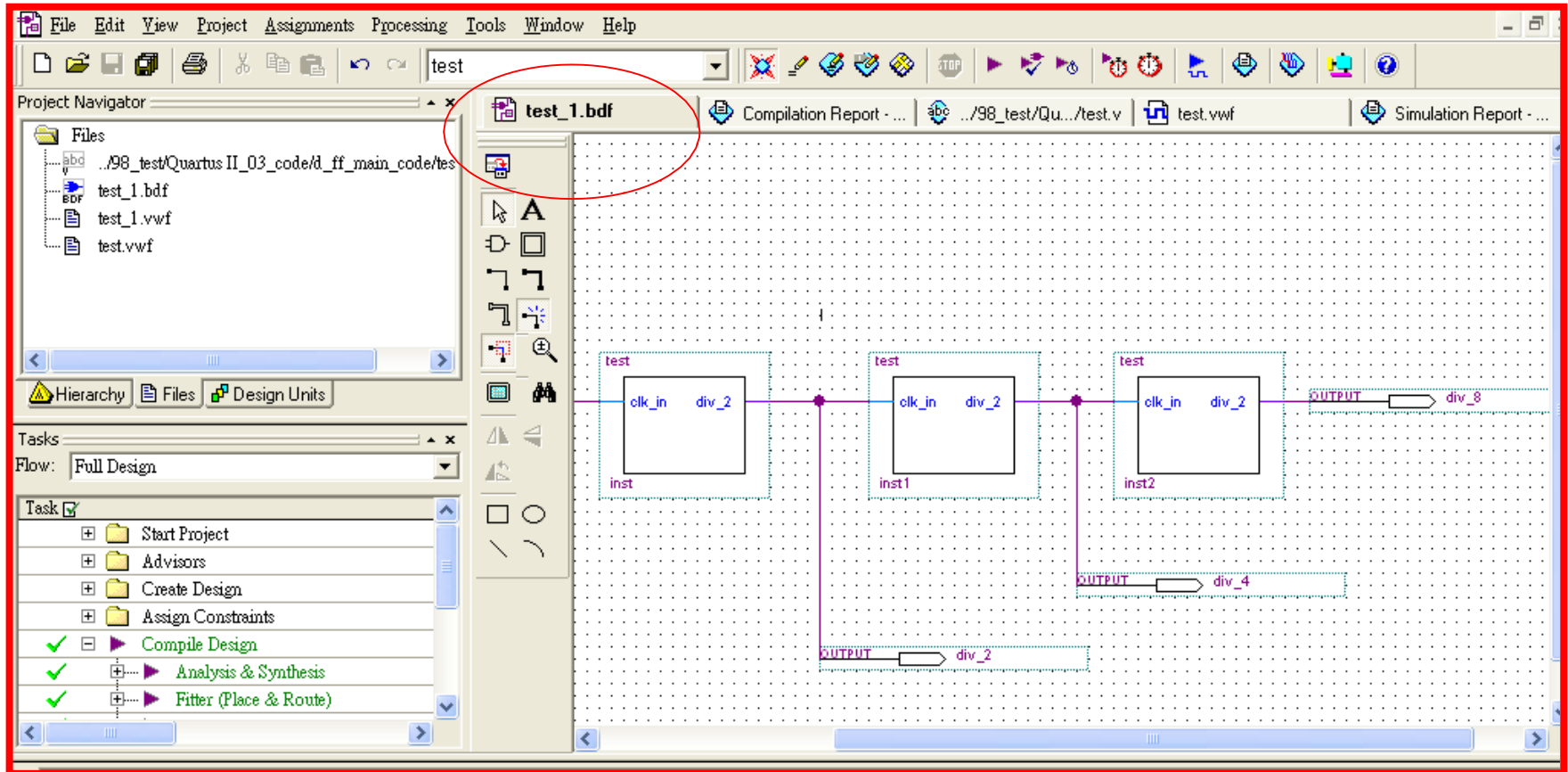
1 div_2 B

Quartus II

Simulator was successful

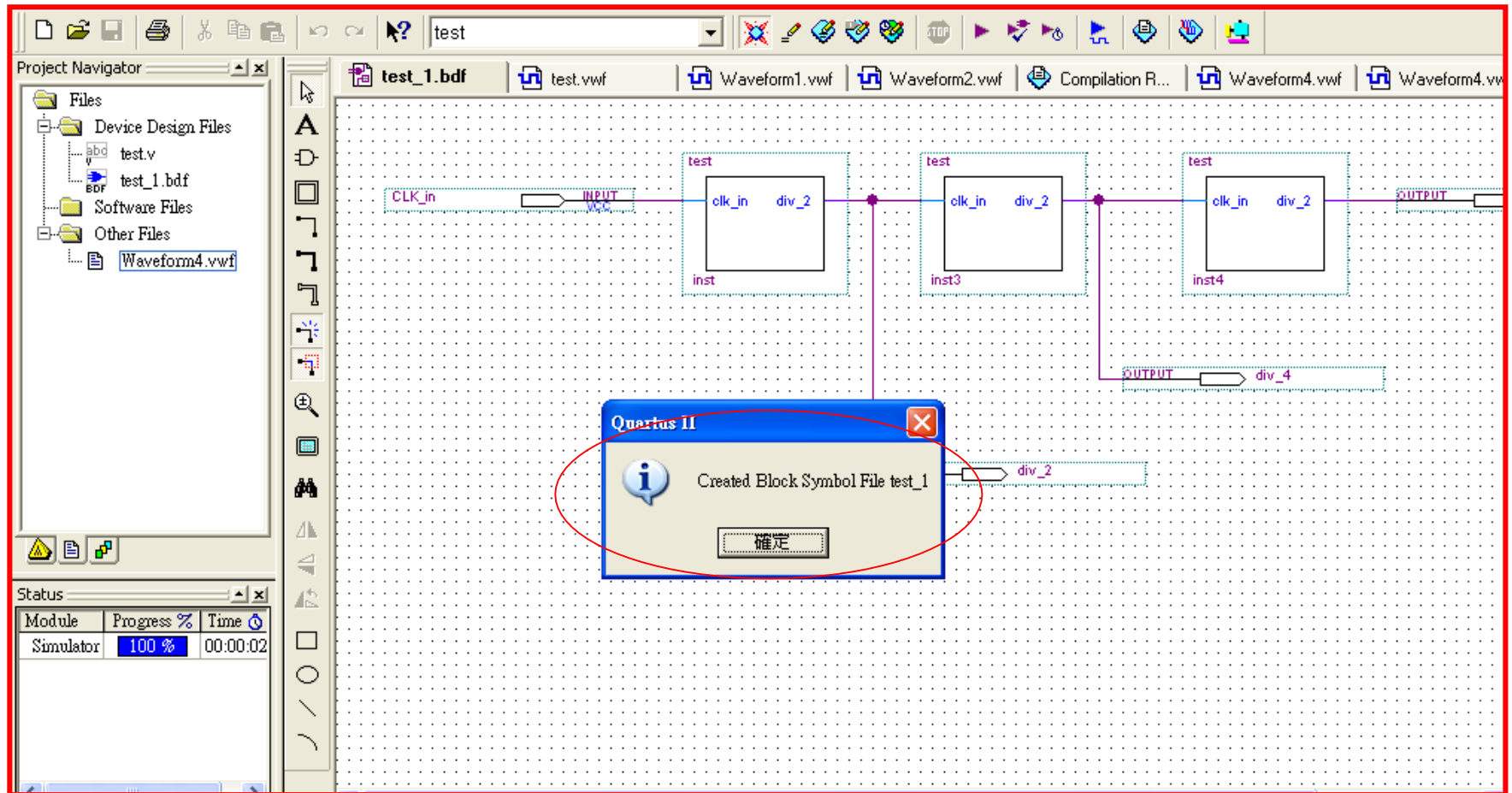
確定

Bottom-Up Design (33)



選定” test_1.bdf” =>New=>Create/Update=>**Create “Symbol” Files** for current file.

Bottom-Up Design (34)



Bottom-Up Design (35)

The screenshot displays the Quartus II IDE interface. The 'New' dialog box is open, showing a tree structure of file types. 'Block Diagram/Schematic File' is selected under the 'Design Files' category. A text box overlay with a blue border contains the text: '呼叫Symbol: => New => Block Diagram/Schematic Files => OK'. The background interface includes the Project Navigator on the left, the Tasks pane below it, and a message window at the bottom showing system information.

Project Navigator:

- Files
 - test_1.bdf
 - test_1.vwf
 - test.vwf

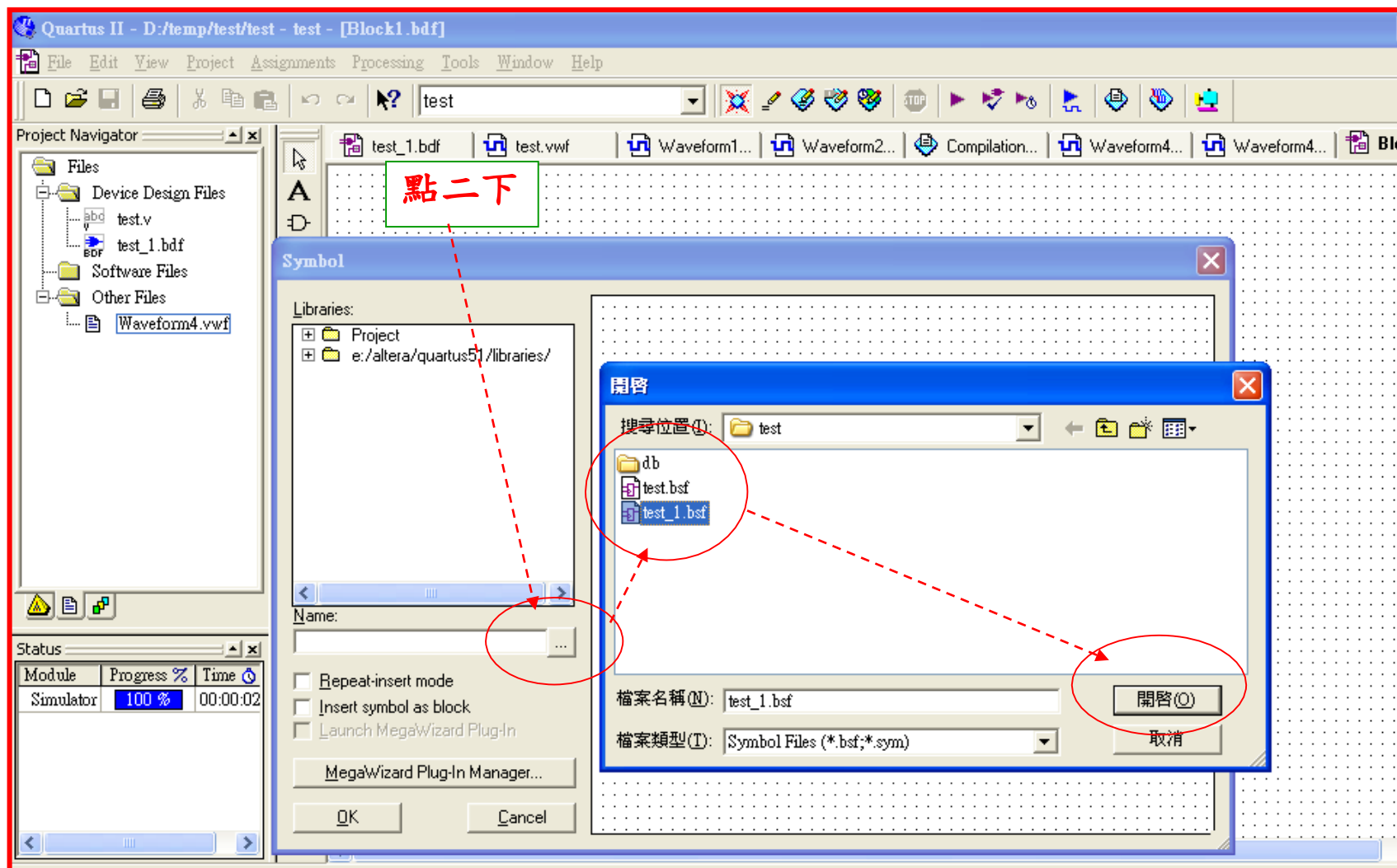
Tasks:

- Flow: Full Design
- Task
 - Start Project
 - Advisors
 - Create Design
 - Assign Constraints
 - Compile Design
 - Analysis & Simulation
 - Fitter (Place and Route)

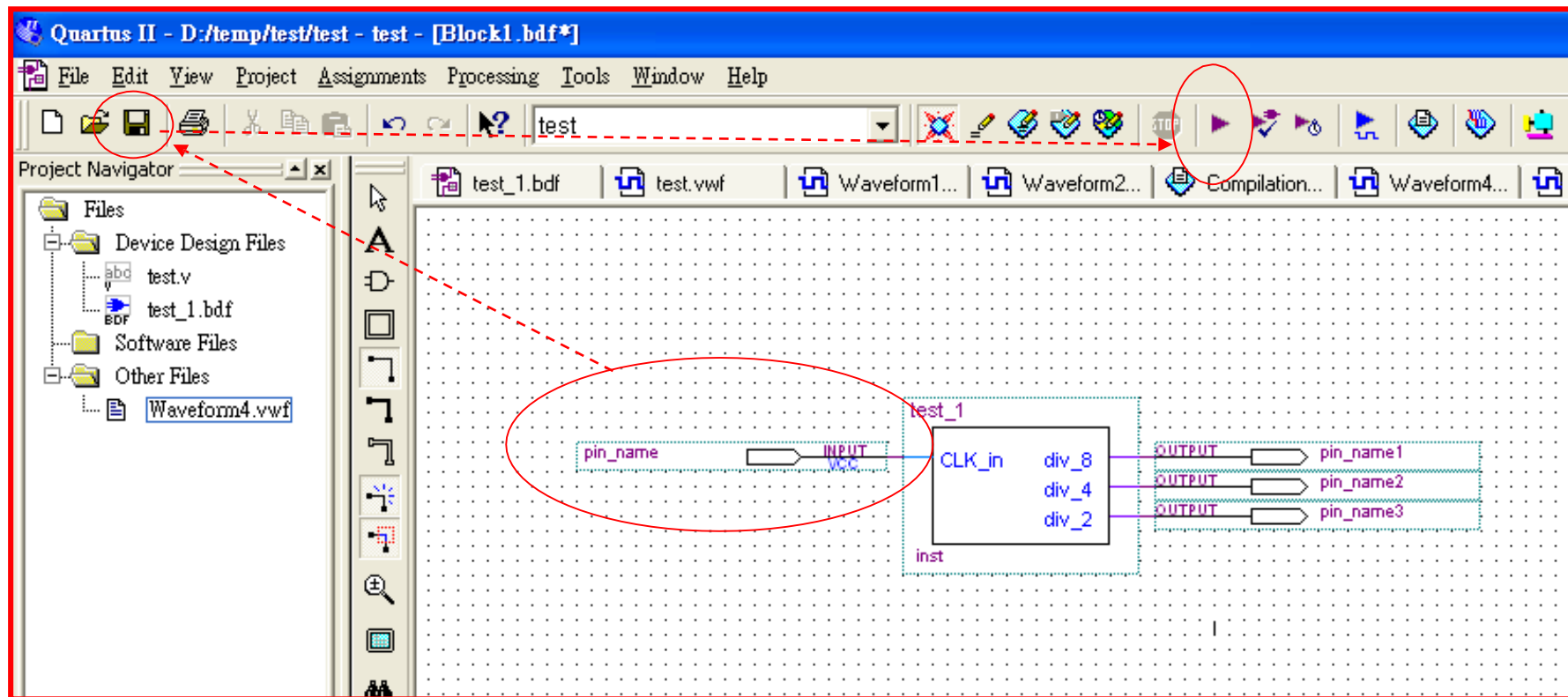
Message Window:

Type	Message
Info	Info: Using vector source file "C:\...
Info	Info: Option to preserve fewer sig...
Info	Info: Simulation partitioned into
Info	Info: Simulation coverage is
Info	Info: Number of transitions in sim...

Bottom-Up Design (36)



Bottom-Up Design (37)



載入輸出入接腳 => 連線 => Save => Compiler

Bottom-Up Design (38)

Project Navigator

- Files
 - test_1.bdf
 - test_1.vwf
 - test.vwf
 - Block2.bdf

Tasks

Flow: Full Design

Task ☒

- Start Project
- Advisors
- Create Design
- Assign Constraints
- ☒ Compile Design
- ☒ Analysis & Synthesis
- ☒ Fitter (Place & Route)

Flow Summary

Flow Status	Successful - Thu Apr 30 09:39:06 2009
Quartus II Version	8.0 Build 215 05/29/2008 SJ Full Version
Revision Name	test
Top-level Entity Name	test_1
Family	Stratix II
Met timing requirements	Yes
Logic utilization	< 1 %
Combinational ALUTs	3 / 12,480 (< 1 %)
Dedicated logic registers	3 / 12,480 (< 1 %)
	3
	4 / 343 (1 %)
	0
	0 / 419,328 (0 %)
	0 / 96 (0 %)
	0 / 6 (0 %)
	0 / 2 (0 %)
Device	EP2S15F484C3
Timing Models	Final

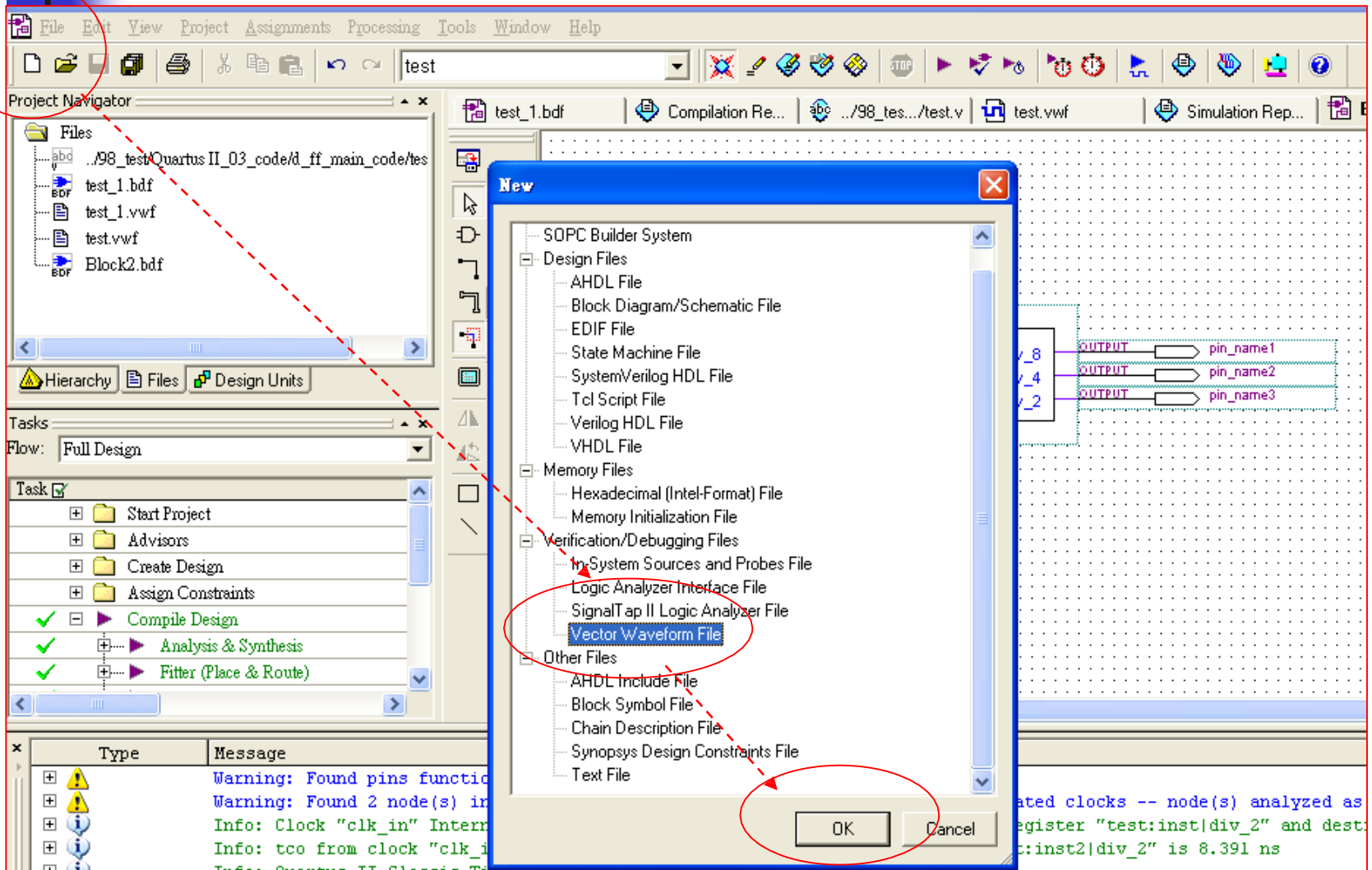
Quartus II

Full Compilation was successful (7 warnings)

确定

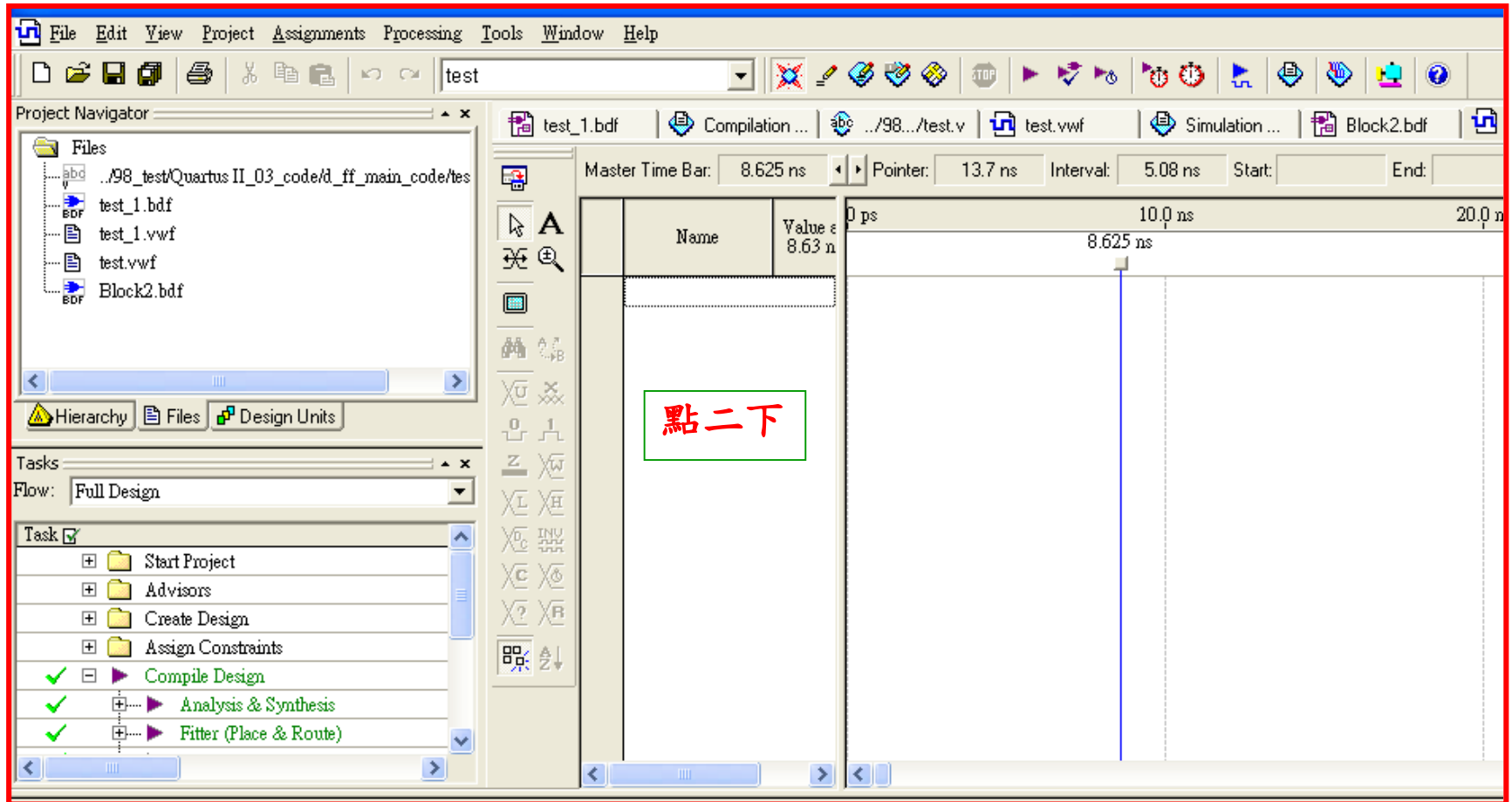
Compiler => OK.

Bottom-Up Design (39)

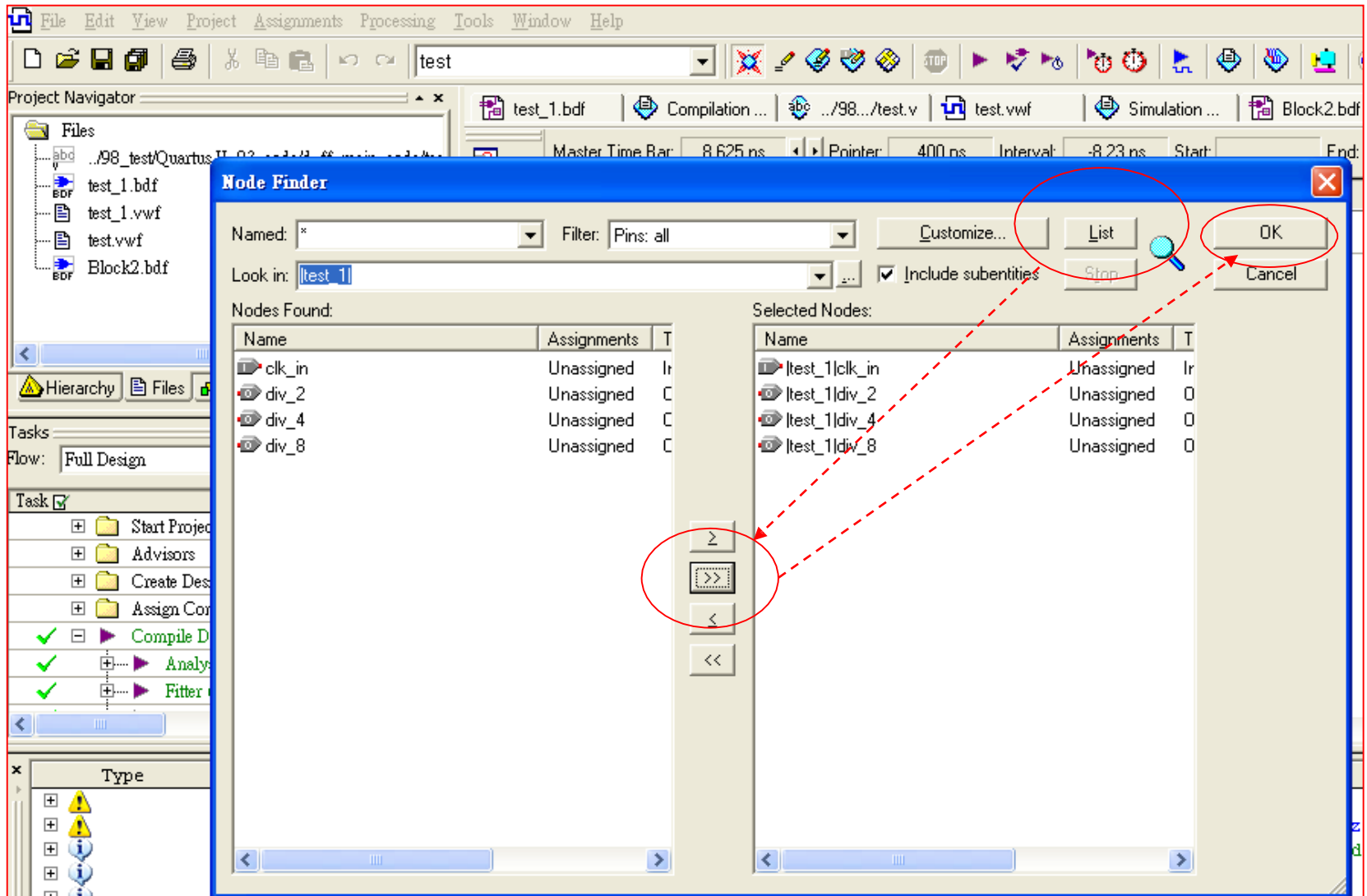


New => Other Files=>Vector Waveform File=>OK.

Bottom-Up Design (40)



Bottom-Up Design (41)

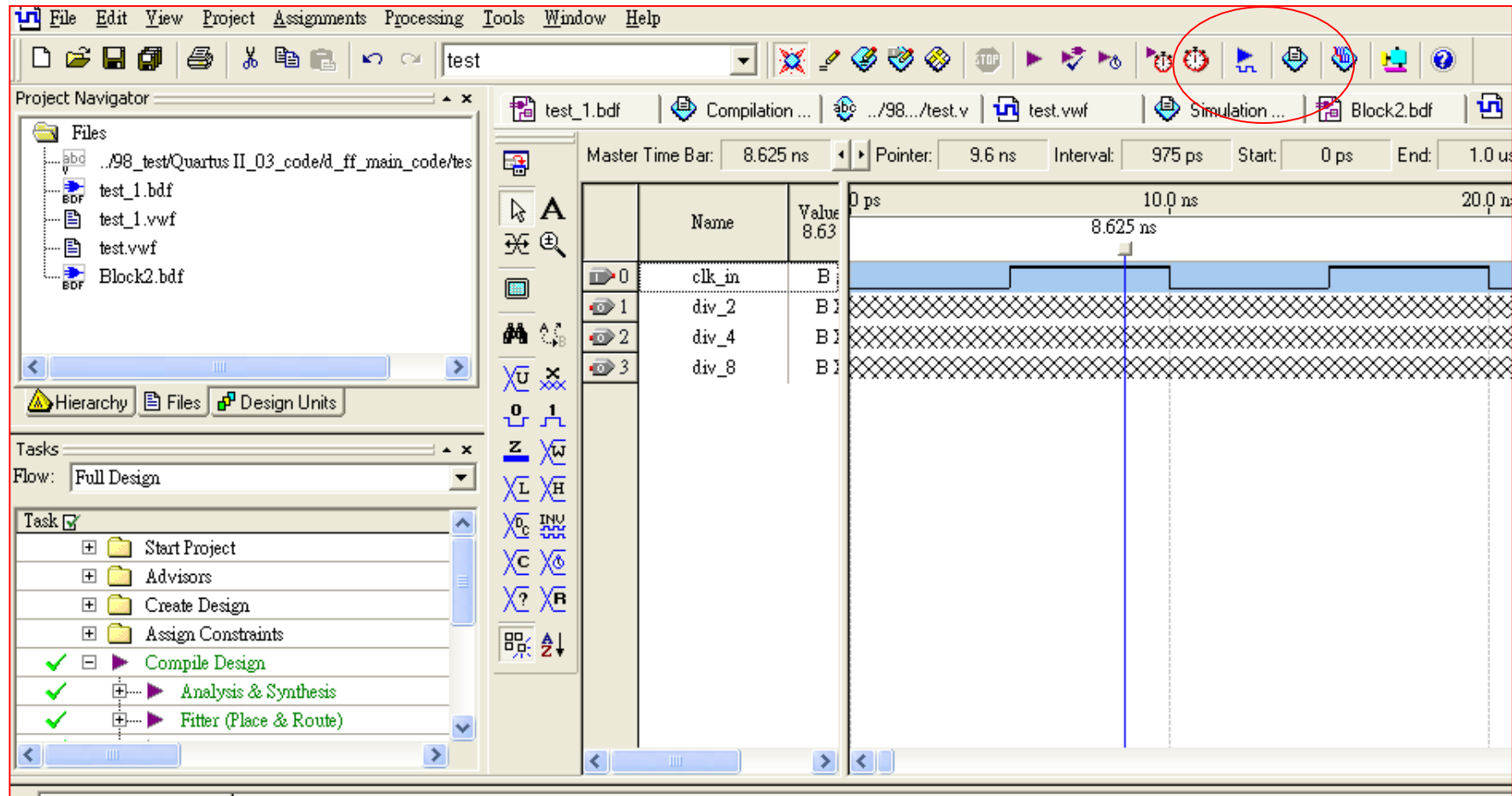


Bottom-Up Design (42)

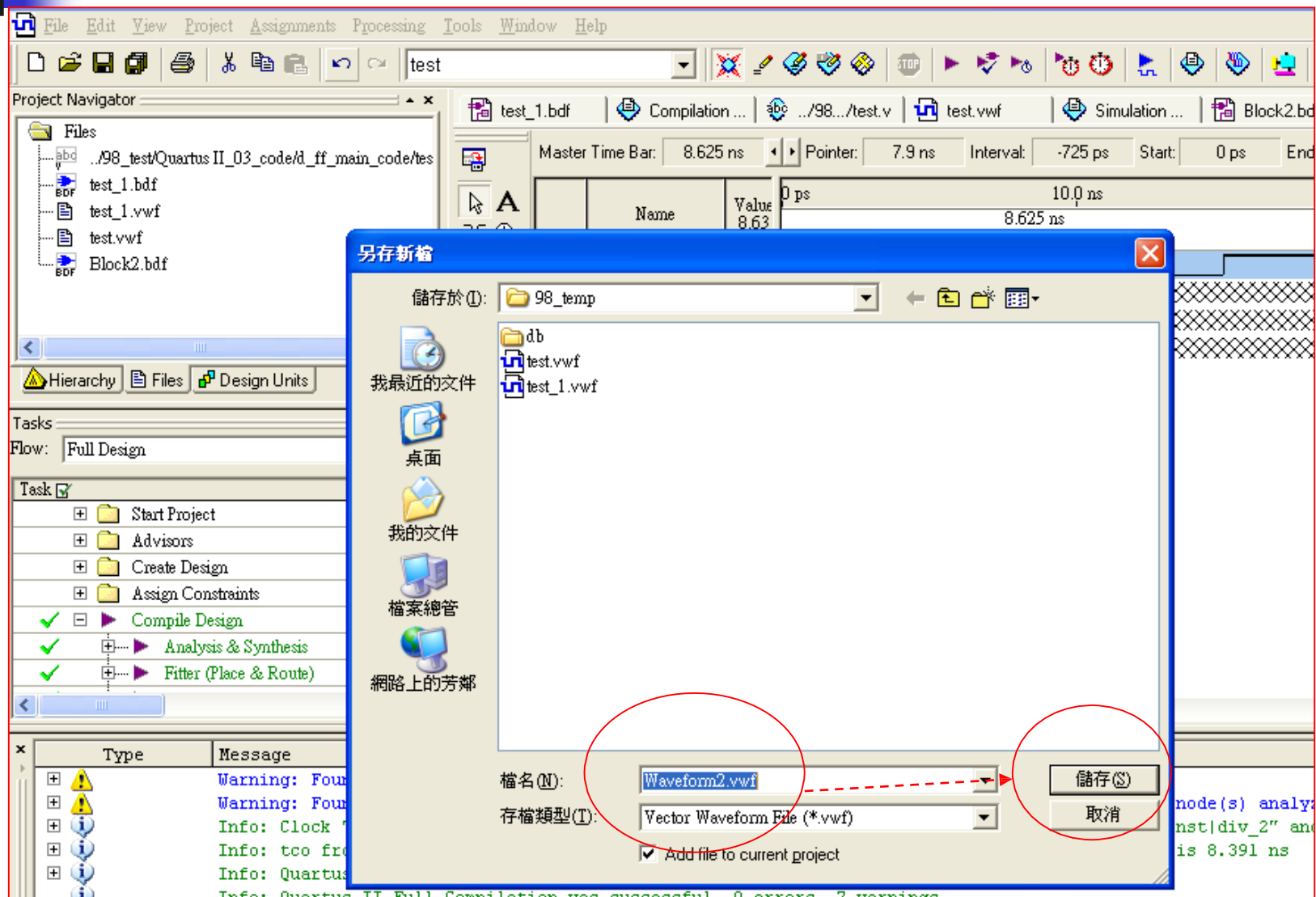
The screenshot displays the Quartus II software interface, illustrating the Bottom-Up Design process. The interface is divided into several panes:

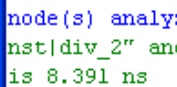
- Project Navigator:** Located on the left, it shows the project files. The "Files" pane lists: test_1.bdf, test_1.vwf, test.vwf, and Block2.bdf. The "Hierarchy" pane shows the project structure.
- Tasks:** Located below the Project Navigator, it shows the design flow. The "Flow" is set to "Full Design". The tasks listed are: Start Project, Advisors, Create Design, Assign Constraints, Compile Design (checked), Analysis & Synthesis (checked), and Fitter (Place & Route) (checked).
- Signal List:** Located in the center, it displays a table of signals. The signals are: clk_in, div_2, div_4, and div_8. The "Name" column is circled in red, and a red dashed arrow points from the "div_2" signal to the "Fitter (Place & Route)" task in the Tasks pane.
- Timing Diagram:** Located on the right, it shows a timing diagram. The "Master Time Bar" is set to 8.625 ns. The "Pointer" is set to 1.9 ns, and the "Interval" is set to -6.73 ns. The diagram shows a signal transition at 8.625 ns, with a vertical cursor at that time.

Bottom-Up Design (43)

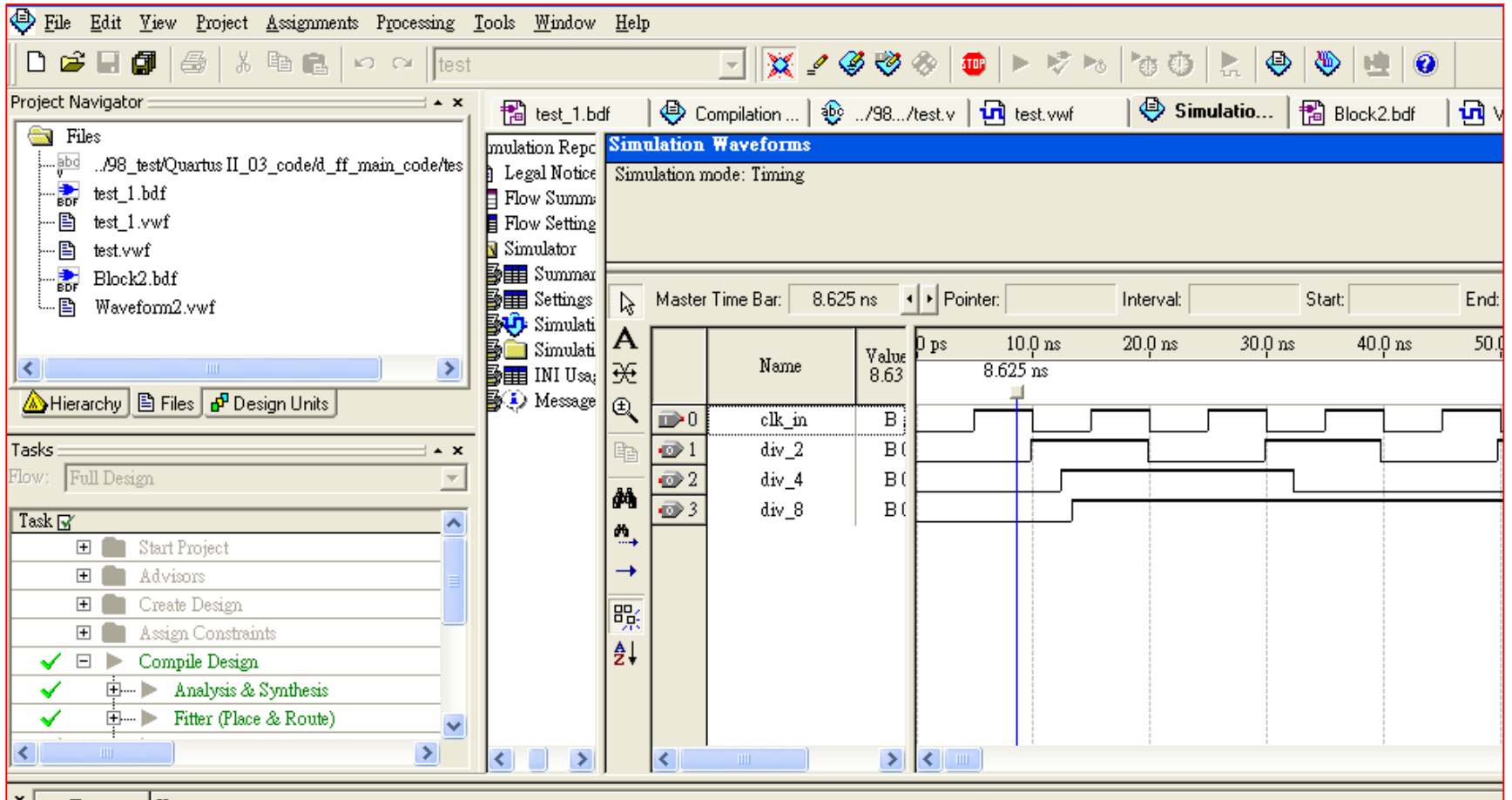


Bottom-Up Design (44)





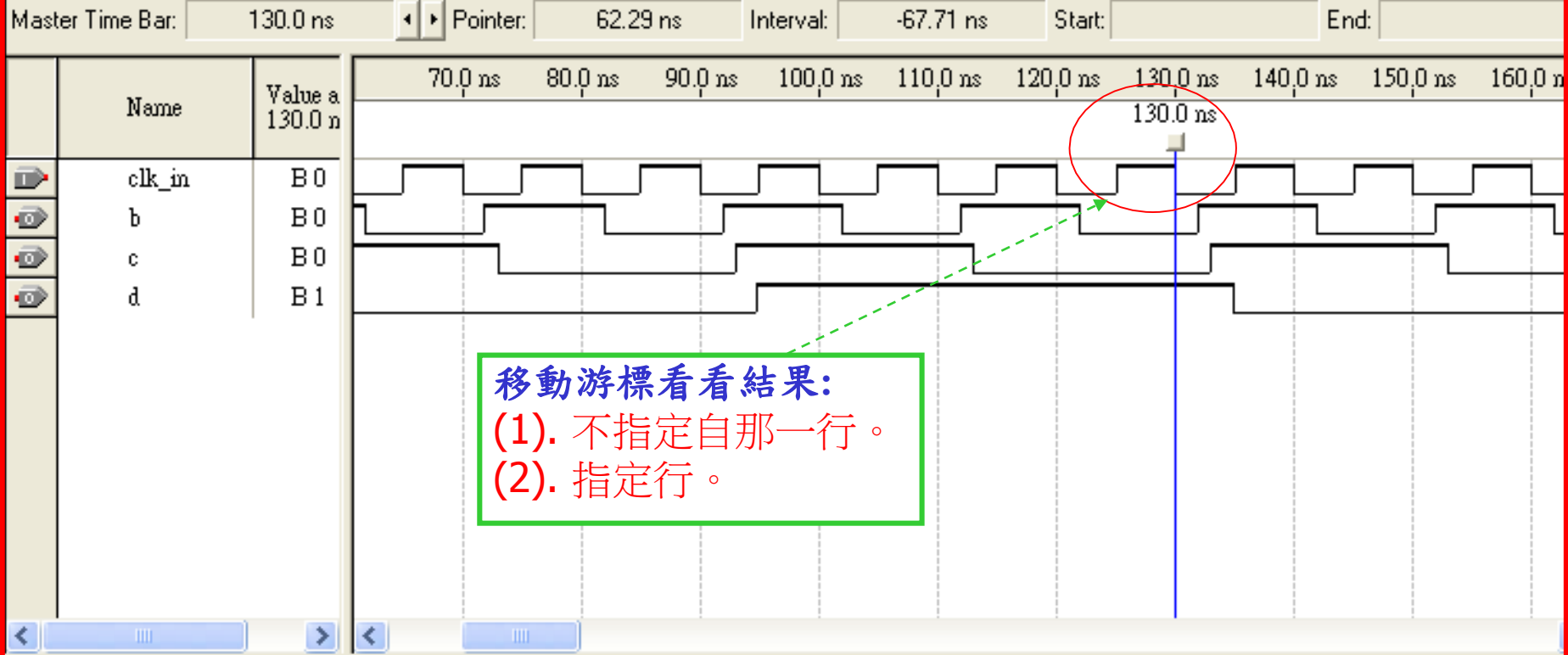
Bottom-Up Design (45)



Bottom-Up Design (46)

Simulation Waveforms

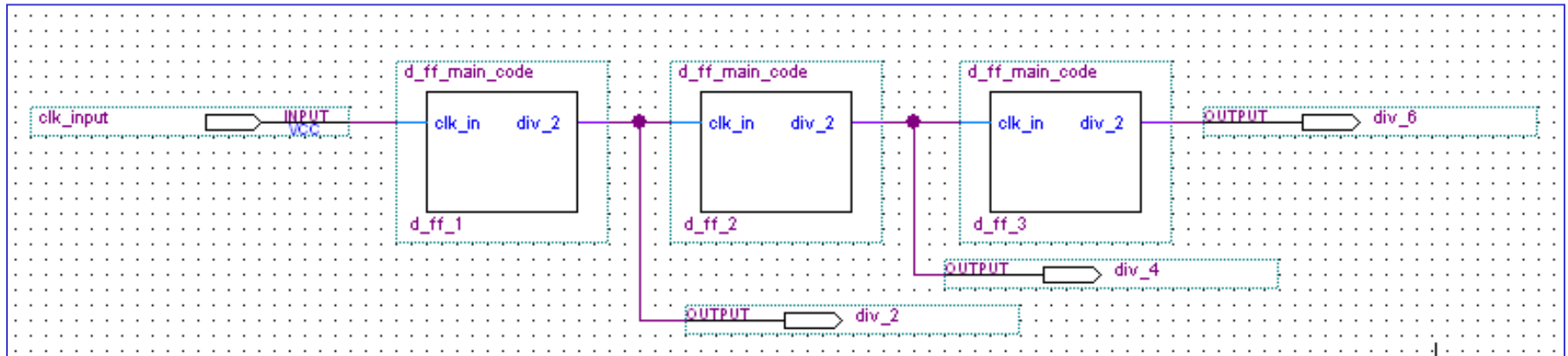
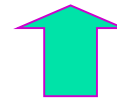
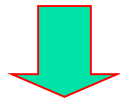
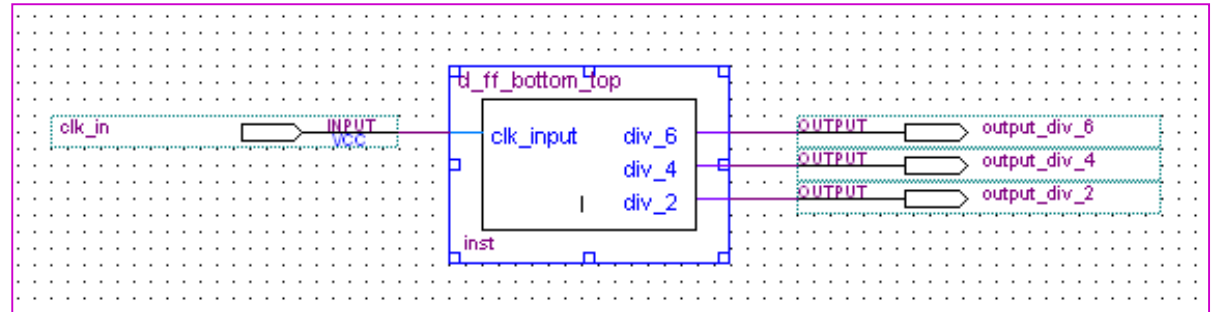
Simulation mode: Timing

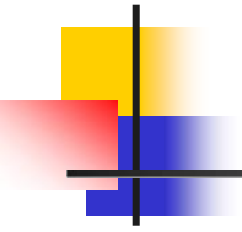


Bottom-Up Design (47)

Output

```
module d_ff_main_code(clk_in,div_2);  
    input clk_in;  
    output div_2;  
  
    reg div_2;  
  
    always@(posedge clk_in)  
    begin  
        div_2 = div_2 ^ 1;  
    end  
endmodule
```



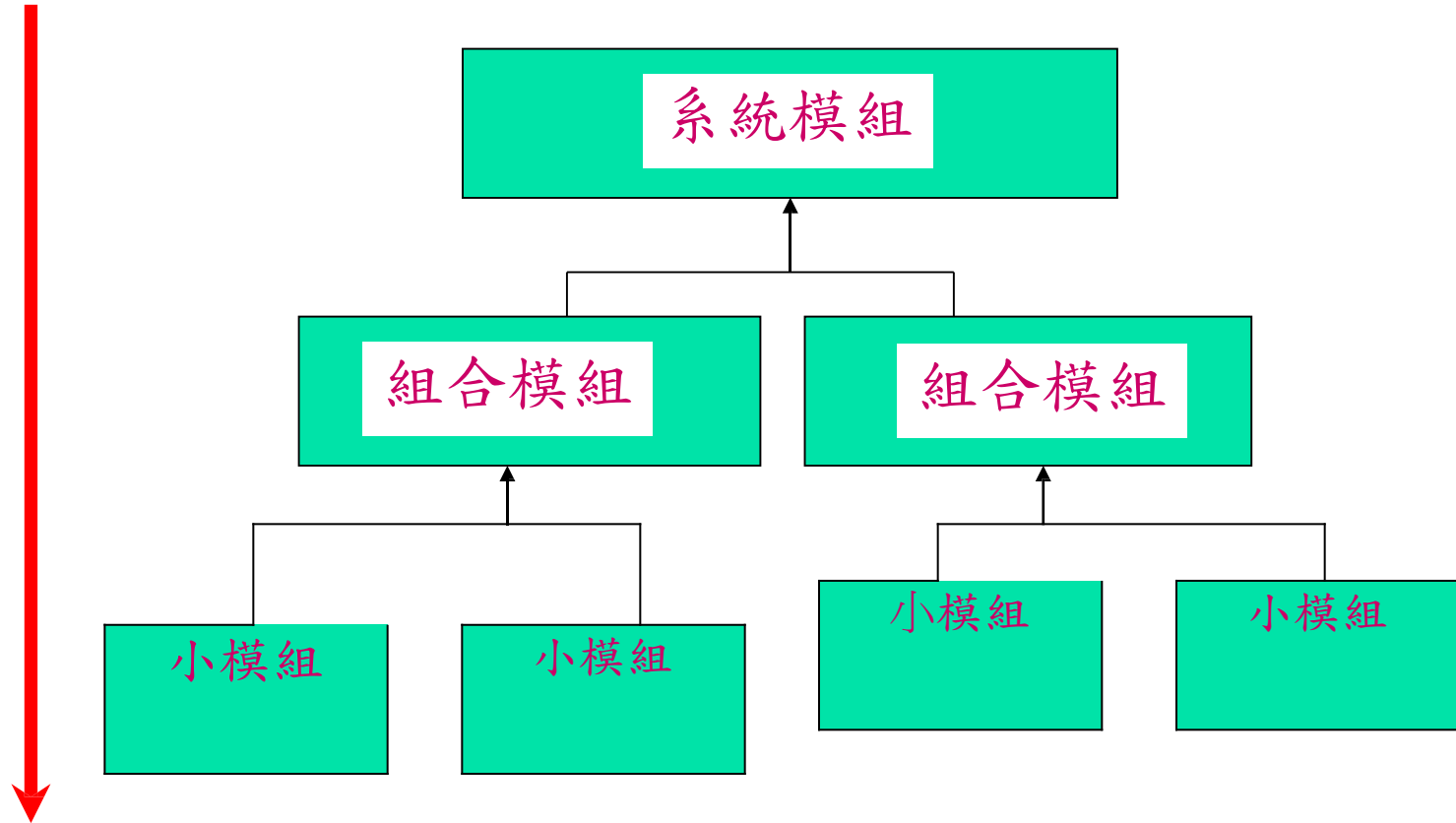


Top-Down Design

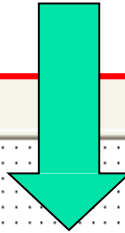
Top-Down
Design Flow

Top-Down Design

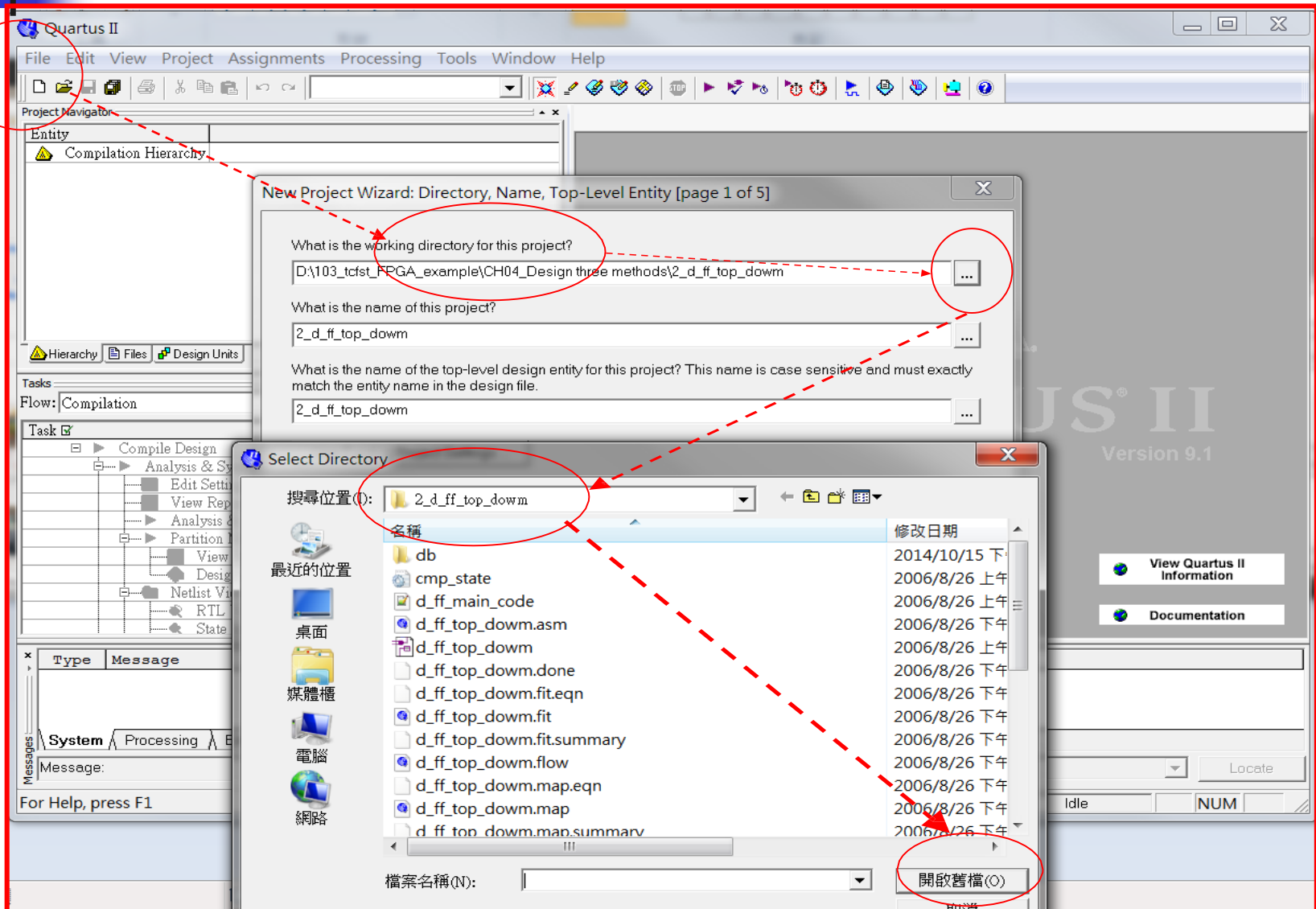
設計的順序



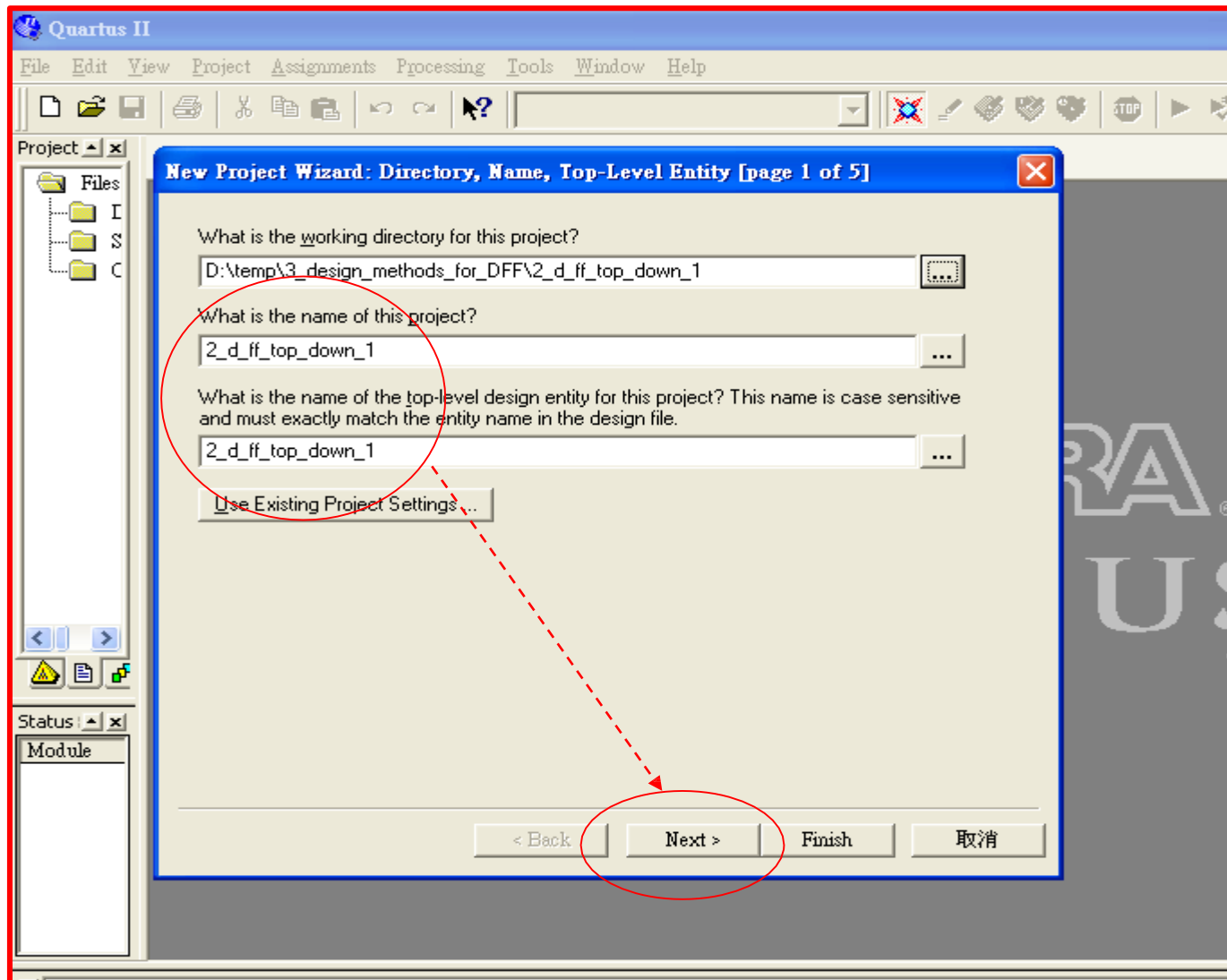
目標



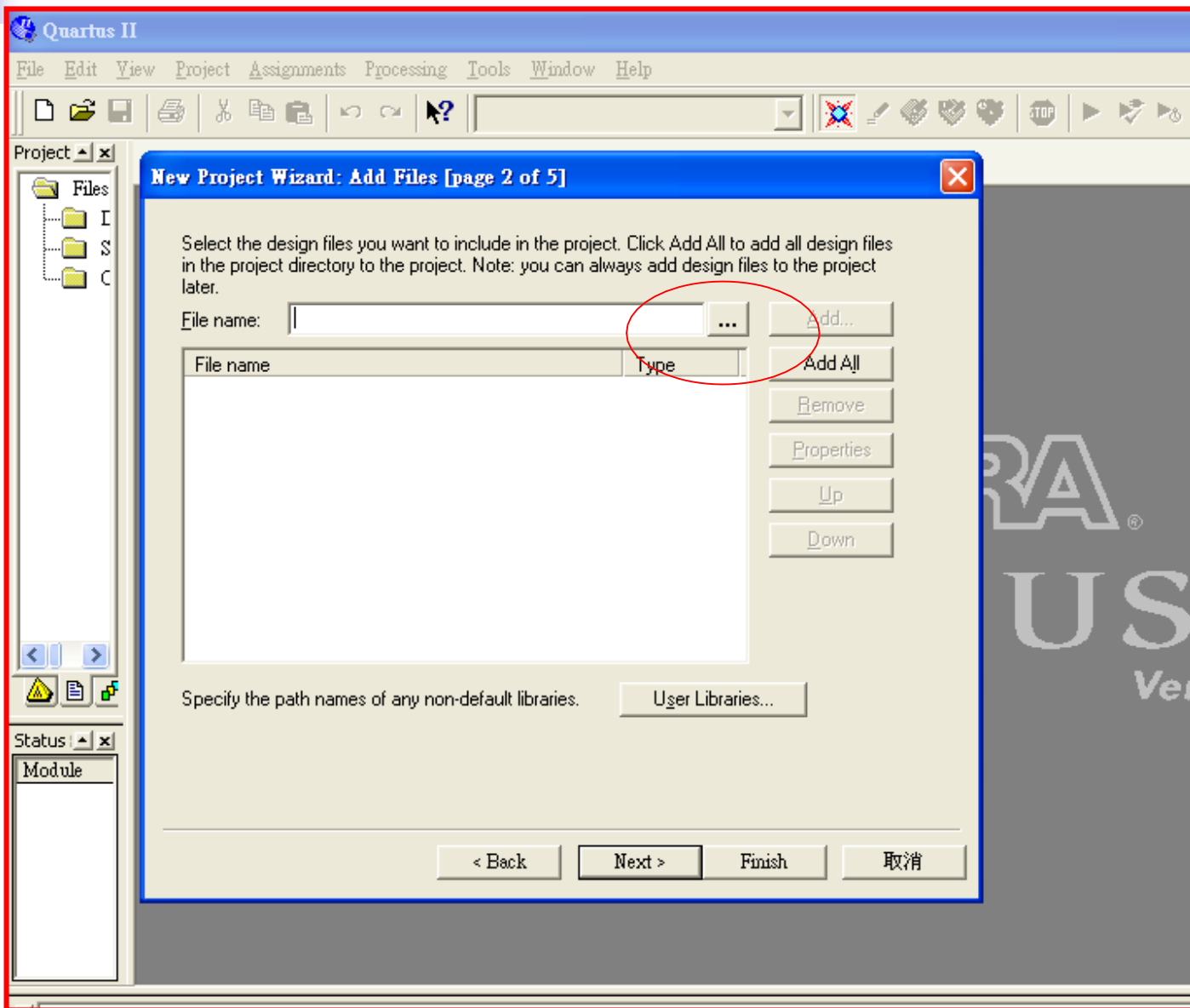
Top-Down Design (2)



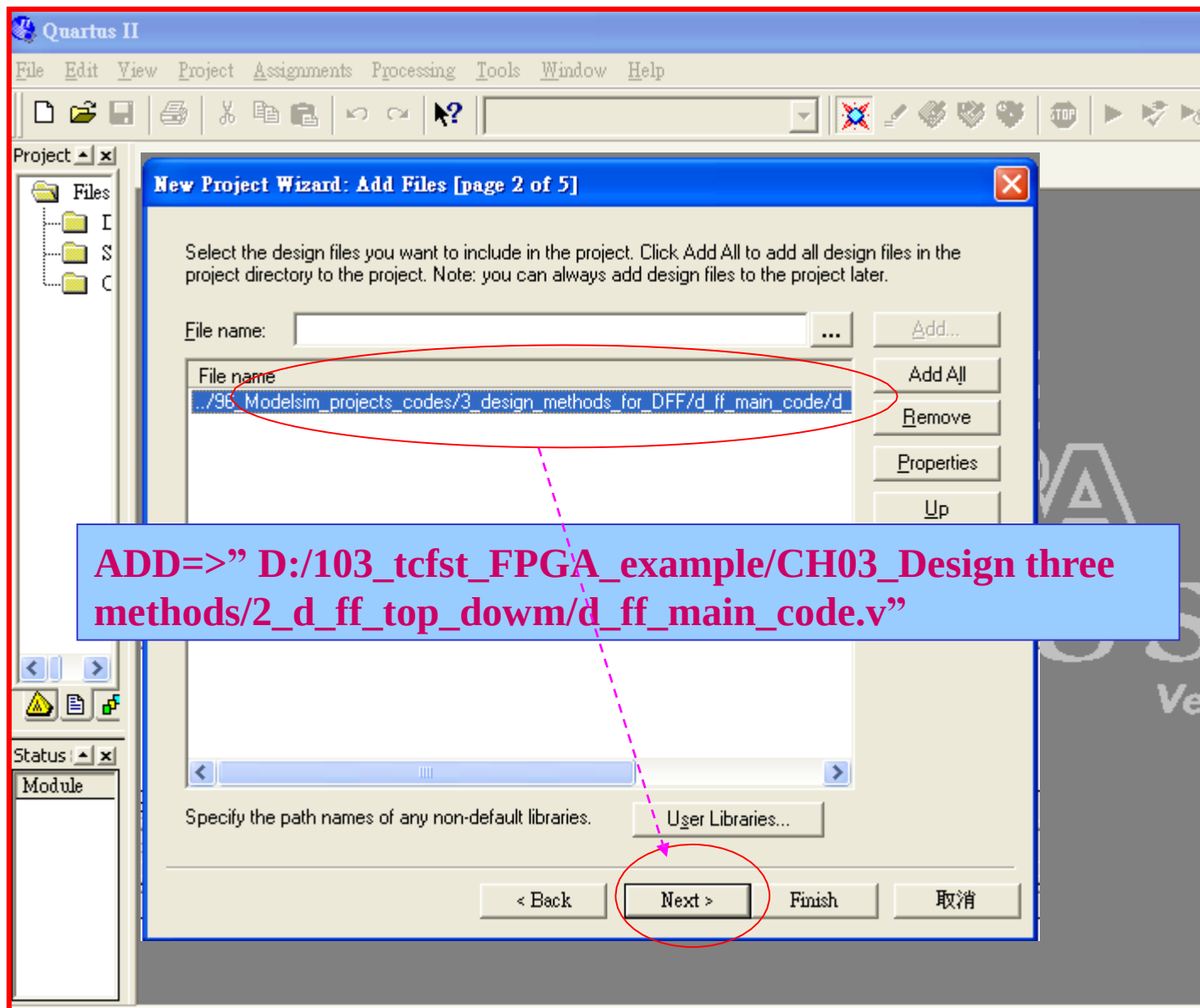
Top-Down Design (3)



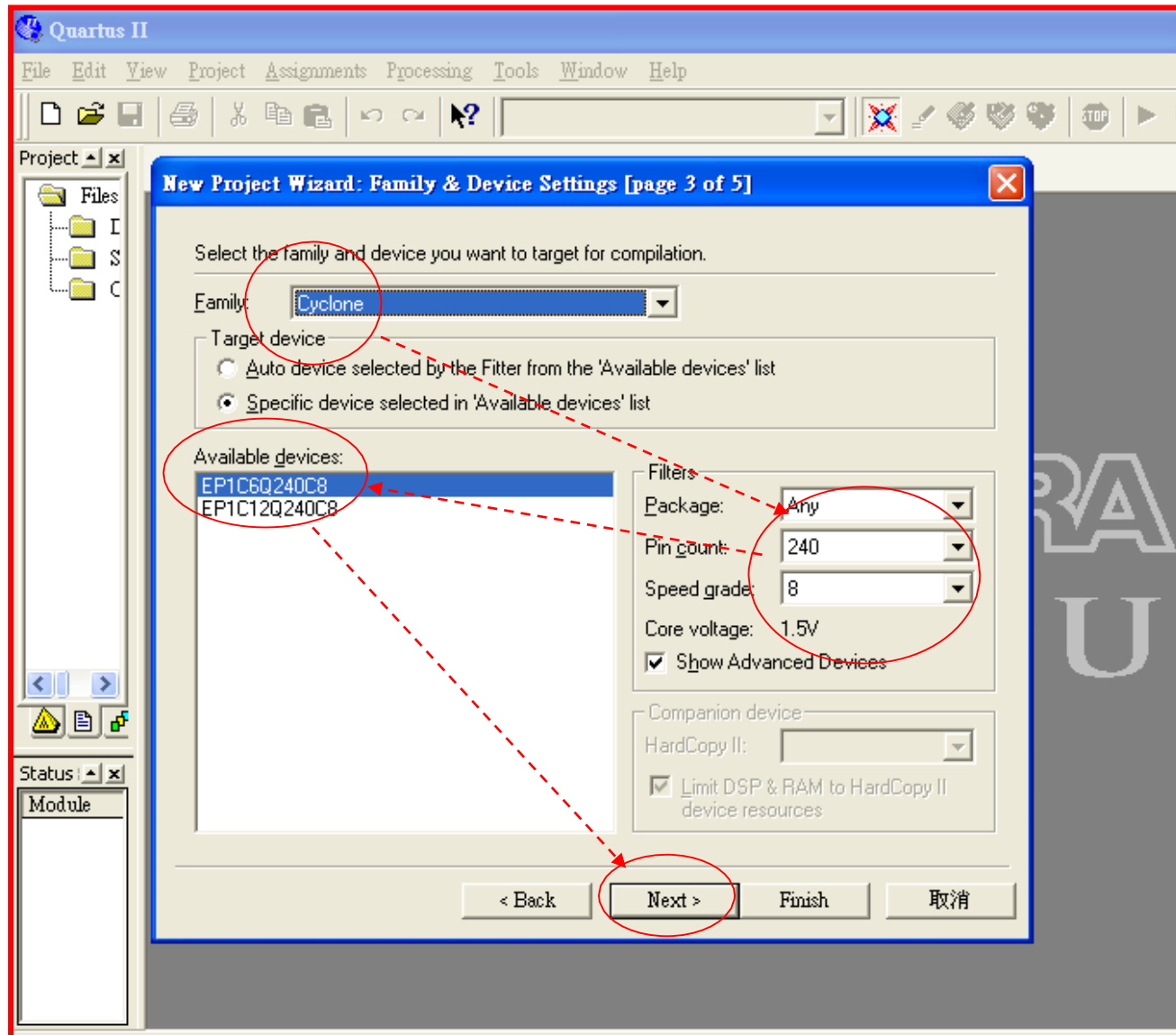
Top-Down Design (4)



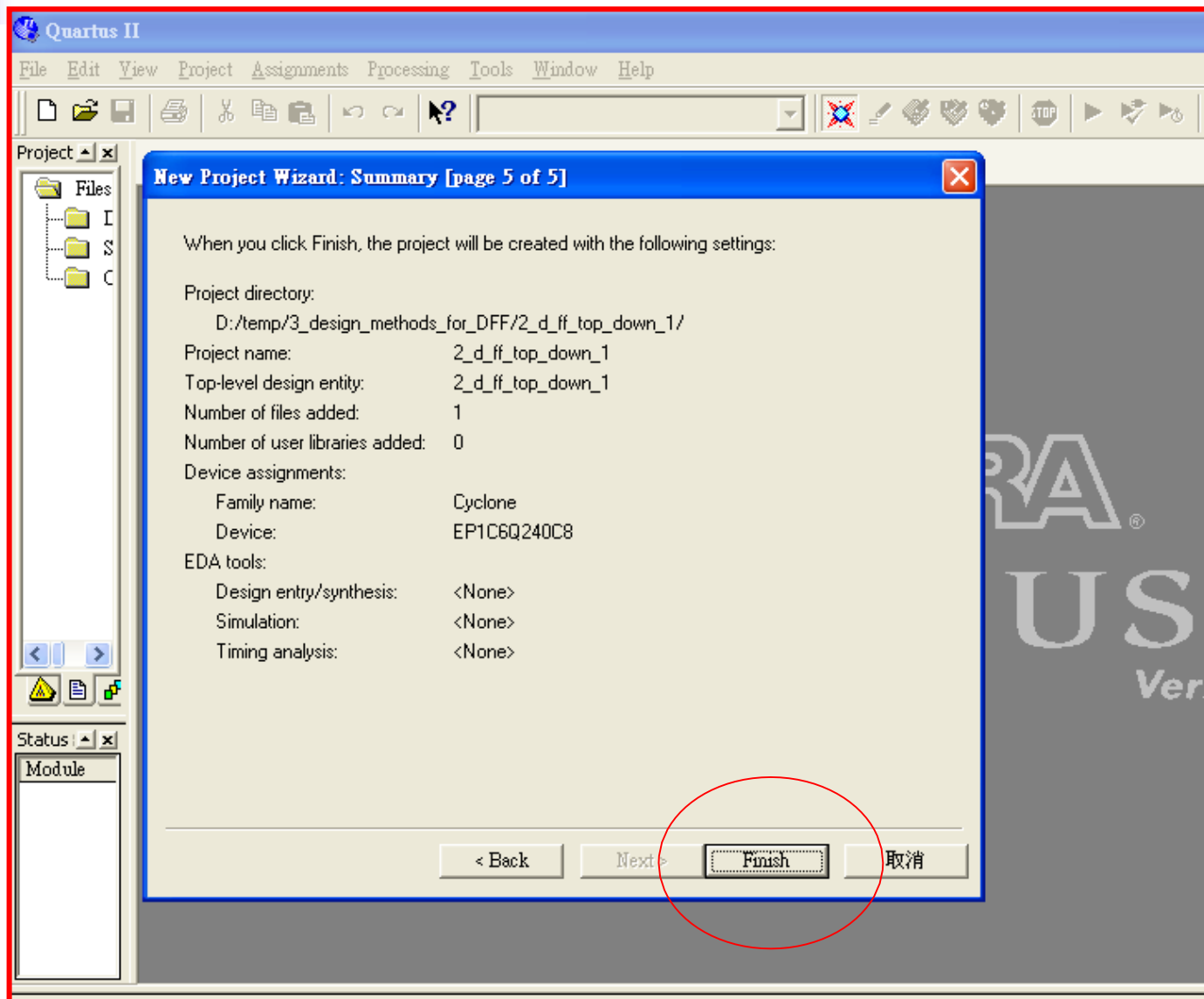
Top-Down Design (5)



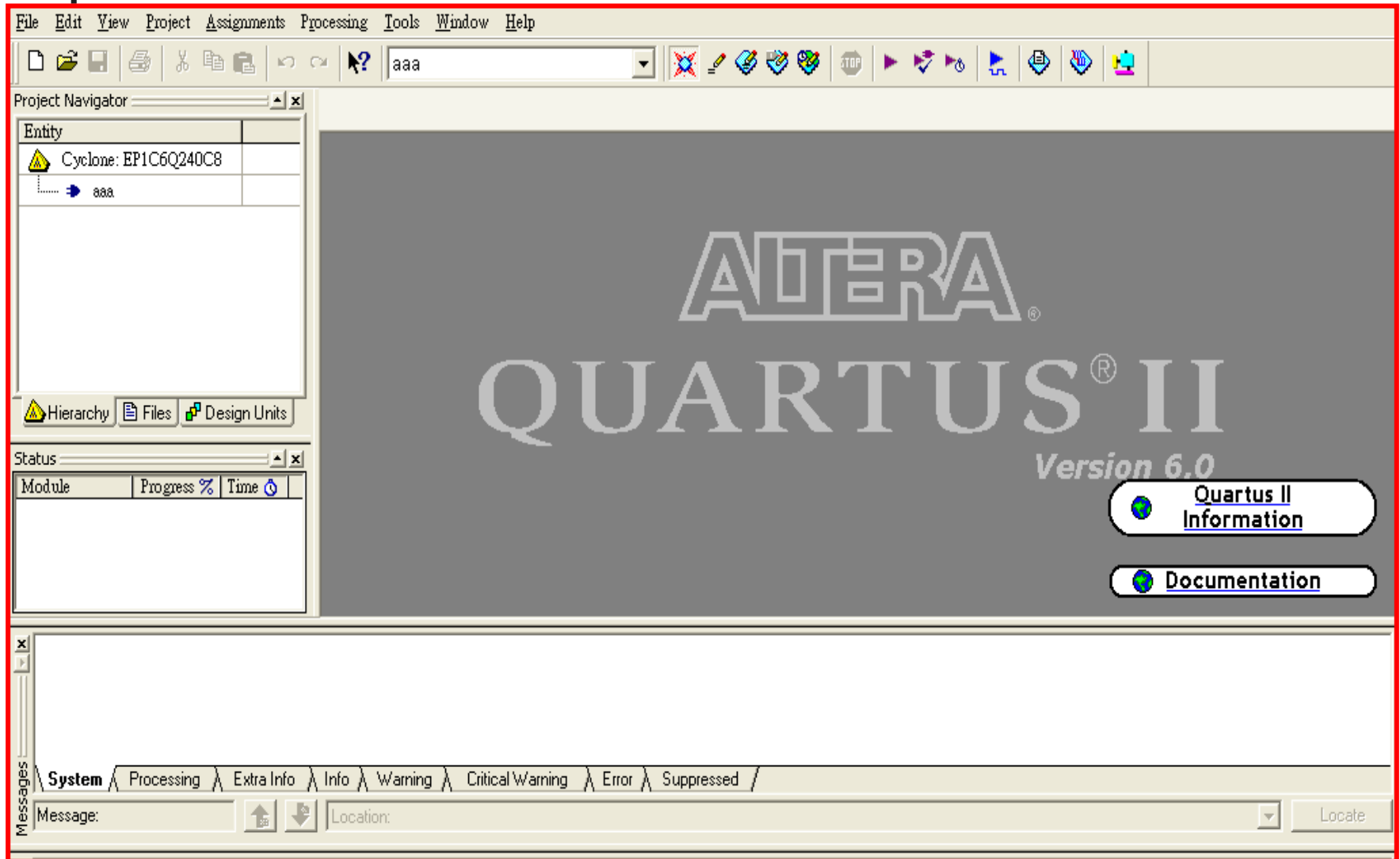
Top-Down Design (6)



Top-Down Design (7)



Top-Down Design (8)



Top-Down Design (9)

Quartus II - D:/temp/3_design_methods_for_DFF/2_d_ff_top_down_1/2_d_ff_top_down_1 - 2_d_ff_top_down_1

File Edit View Project Assignments Processing Tools Window Help

2_d_ff_top_down_1

Project Explorer: Files, I, S, C

Block1.bdf*

(1)

(2)

block_name

I/O	Type

inst

(1). File => New => Design Files => Block diagram/schematic file.
(2). Block => 選定適當大小 => 出現” Block 圖框” .

Top-Down Design (10)

選擇圖框=>右鍵=>Block Properties=>General=>改名稱" d_ff_main_code".

Block Properties

General I/Os Parameters Format

Name: block_name

Instance name: inst

```
ln #
1 module d_ff_main_code(clk_in, div_2);
2
3   input clk_in;
4   output div_2;
5
6   reg div_2;
7
8   always@(posedge clk_in)
9   begin
10     div_2 = div_2 ^ 1;
11   end
12
13 endmodule
14
```

確定 取消

d_ff_main_code.v

Top-Down Design (11)

Block Properties

General I/Os Parameters Format

I/O

Name: clk_in

Type: INPUT

Add

Delete

Existing block I/Os:

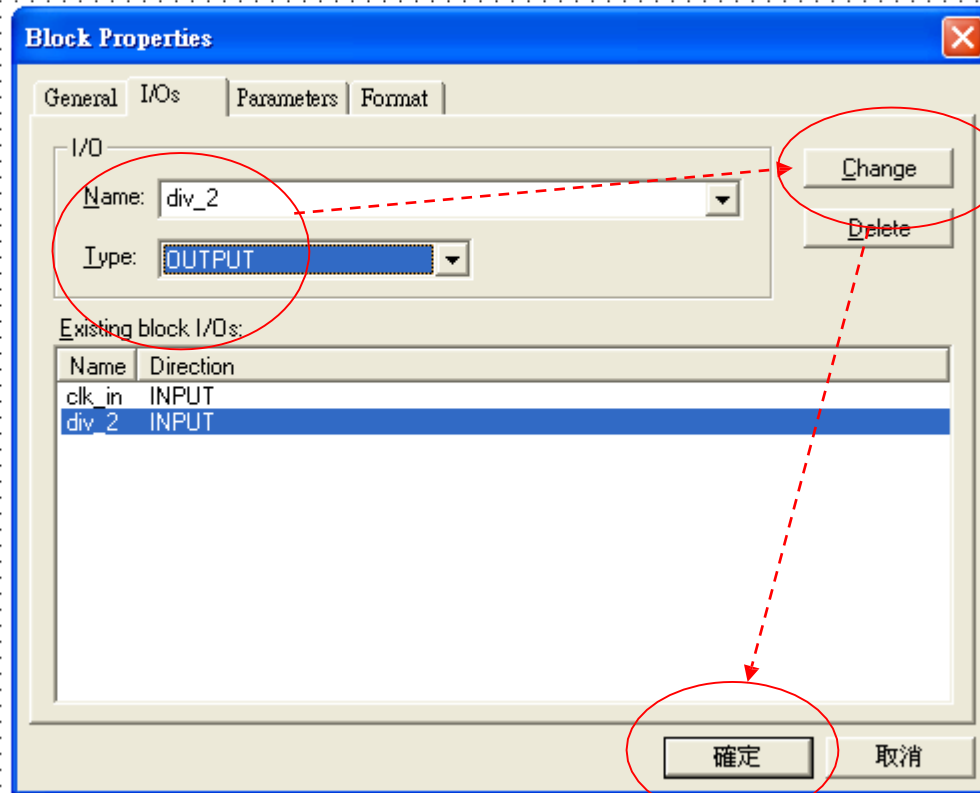
確定 取消

```
ln #
1 module d_ff_main_code (clk_in, div_2);
2
3 input clk_in;
4 output div_2;
5
6 reg div_2;
7
8 always@(posedge clk_in)
9 begin
10     div_2 = div_2 ^ 1;
11 end
12
13 endmodule
14
```

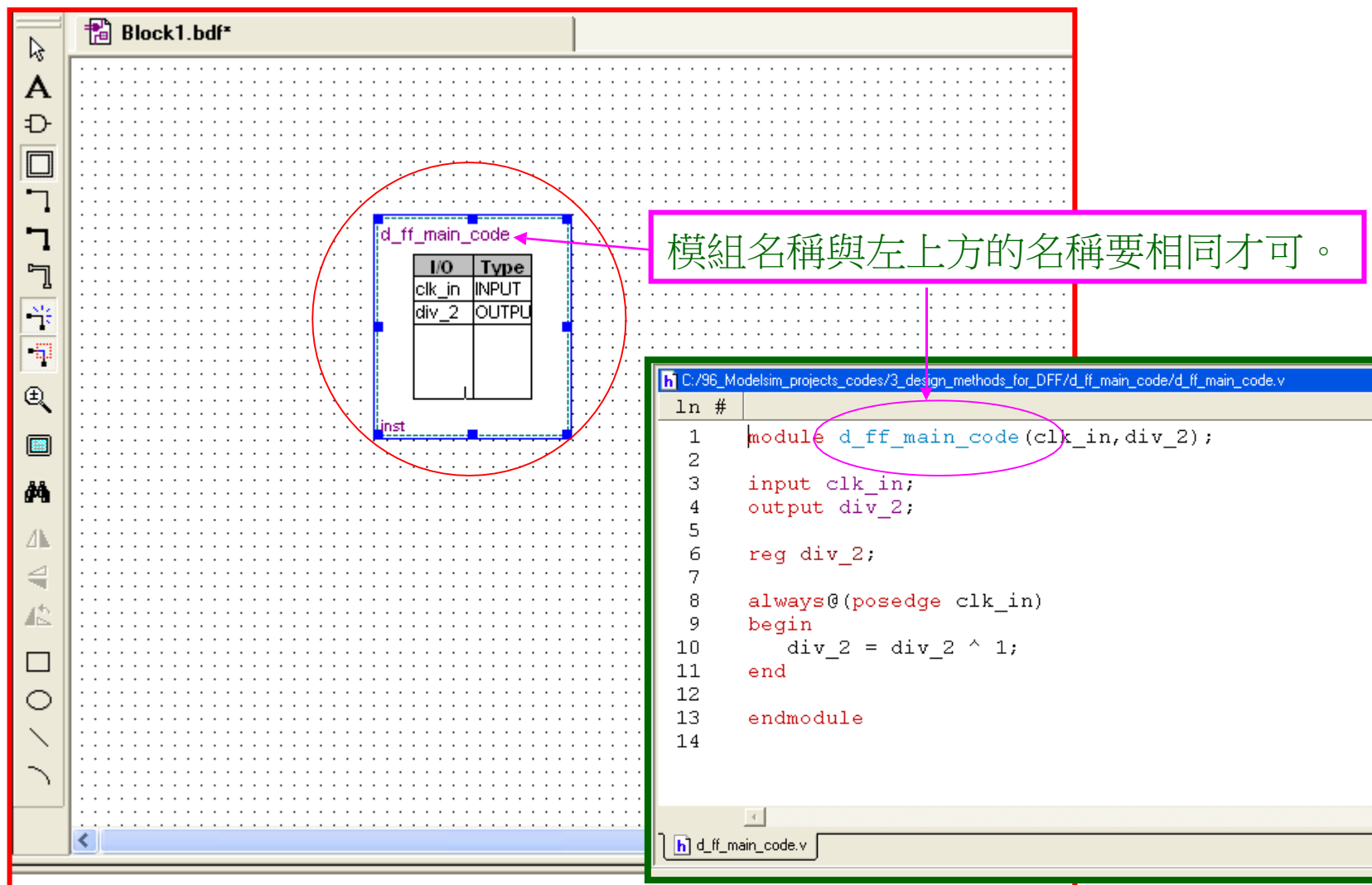
選擇I/Os=>鍵入" clk_in"=>選擇" Type"為" INPUT"=>Ad

Top-Down Design (12)

鍵入" div_2"=>選擇" Type"為" OUTPUT"=>Add =>確定



Top-Down Design (13)



Block1.bdf*

d_ff_main_code

I/O	Type
clk_in	INPUT
div_2	OUTPUT

inst

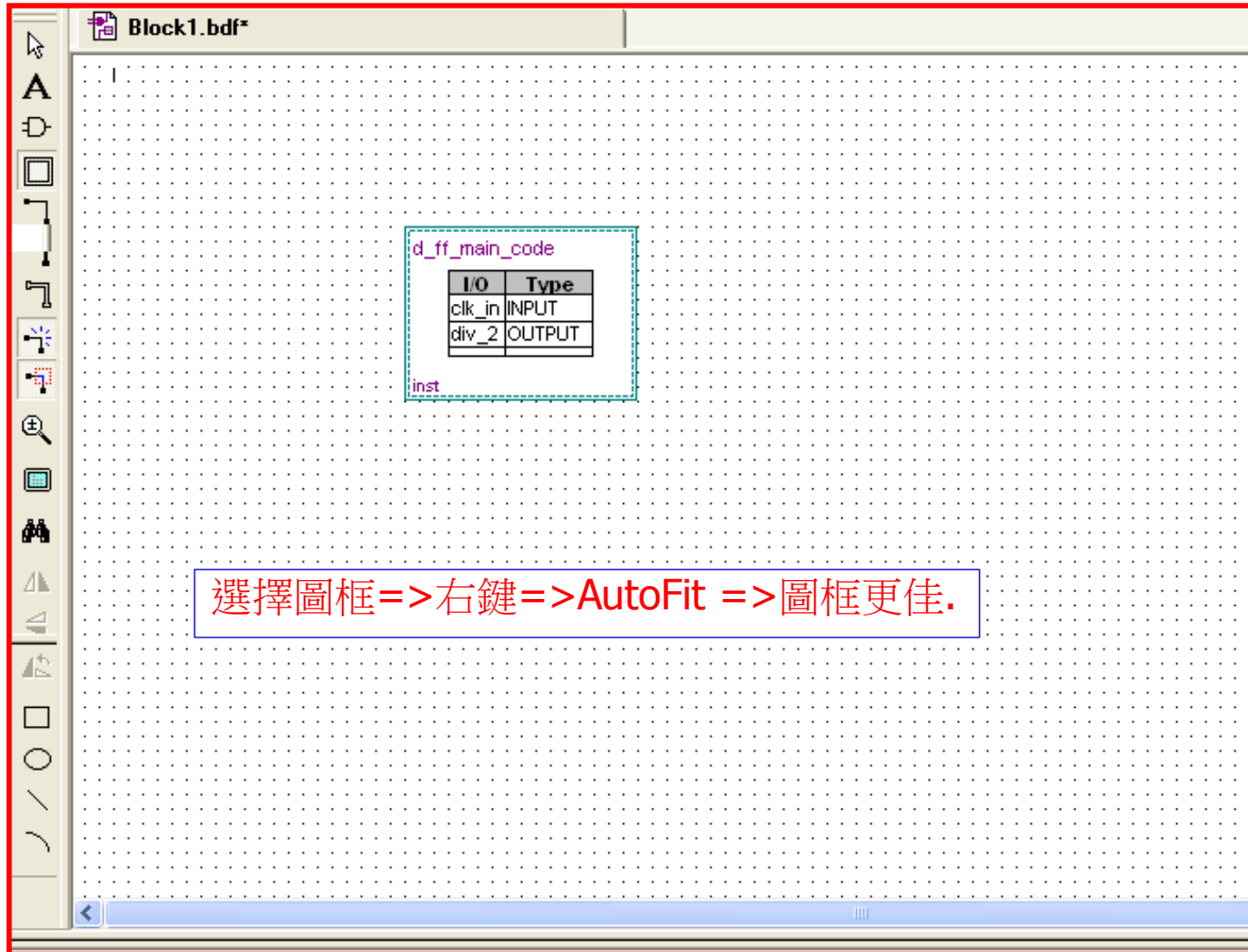
模組名稱與左上方的名稱要相同才可。

C:/96_Modelsim_projects_codes/3_design_methods_for_DFF/d_ff_main_code/d_ff_main_code.v

```
ln #
1 module d_ff_main_code(clk_in, div_2);
2
3   input clk_in;
4   output div_2;
5
6   reg div_2;
7
8   always@(posedge clk_in)
9   begin
10     div_2 = div_2 ^ 1;
11   end
12
13   endmodule
14
```

d_ff_main_code.v

Top-Down Design (14)



Top-Down Design (15)

The screenshot shows a logic design software interface. The main workspace displays a block named `d_ff_main_code` with the following I/O table:

I/O	Type
clk_in	INPUT
div_2	OUTPUT

Below the table, the block is labeled `inst`. A blue text box above the block states: "Device Design Files => 右鍵 => ADD => '.../d_ff_main_code.v'". A green text box to the right of the block says: "點二下看其內含的Code。". A large green arrow points from the block to the code editor window below.

The code editor window shows the following Verilog code:

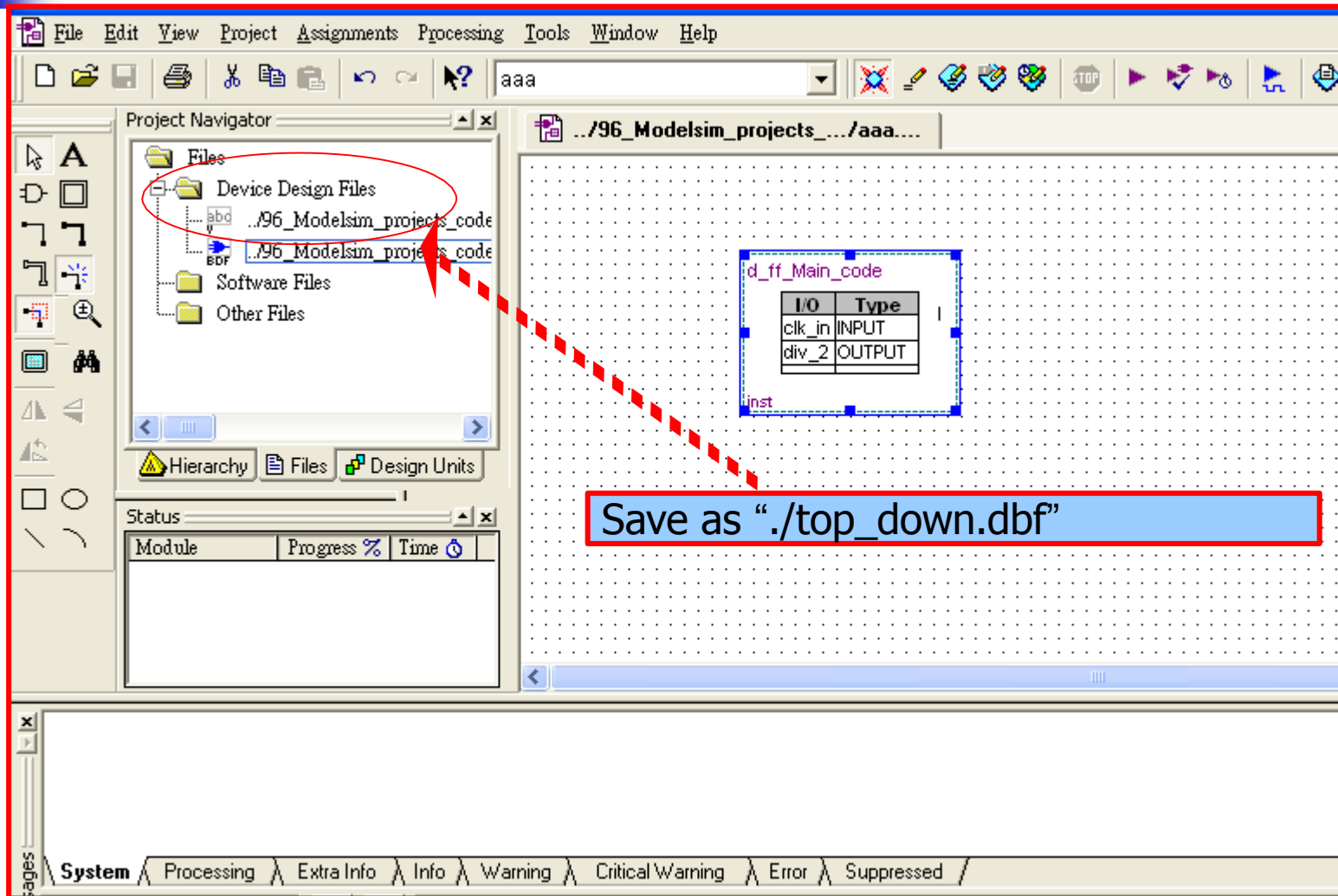
```
1 module d_ff_main_code(clk_in, div_2);
2
3 input  clk_in;
4 output div_2;
5
6 reg div_2;
7
8 always@ (posedge clk_in)
9 begin
10     div_2 = div_2 ^ 1;
11 end
12 endmodule
13
14
```

The Project Navigator on the left shows the file structure:

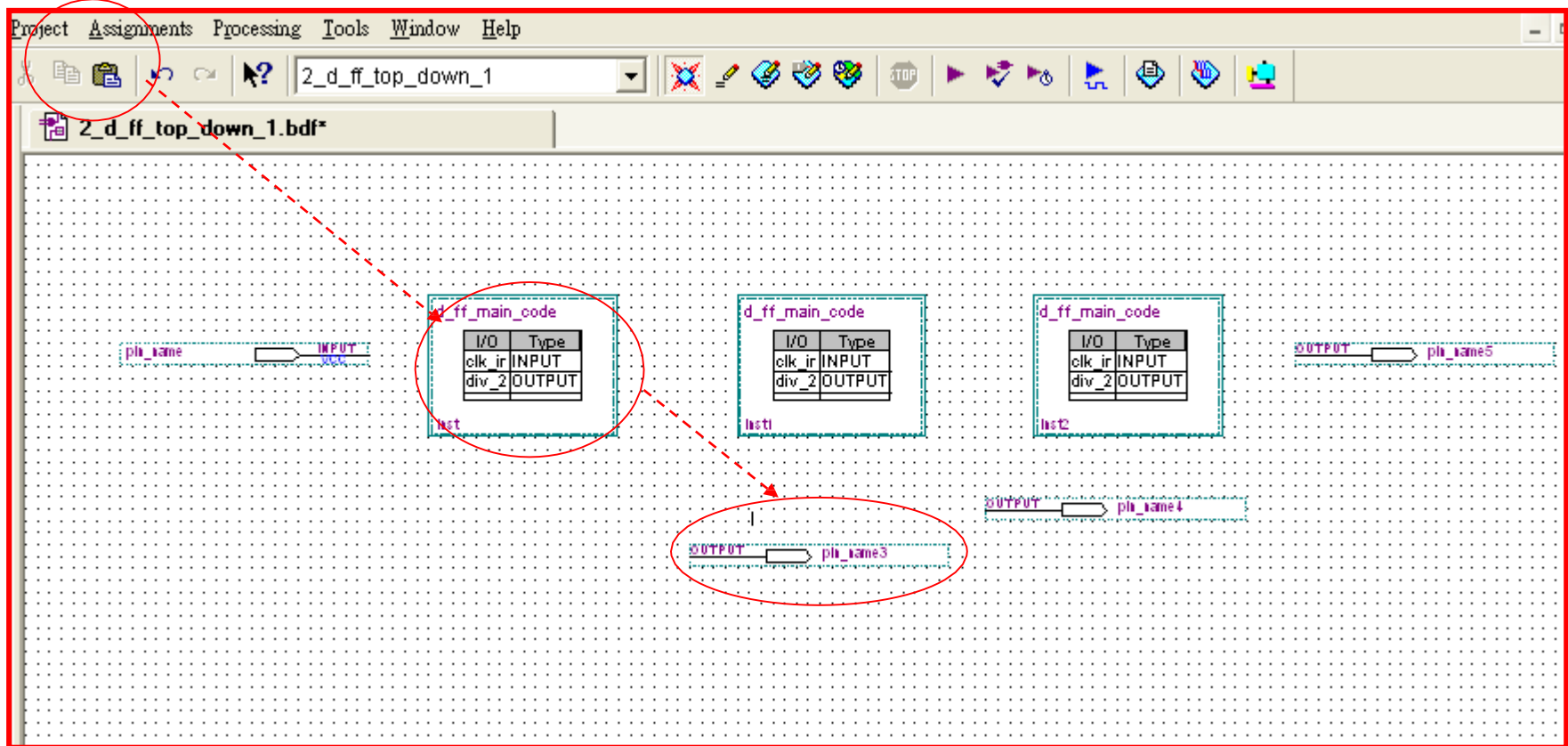
- Files
 - Device Design Files
 - ..96_Modelsim_projects code
 - Software Files
 - Other Files

The Status bar at the bottom shows "Module", "Progress %", and "Time".

Top-Down Design (16)

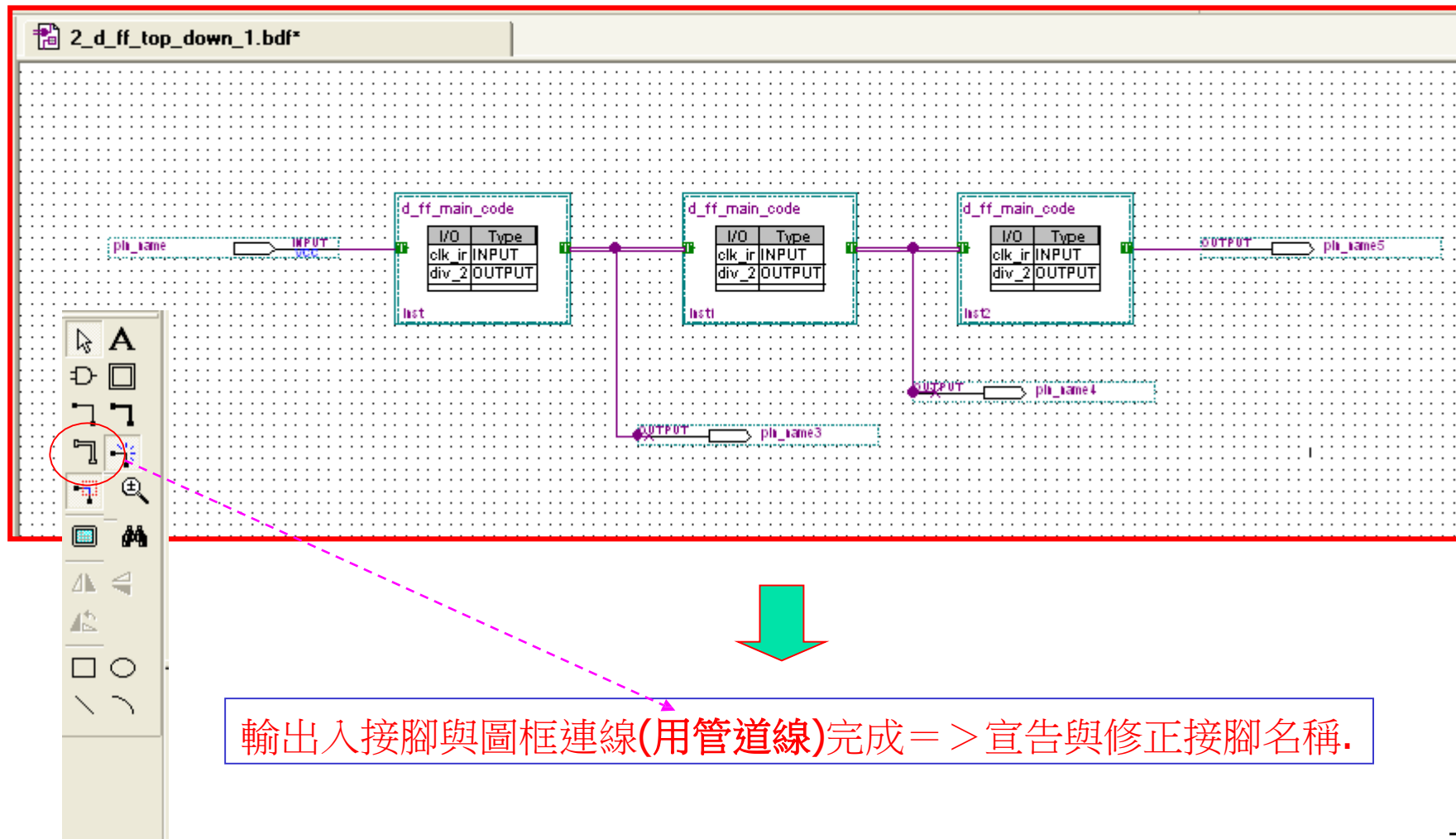


Top-Down Design (17)



Copy 2 Figures => Insert "Input and Output Pins" => 連線

Top-Down Design (18)



Top-Down Design (19)

2_d_ff_top_down_1.bdf*

Manner Properties

General | Mappings

Block name: d_ff_main_code

Block instance name: inst

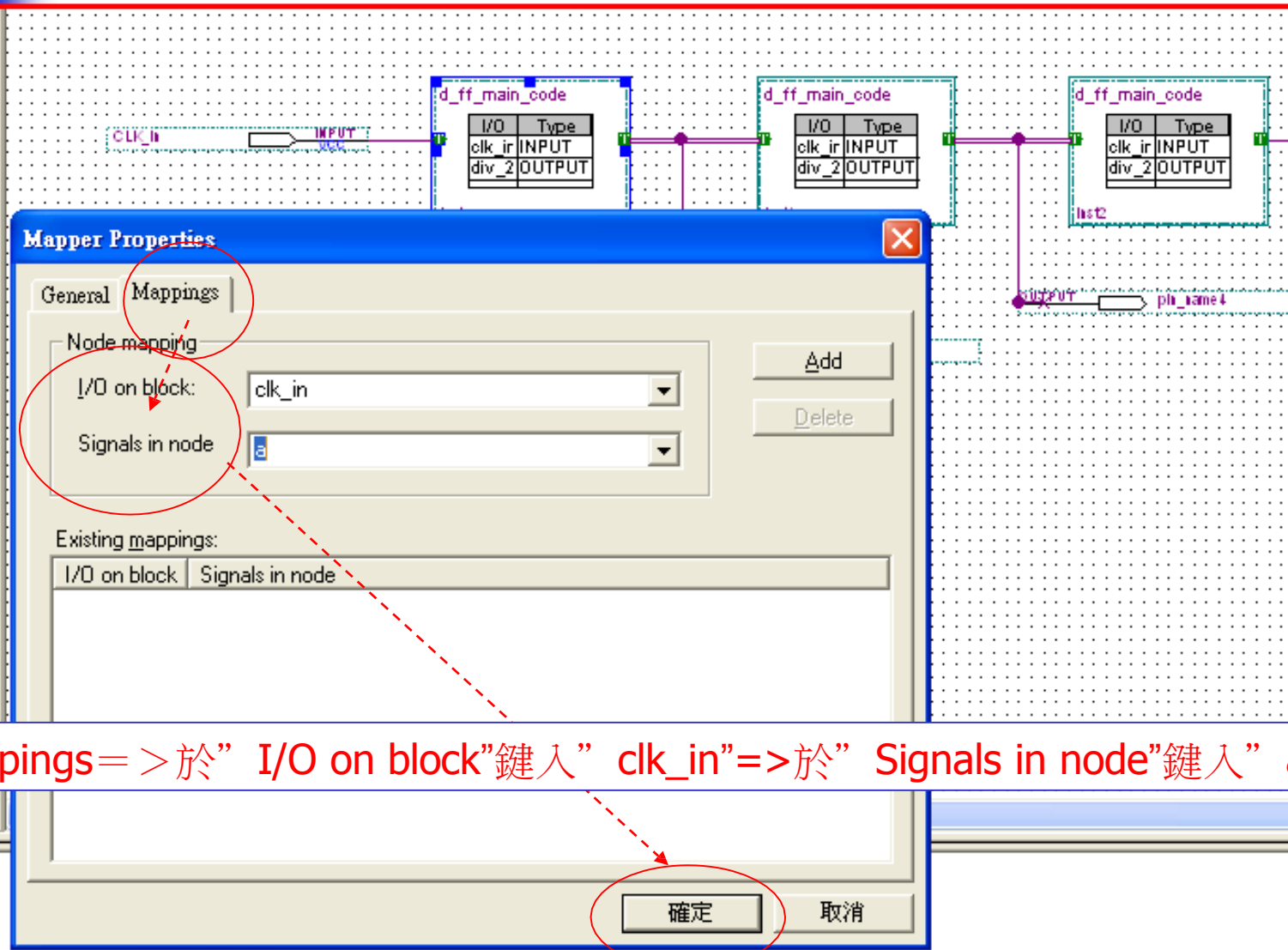
Type: INPUT

確定 取消

(1).修正輸入接腳為” clk_in”.

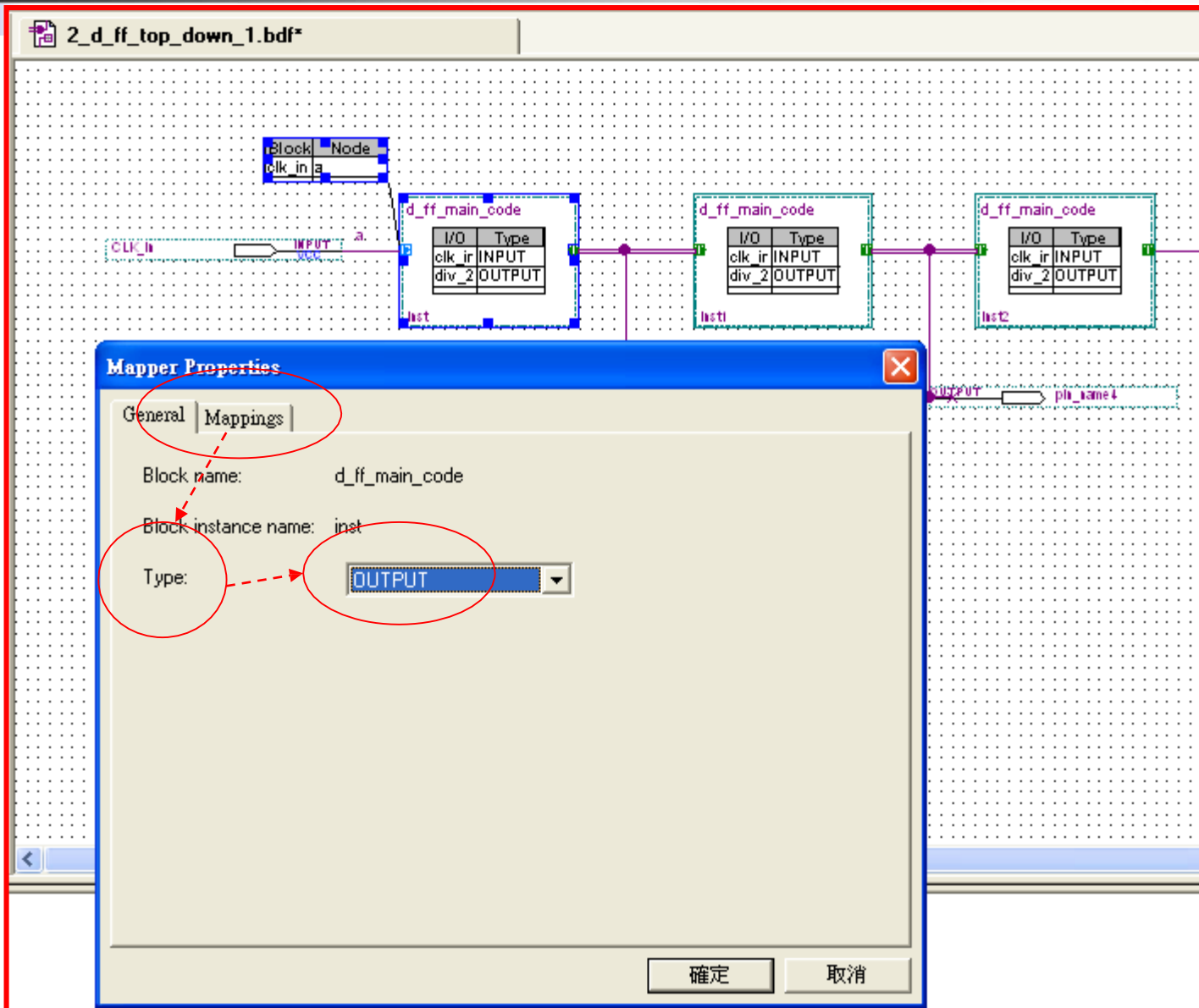
(2).於圖框輸入端出現” ◀▶ ” 雙端輸出入 => 點二下 => General => Type => 選” input”

Top-Down Design (20)



選Mappings=>於” I/O on block”鍵入” clk_in”=>於” Signals in node”鍵入” a”=>確定。

Top-Down Design (21)

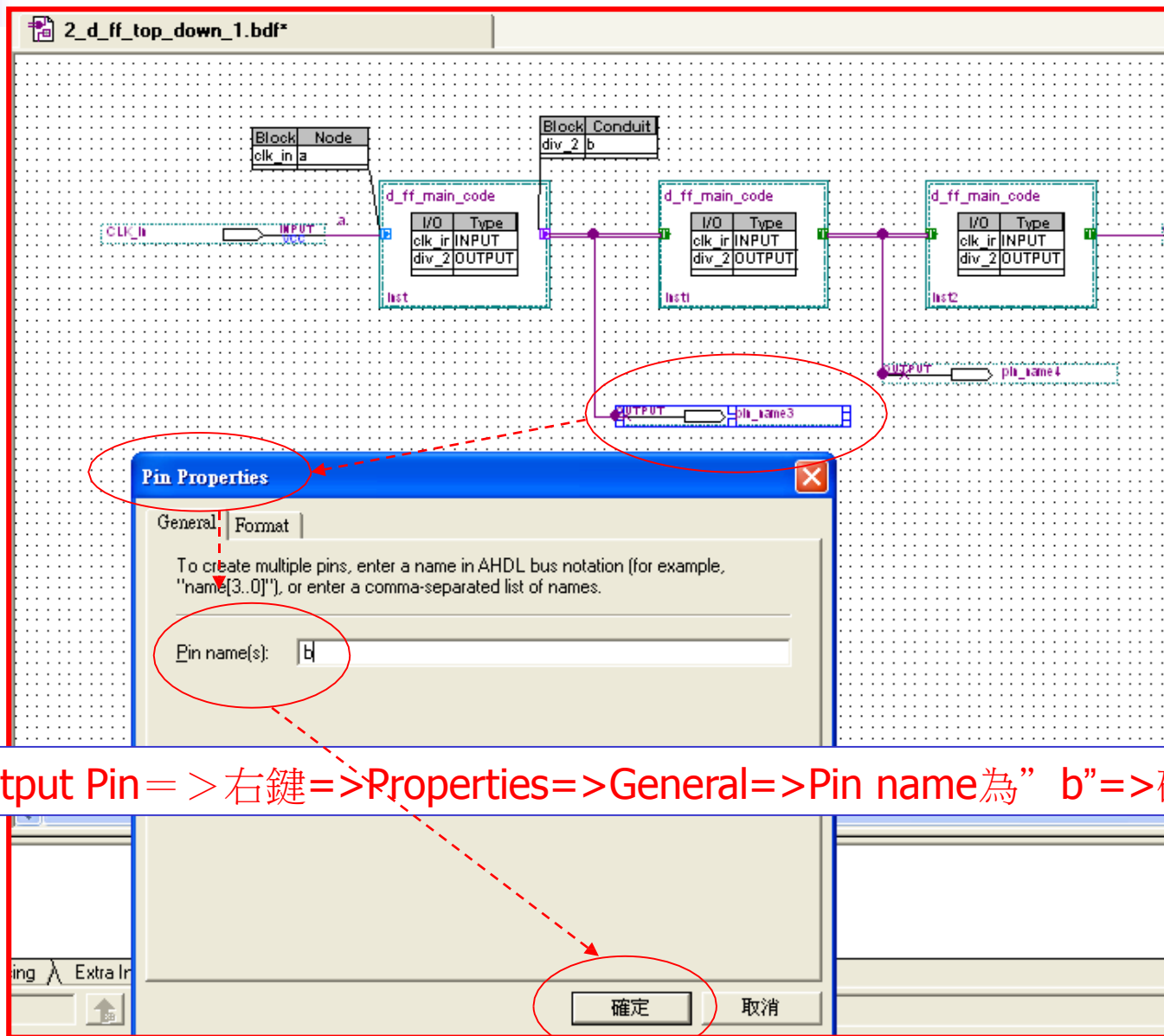


Top-Down Design (22)

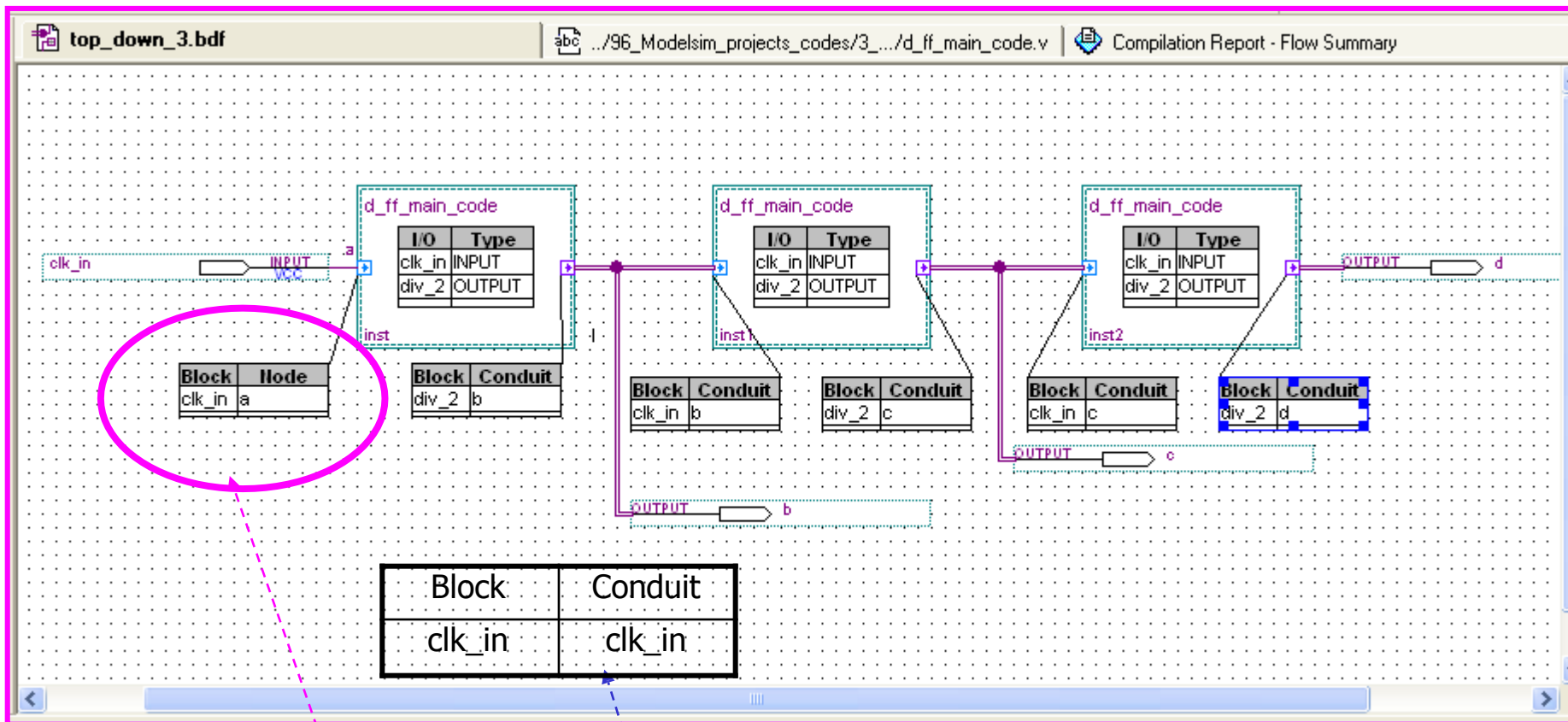
The screenshot shows the 'Mapper Properties' dialog box with the 'Mappings' tab selected. The 'Conduit mapping' section has 'I/O on block' set to 'div_2' and 'Signals in conduit' set to 'b'. The 'Existing mappings' table is empty. The '確定' (OK) button is circled in red. A red dashed arrow points from the text box below to the '確定' button.

選Mappings=>於” I/O on block”鍵入” div_2”=>於” Signals in node”鍵入” b”=>確定

Top-Down Design (23)



Top-Down Design (24)



- (1). 輸出入接腳的名稱若與節點名稱相同，使用”單位元接線（實線）”或”管線”均可。
- (2). 若輸出入接腳的名稱若與節點名稱不同，僅能使用”管線”。
- (3). 嚐試使用管線或位元連線看看結果如何?(單位元均可適用,惟多位元會有誤).

Top-Down Design (25)

The screenshot displays the ModelSim IDE interface for a project named 'top_down'. The Project Navigator on the left shows the file hierarchy, with 'top_down_3.bdf' selected under 'Device Design Files'. The main workspace shows a block diagram of a D flip-flop component named 'd_ff_main_code'. The component has two inputs: 'clk_in' (INPUT) and 'div_2' (OUTPUT). The diagram shows a connection from 'clk_in' to the 'div_2' output. The Status window at the bottom shows the compilation progress for the module.

Module	Progress %	Time
Full Compilation	100 %	00:00:14
Analysis & Synthesis	100 %	00:00:02
Fitter	100 %	00:00:07
Assembler	100 %	00:00:04

Top-Down Design (26)

The screenshot displays the Quartus II IDE interface. The top toolbar includes icons for file operations, compilation, and simulation. The 'Project Navigator' on the left shows the project structure, with 'top_down_3.bdf' selected under 'Device Design Files'. The main window displays the 'Flow Summary' for the compilation of 'top_down_3.bdf'. The summary table indicates a successful compilation on Monday, September 01, 2008, at 23:59:36. The compilation used Quartus II Version 6.0 Build 178 (04/27/2006 SJ Full Version) for the 'top_down' revision. The top-level entity is 'top_down_3', and the target device is 'EP1C6Q240C8'. The compilation was performed using 'Final' timing models, and the timing requirements were met. The total logic elements used are 3 out of 5,980, which is less than 1%.

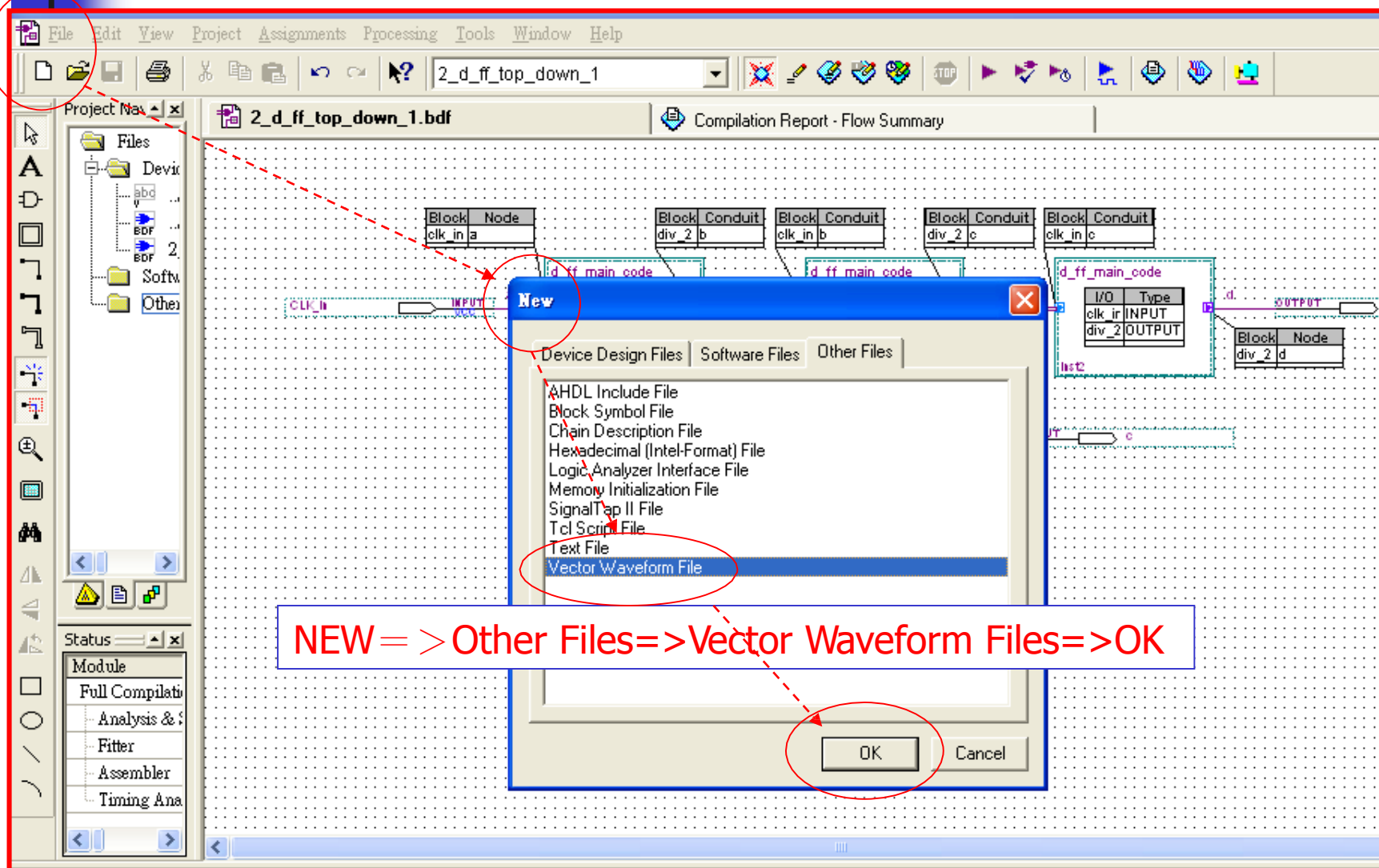
A 'Quartus II' dialog box is overlaid on the flow summary, displaying an information icon and the message: 'Full Compilation was successful (3 warnings)'. The dialog box has a '確定' (OK) button at the bottom.

The status bar at the bottom shows the compilation progress, with the 'Processing' tab selected. The status bar also displays the following information:

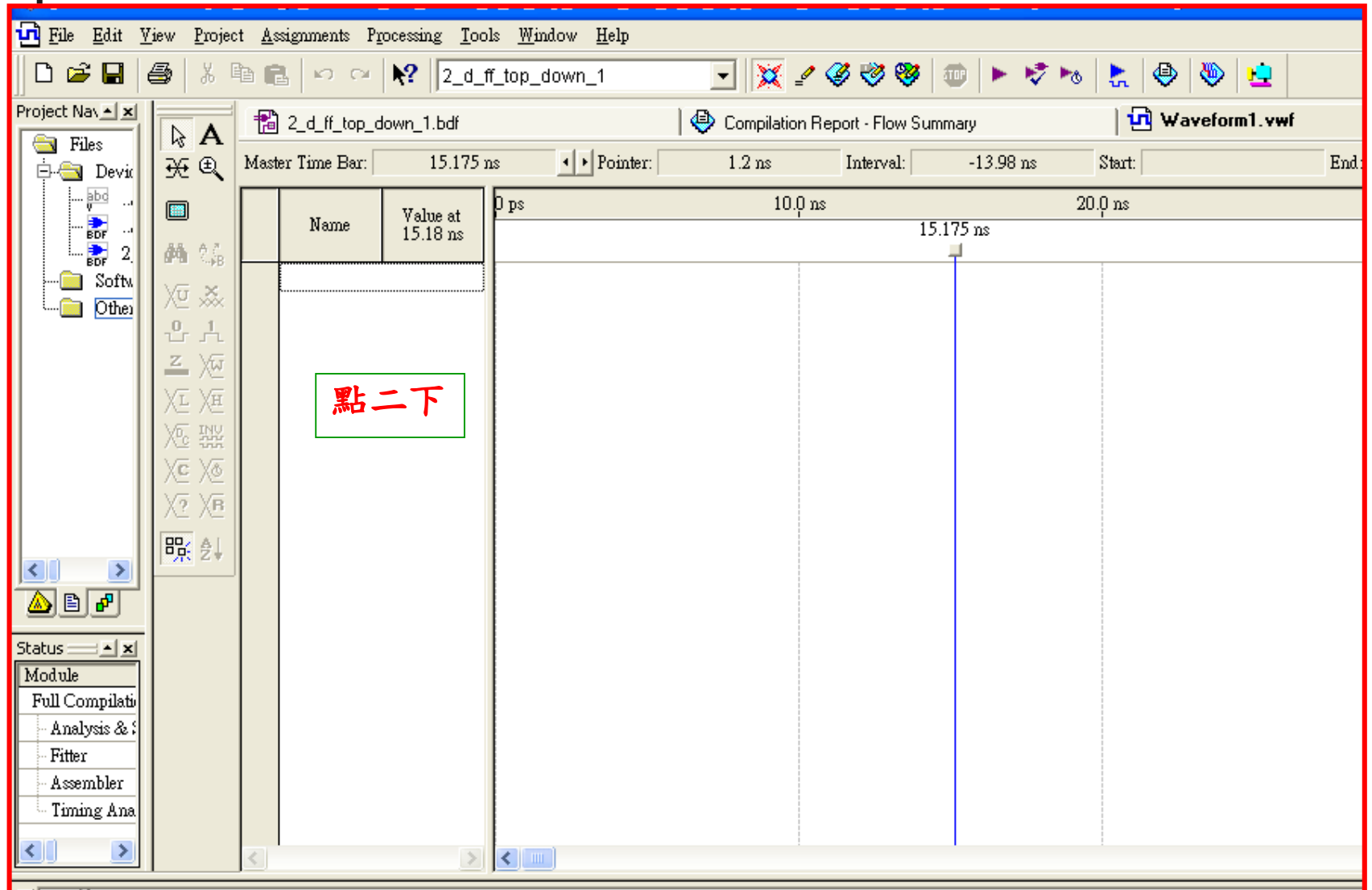
- Info: tco from clock "clk_in" to destination pin "d" through register "d_ff_main_code:inst2|div_2" is 10.528 ns
- Info: Quartus II Timing Analyzer was successful. 0 errors, 2 warnings
- Info: Quartus II Full Compilation was successful. 0 errors, 3 warnings

The status bar also includes a tabbed interface for viewing compilation results, with the following tabs: System, Processing, Extra Info, Info, Warning, Critical Warning, Error, and Suppressed.

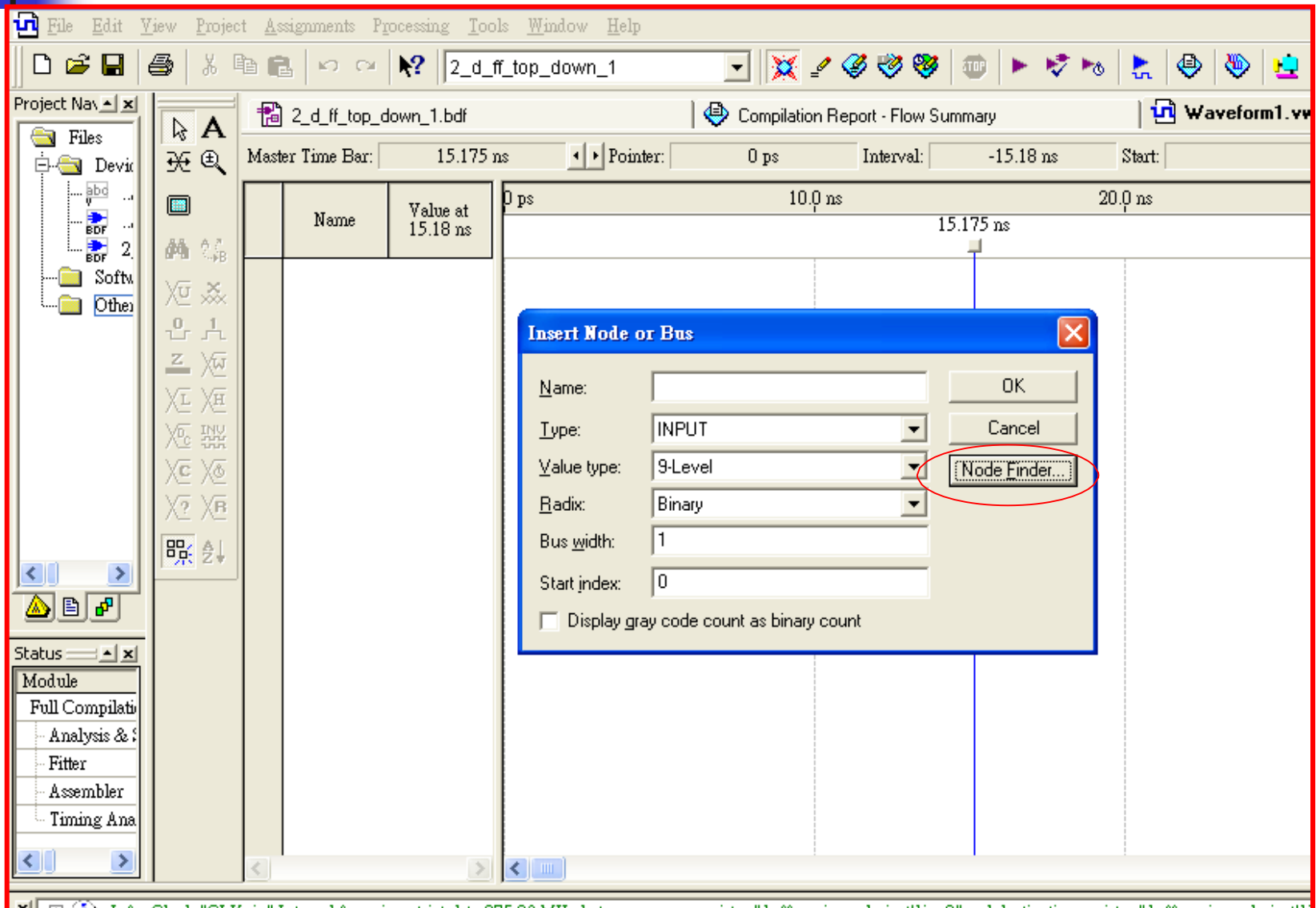
Top-Down Design (27)



Top-Down Design (28)



Top-Down Design (29)



Top-Down Design (30)

The screenshot shows the Node Finder dialog box in a logic design software. The dialog is used to search for nodes in a project. The 'Look in' field is set to '2_d_ff_top_down_1'. The 'Filter' is set to 'Pins: all'. The 'Include subentities' checkbox is checked. The 'Nodes Found' table lists the following nodes:

Name	Assignments	T
b	Unassigned	C
c	Unassigned	C
CLK_in	Unassigned	Ir
Output	Unassigned	C

The 'Selected Nodes' table lists the following nodes:

Name	Assignments	T
l2_d_ff_top_down_1lb	Unassigned	0
l2_d_ff_top_down_1lc	Unassigned	0
l2_d_ff_top_down_1CLK_in	Unassigned	Ir
l2_d_ff_top_down_1Output	Unassigned	0

Red dashed arrows indicate the workflow: from the 'Filter' field to the 'List' button, from the 'List' button to the 'Selected Nodes' table, and from the 'Nodes Found' table to the '>>' button. The '>>' button is used to move selected nodes from the 'Nodes Found' table to the 'Selected Nodes' table.

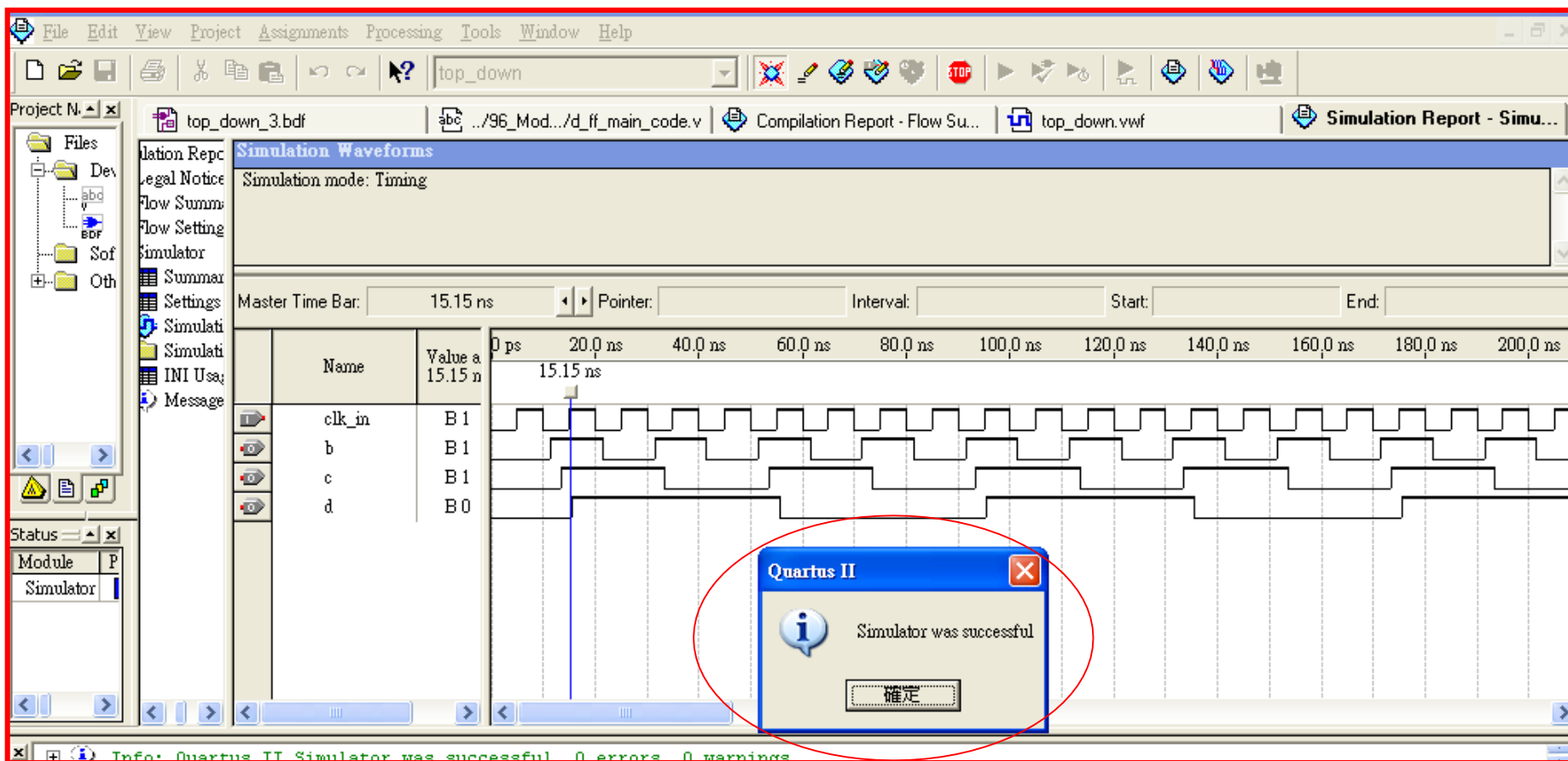
Top-Down Design (31)

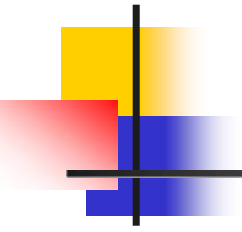
若要將CLK_in往上移，必須在此步驟先移上，模擬後即不可移動。

Save as=>

- (1).名稱與top_down.bdf相同，即為top_down.vwf (注意路徑)；若仍不可行時，
- (2)名稱與路徑名稱相同即可。

Top-Down Design (32)



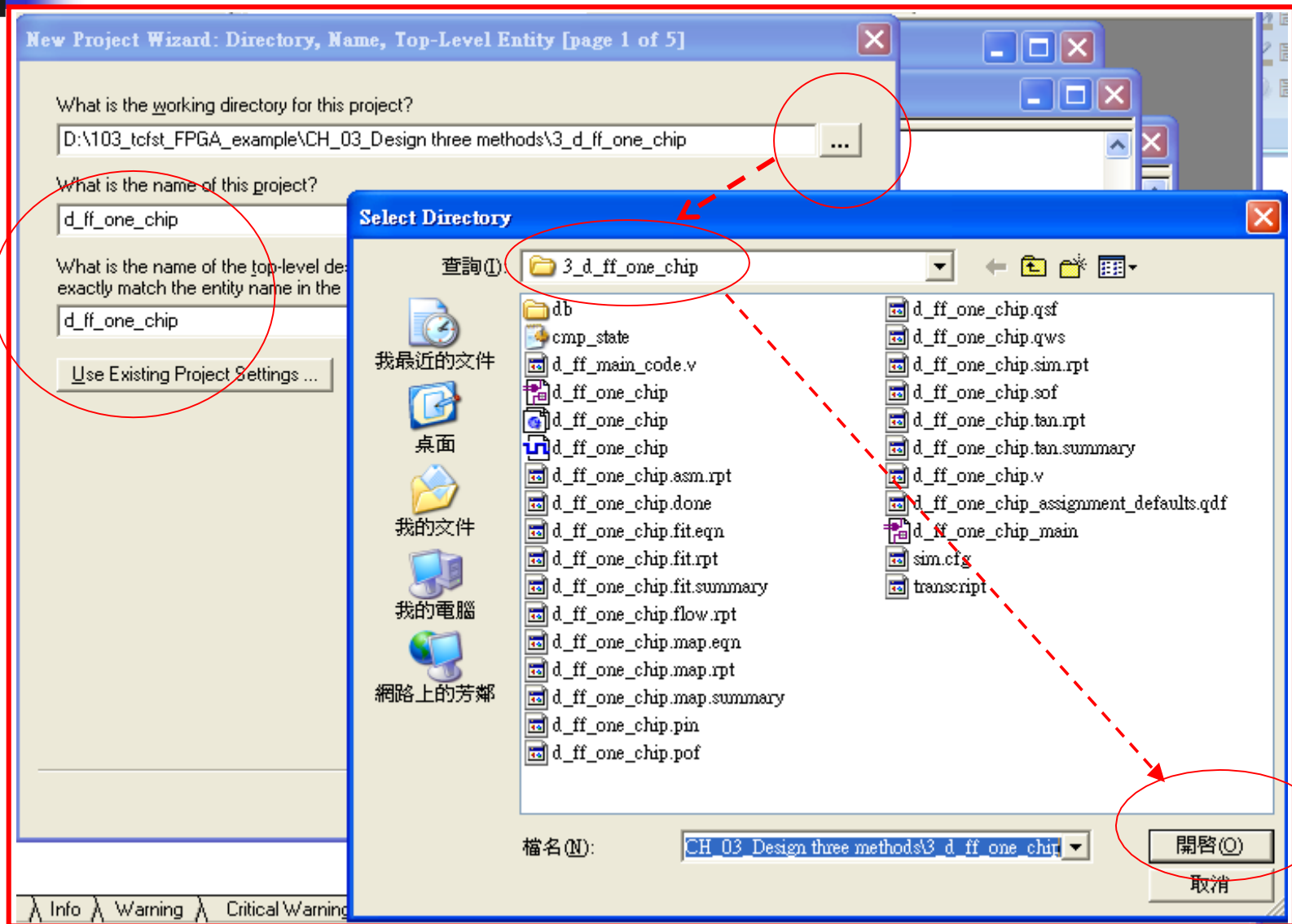


One-Chip Design

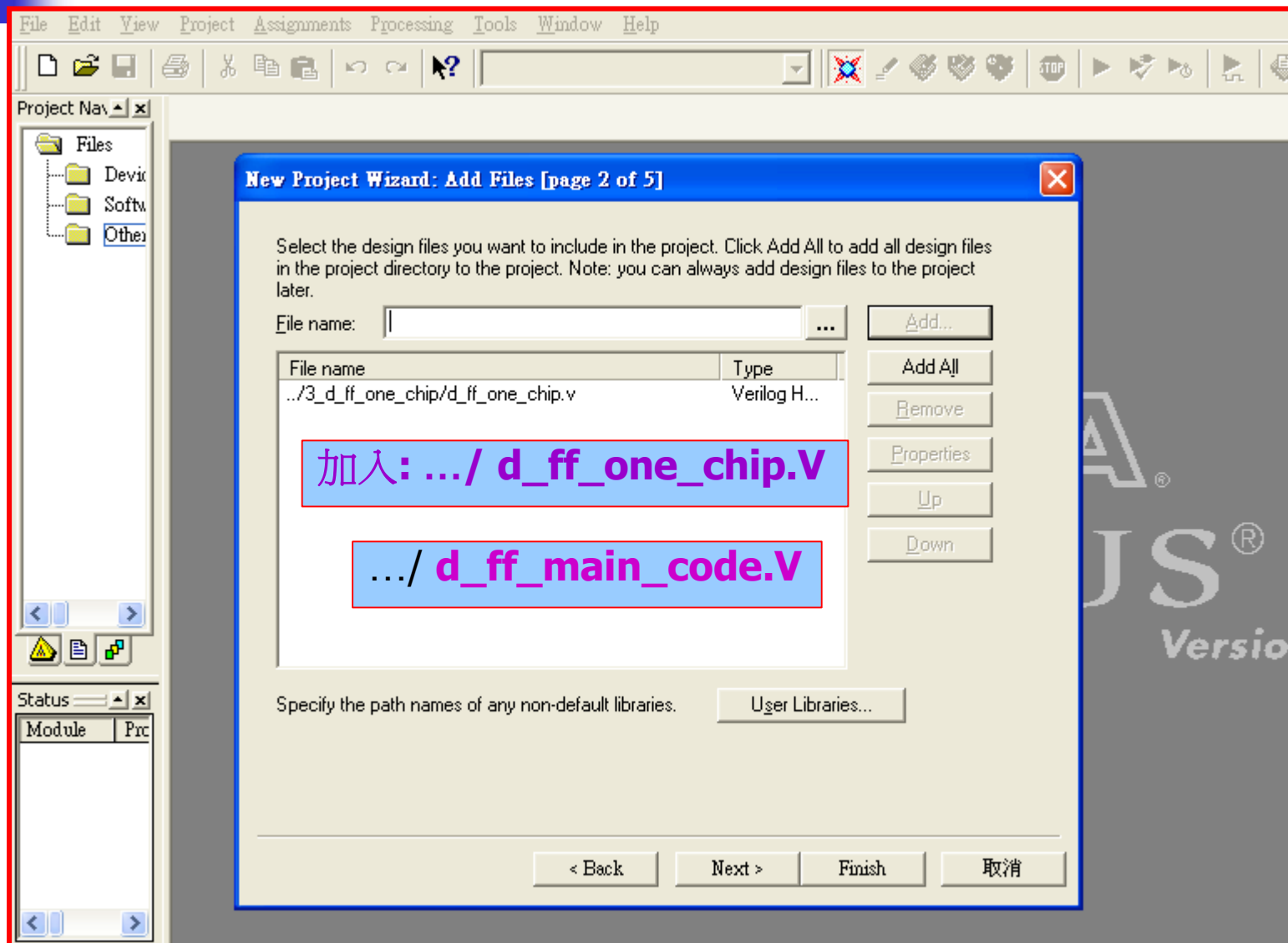
One-Chip

Design Flow

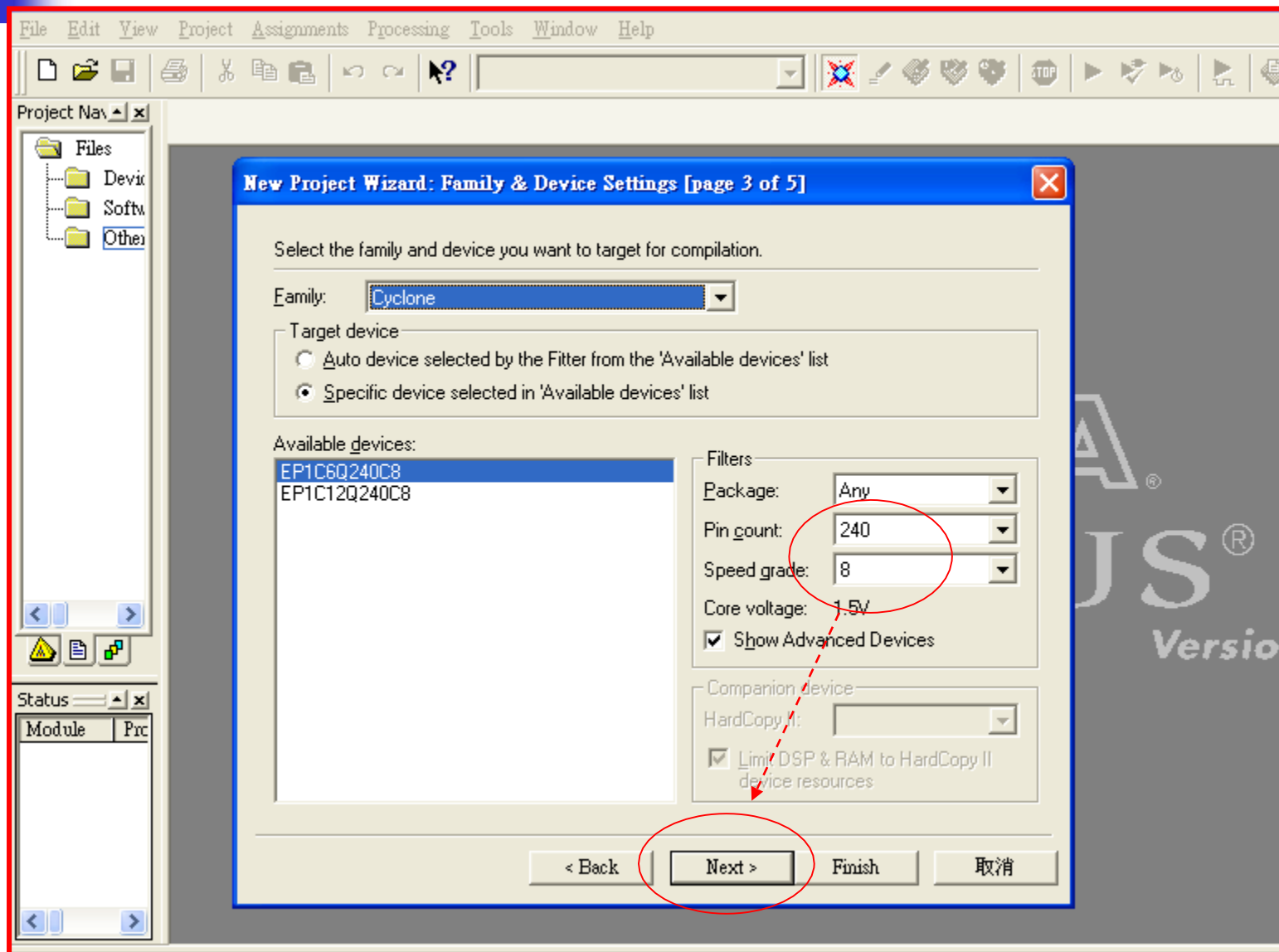
One-Chip Design (1)



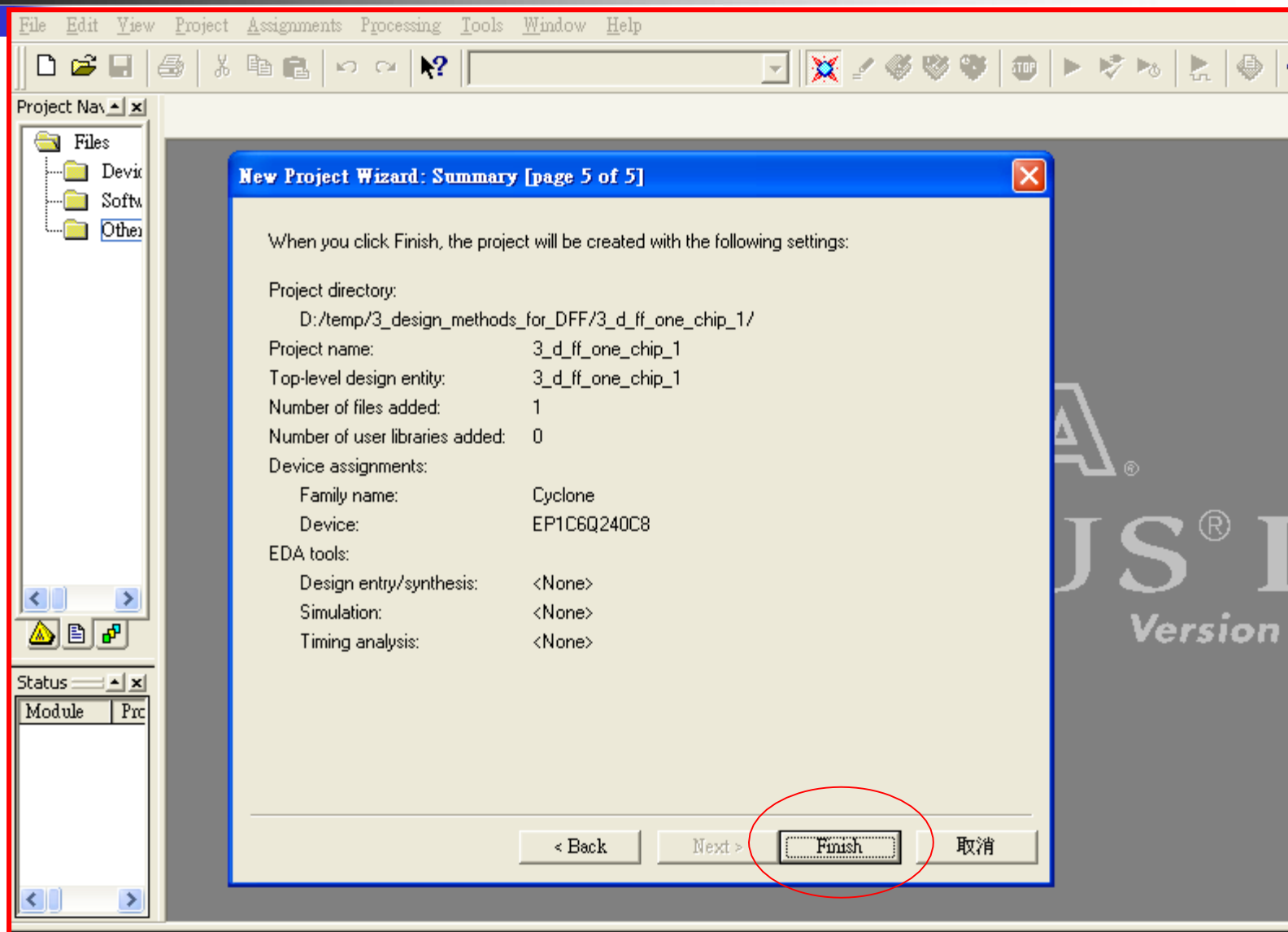
One-Chip Design (2)



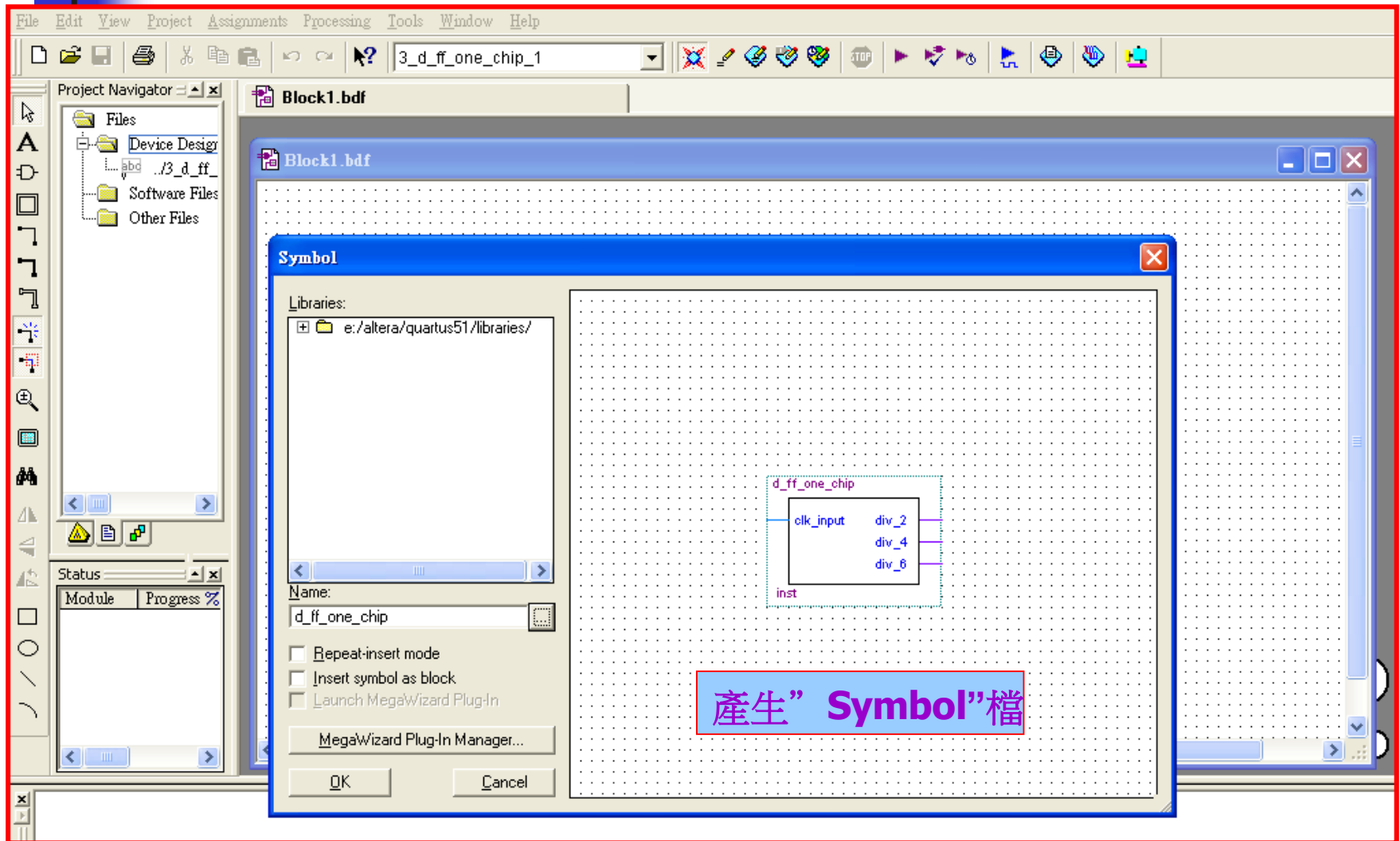
One-Chip Design (3)



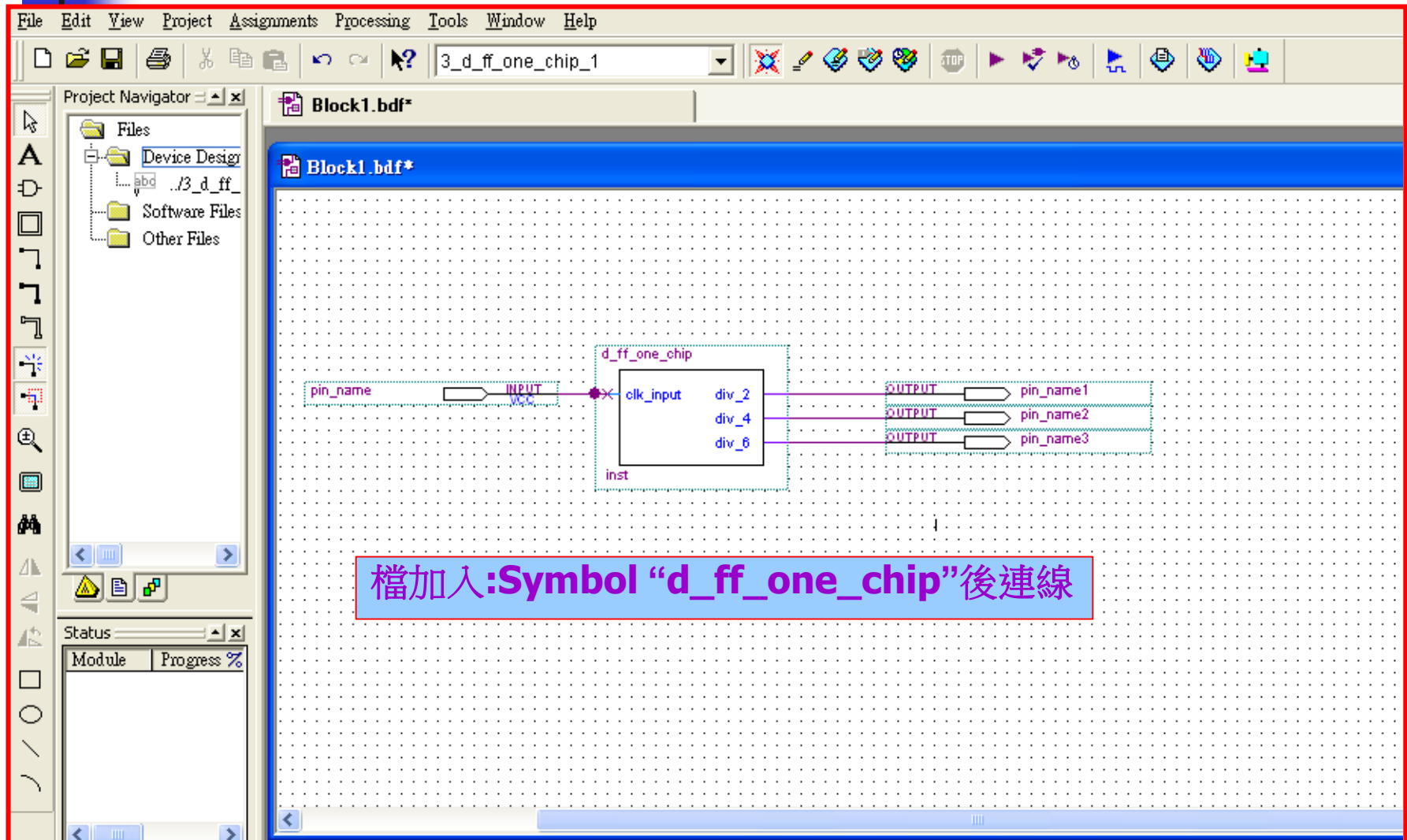
One-Chip Design (4)

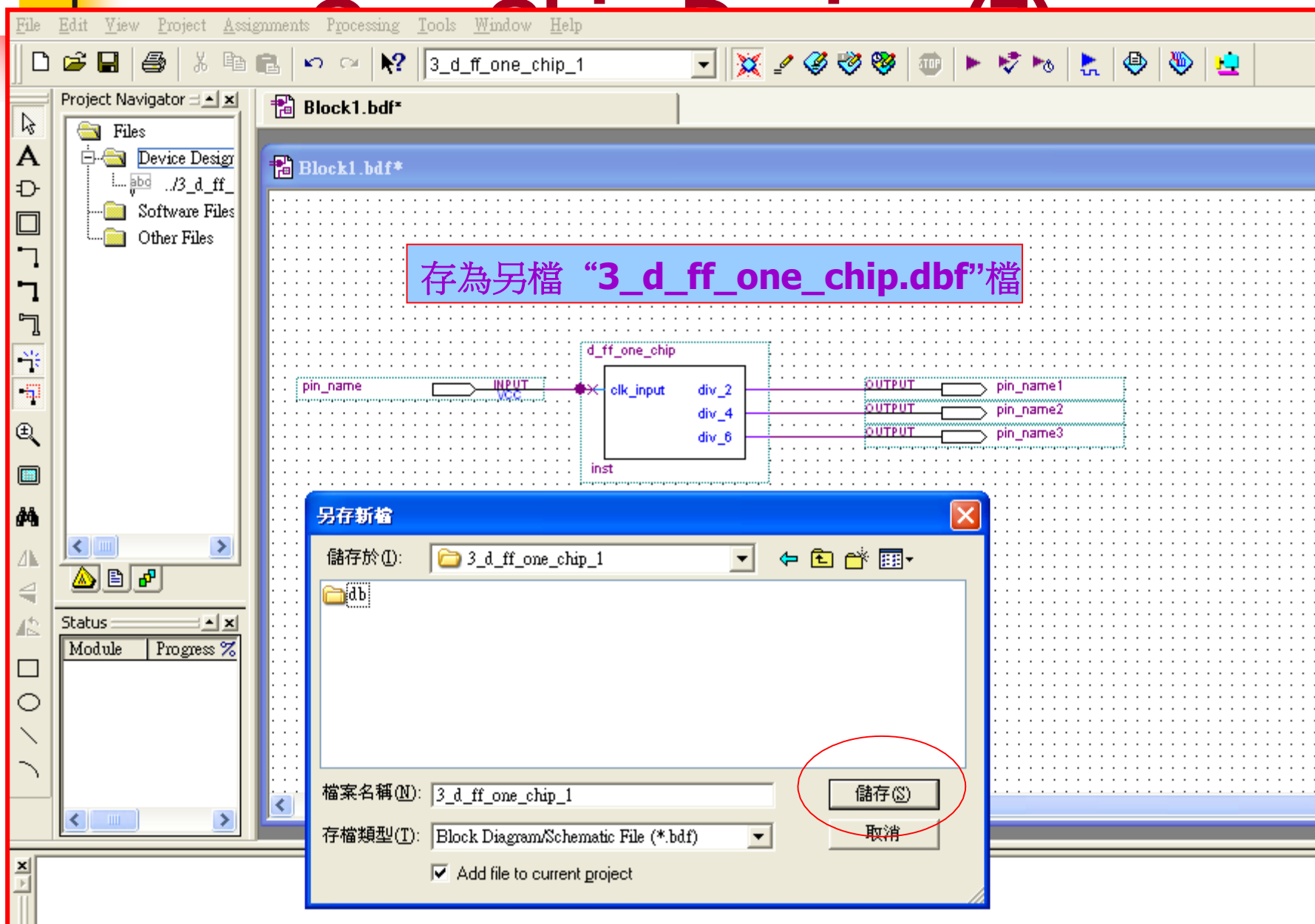


One-Chip Design (5)

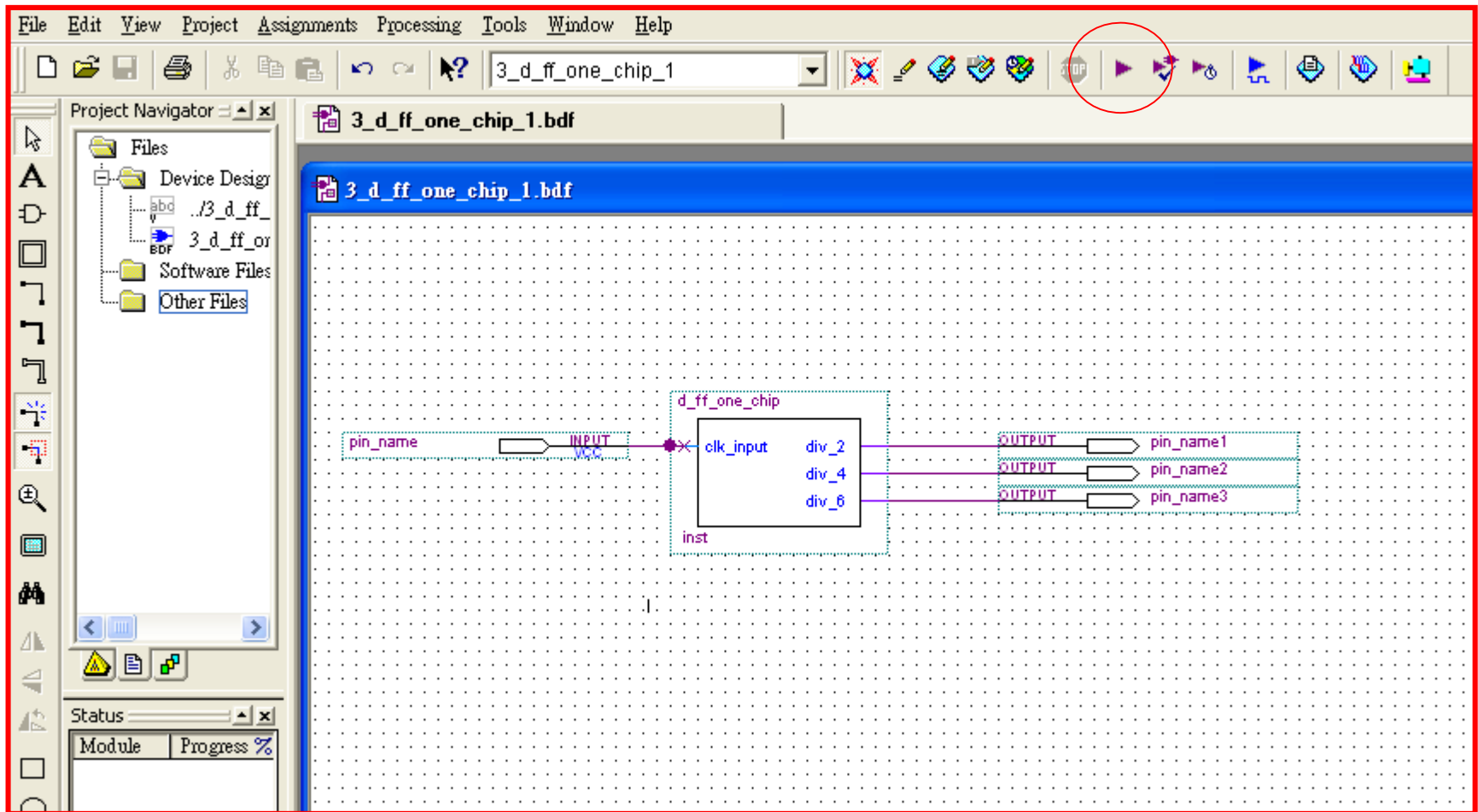


One-Chip Design (6)





One-Chip Design (8)



One-Chip Design (9)

The screenshot displays the Quartus II IDE interface. The top menu bar includes File, Edit, View, Project, Assignments, Processing, Tools, Window, and Help. The toolbar contains various icons for file operations and compilation. The Project Navigator on the left shows a tree structure with 'Files', 'Device Design', 'Software Files', and 'Other Files'. The main workspace shows a schematic diagram with components like '3_d_ff_one_chip_1.bdf' and 'd_ff_one_chip'. A 'Compilation Report - Flow Summary' window is open, showing a list of reports: Legal Notice, Flow Summary, Flow Settings, Flow Elapsed Time, Flow Log, and Analysis & Synthesis. A 'Quartus II' error dialog box is displayed in the foreground, stating 'Full Compilation was NOT successful (1 errors)' with a '確定' (OK) button. The status bar at the bottom shows 'Module Full Compilation Analysis & Synthesis Fitter'. A red box with the text 'Why?' is overlaid on the bottom right of the error dialog.

File Edit View Project Assignments Processing Tools Window Help

3_d_ff_one_chip_1

Project Navigator

Files

Device Design

Software Files

Other Files

3_d_ff_one_chip_1.bdf

Compilation Report - Flow Summary

3_d_ff_one_chip_1.bdf

d_ff_one_chip

pin_name

Compilation Report - Flow Summary

Flow Summary

Compilation Report

Legal Notice

Flow Summary

Flow Settings

Flow Elapsed Time

Flow Log

Analysis & Synthesis

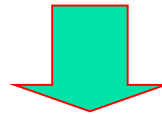
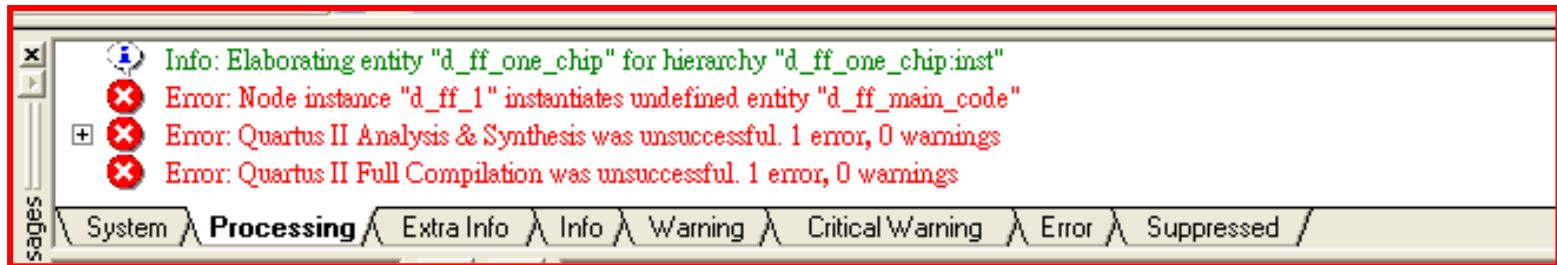
Quartus II

Full Compilation was NOT successful (1 errors)

確定

Why?

One-Chip Design (10)



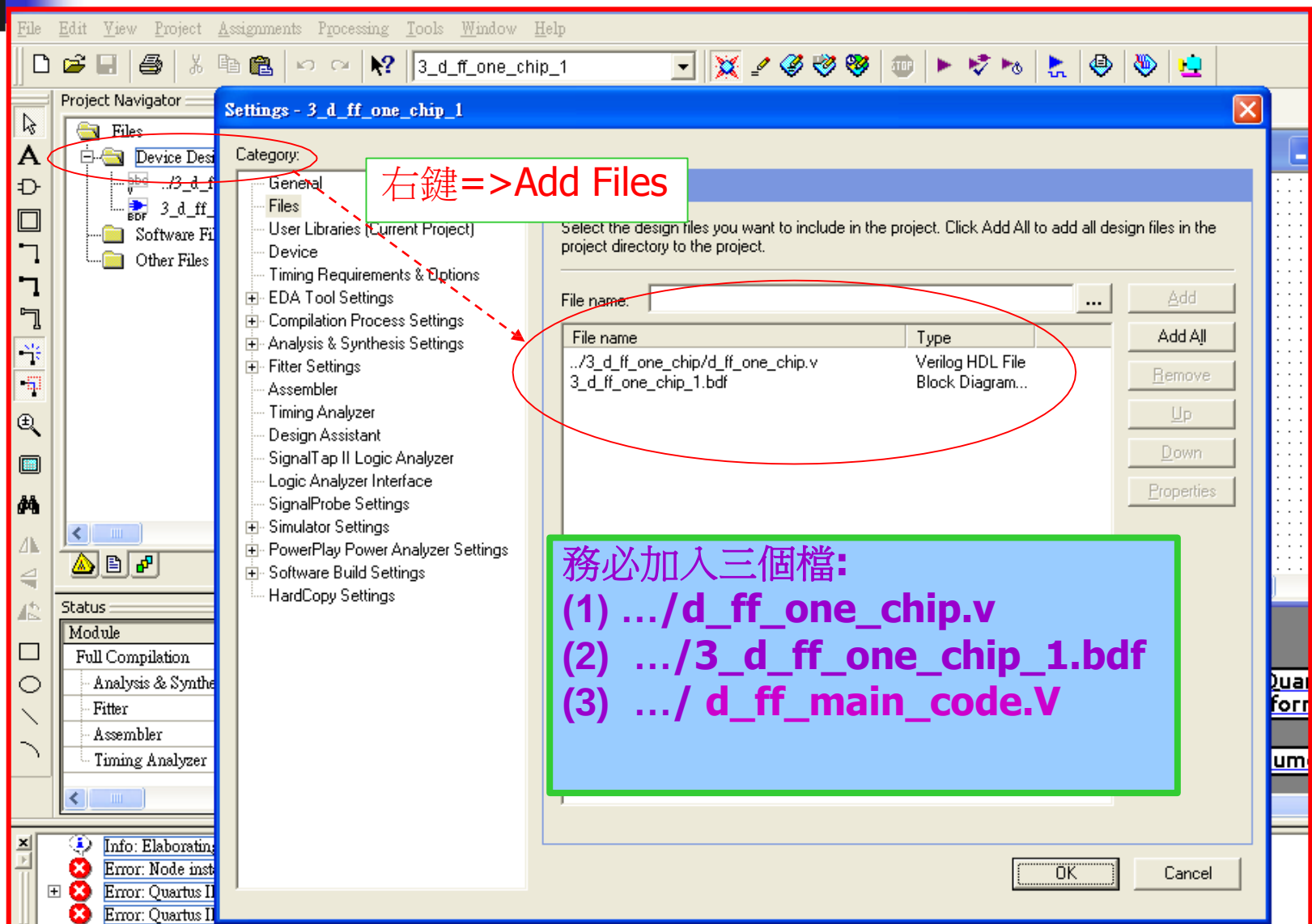
Error: Node instance "d_ff_1" instantiates undefined entity "d_ff_main_code" “
=> 表示別名” d_ff_1” 所對應的” d_ff_main_code” 主程式未見,必項加入.

Error: Quartus II Analysis & Synthesis was unsuccessful. 1 error, 0 warnings Error:
Processing ended: Mon Aug 28 16:16:37 2006

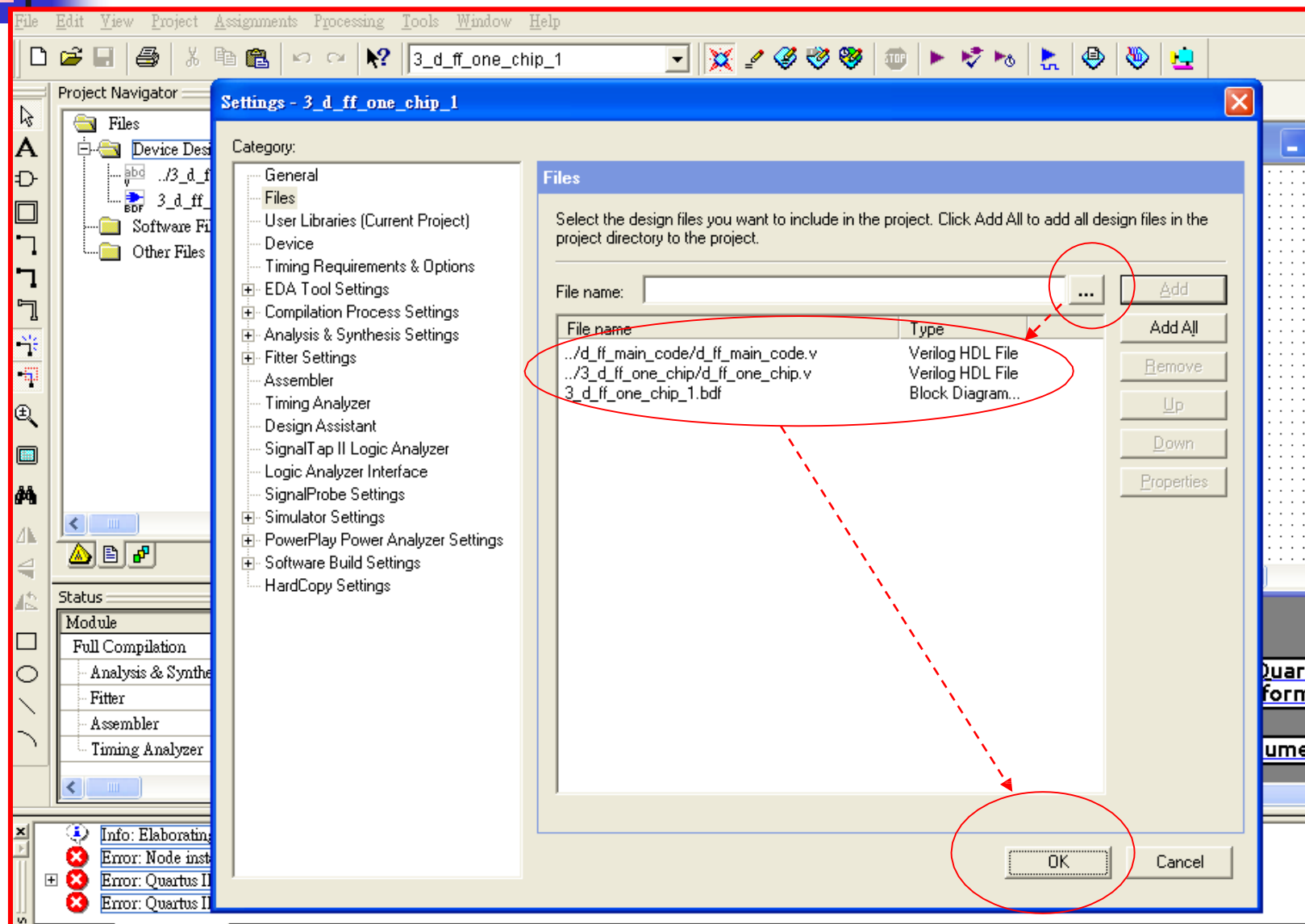
Error: Elapsed time: 00:00:01

Error: Quartus II Full Compilation was unsuccessful. 1 error, 0 warnings

One-Chip Design (11)



One-Chip Design (12)



One-Chip Design (13)

File Edit View Project Assignments Processing Tools Window Help

3_d_ff_one_chip_1

Project Navigator

Files

- Device Design Files
 - abc .d_ff_main_coc
 - abc .3_d_ff_one_cl
 - BDF 3_d_ff_one_chip
- Software Files
- Other Files

3_d_ff_one_chip_1.bdf

Compilation Report - Flow Summary

3_d_ff_one_chip_1.bdf

d_ff_one_chip

pin_name INPUT VCC

clk_input div_2

div_4

OUTPUT pin_name1

OUTPUT pin_name2

Compilation Report - Flow Summary

Flow Summary

Flow Status Successful - Mon Aug 28 16:24:54 2006

Build 216 03/06/2006 SP 2 SJ Full Version

Quartus II

Full Compilation was successful (3 warnings)

確定

Total logic elements 375,980 (< 1 %)

Total pins 4 / 185 (2 %)

Total virtual pins 0

Total memory bits 0 / 92,160 (0 %)

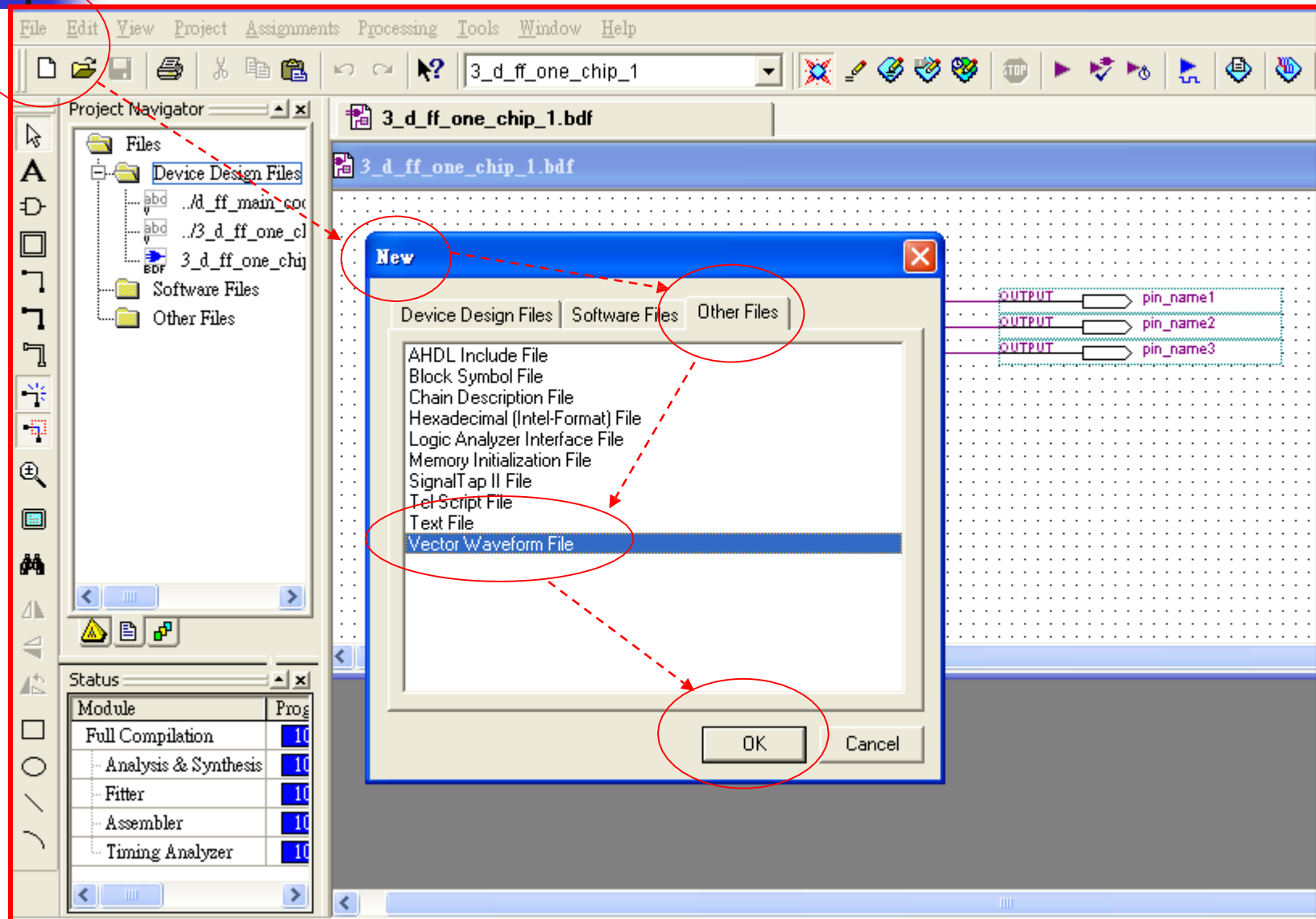
Total PLLs 0 / 2 (0 %)

Status

Module	Prog
Full Compilation	100
Analysis & Synthesis	100
Fitter	100
Assembler	100
Timing Analyzer	100

Documentation

One-Chip Design (14)



One-Chip Design (15)

File Edit View Project Assignments Processing Tools Window Help

3_d_ff_one_chip_1

Project Navigator

- Files
 - Device Design Files
 - ..d_ff_main_coc
 - ..3_d_ff_one_cl
 - 3_d_ff_one_chip
 - Software Files
 - Other Files

3_d_ff_one_chip_1.bdf

Waveform1.vwf

Master Time Bar: 13.325 ns Pointer: 24.34 ns Interval: 11.02 ns Start: End:

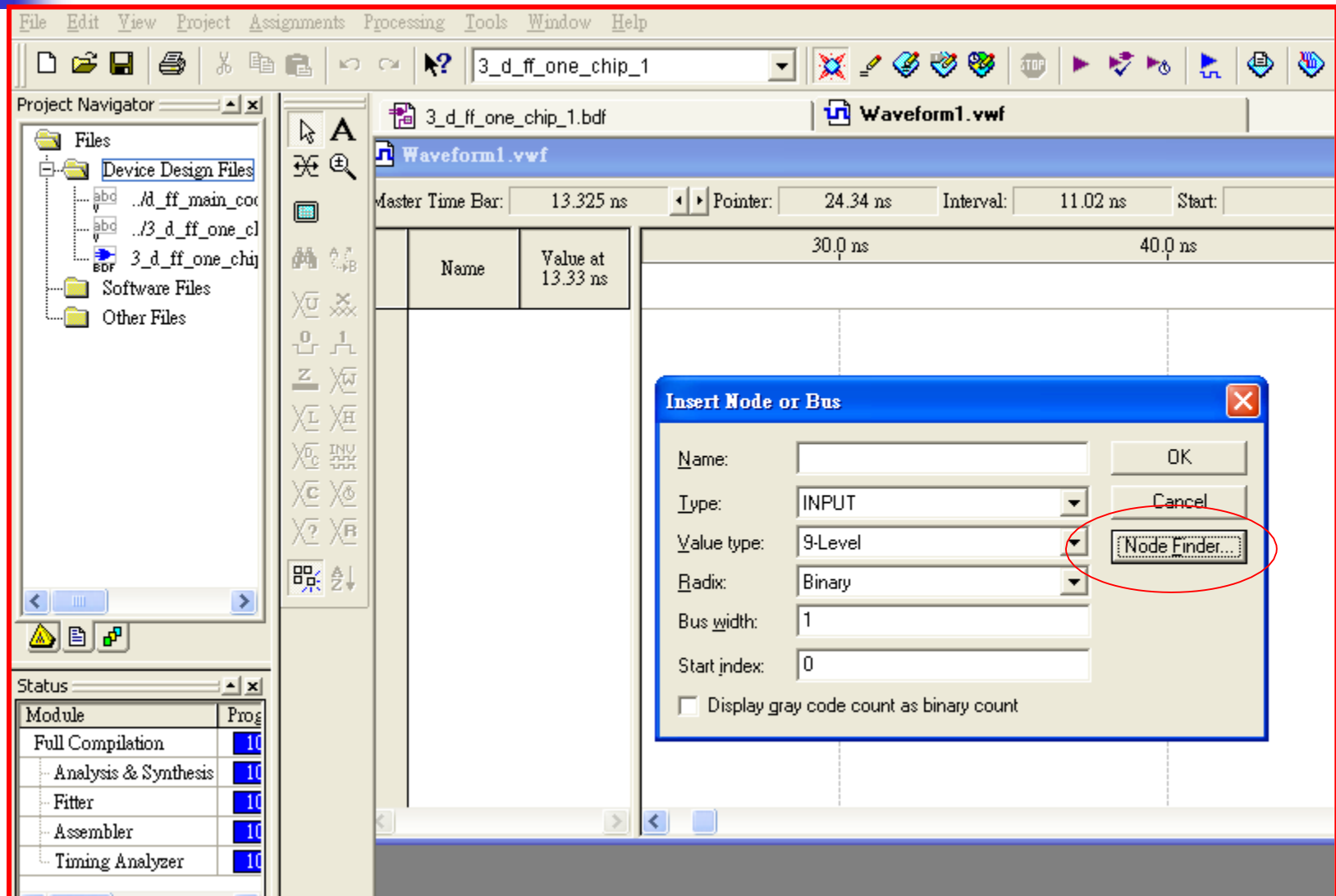
Name	Value at 13.33 ns	30.0 ns	40.0 ns	50.0 ns
點二下				

Status

Module	Prog
Full Compilation	10
Analysis & Synthesis	10
Fitter	10
Assembler	10
Timing Analyzer	10

Documentation

One-Chip Design (16)



One-Chip Design (17)

The screenshot shows the Node Finder dialog box in a logic design tool. The dialog is used to search for nodes in the design. The "Named:" field is set to "*", and the "Filter:" is set to "Pins: all". The "Look in:" field is set to "3_d_ff_one_chip_1". The "Nodes Found:" table lists four nodes: pin_name, pin_name1, pin_name2, and pin_name3, all with "Unassigned" assignments. The "Selected Nodes:" table lists the same four nodes, but with specific assignments: pin_name is assigned to "3_d_ff_one_chip_1", pin_name1 to "3_d_ff_one_chip_1", pin_name2 to "3_d_ff_one_chip_1", and pin_name3 to "3_d_ff_one_chip_1". The "List" button is circled in red, and the "OK" button is also circled in red. Red dashed arrows indicate the flow of the process: from the "List" button to the "Selected Nodes" table, and from the "pin_name2" entry in the "Nodes Found" table to the ">>" button. The ">>" button is also circled in red.

File Edit View Project Assignments Processing Tools Window Help

3_d_ff_one_chip_1

Project Navigator

Files

- Device Design Files
- 3_d_ff_one_chip_1.bdf
- Waveform1.vwf

Master Time Bar: 13.325 ns Pointer: 24.34 ns Interval: 11.02 ns Start: End:

Node Finder

Named: * Filter: Pins: all Customize... List Stop

Look in: 3_d_ff_one_chip_1 Include subentities

Nodes Found:

Name	Assignments	T
pin_name	Unassigned	Ir
pin_name1	Unassigned	C
pin_name2	Unassigned	C
pin_name3	Unassigned	C

Selected Nodes:

Name	Assignments	T
3_d_ff_one_chip_1pin_name	Unassigned	Ir
3_d_ff_one_chip_1pin_name1	Unassigned	0
3_d_ff_one_chip_1pin_name2	Unassigned	0
3_d_ff_one_chip_1pin_name3	Unassigned	0

Module

- Full Compilation
- Analysis & Synthesis
- Fitter
- Assembler
- Timing Analysis

Status

Info: top from clock "pin_name" to destination pin "pin_name3" through register "3_d_ff_one_chip_inst1 3_d_ff_main_code 3_d_ff_3div_2" is 10.304 ns

One-Chip Design (18)

The screenshot displays the Quartus II software interface during a simulation. The main window is titled "Simulation Report - Simulation Waveforms". The "Simulation mode: Timing" is selected. The "Master Time Bar" shows a value of 13.325 ns. The "Pointer" is positioned at 10.0 ns. The "Interval" is set to 13.325 ns. The "Start" and "End" fields are empty. The "Name" and "Value at 13.33 ns" columns are visible, showing values for pin_... (B 0, B 1, B 1, B 0). The "Status" window at the bottom left shows the "Simulator" module with a progress of 100% and a time of 00:01. A "Quartus II" dialog box is open in the foreground, displaying the message "Simulator was successful" and a button labeled "確定" (OK).

File Edit View Project Assignments Processing Tools Window Help

3_d_ff_one_chip_1.bdf 3_d_ff_one_chip_1.vwf Simulation Report - Simulation W

Simulation Report - Simulation Waveforms

Simulation mode: Timing

Master Time Bar: 13.325 ns Pointer: Interval: Start: End:

0 ps 10.0 ns 13.325 ns

Name	Value at 13.33 ns
pin_...	B 0
pin_...	B 1
pin_...	B 1
pin_...	B 0

Quartus II

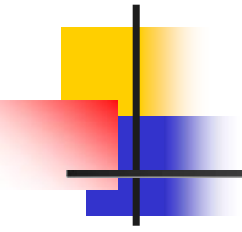
Simulator was successful

確定

Module Progress % Time

Simulator	100 %	00:01
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Documentati



One-Chip Design (19)

```
module d_ff_one_chip(clk_input, div_2, div_4, div_6);
input    clk_input;
output   div_2;
output   div_4;
output   div_6;
wire     a;
wire     b;

assign   div_2 = a;
assign   div_4 = b;

// by name
d_ff_main_code d_ff_1(.clk_in(clk_input),.div_2(a));

d_ff_main_code d_ff_2(.clk_in(a),.div_2(b));

d_ff_main_code d_ff_3(.clk_in(b),.div_2(div_6));

endmodule
```

單晶片之原程式
(By Name)

One-Chip Design (20)

```
module d_ff_one_chip(clk_input, div_2, div_4, div_6);
input    clk_input;
output   div_2;
output   div_4;
output   div_6;
wire     a;
wire     b;

assign   div_2 = a;
assign   div_4 = b;

// by name
d_ff_main_code d_ff_1(clk_input,a);

d_ff_main_code d_ff_2(a,b);

d_ff_main_code d_ff_3(b,div_6));

endmodule
```

單晶片之修正程式
(By Order)



One-Chip Design (21)

The screenshot displays the ABC software interface. The top menu bar includes File, Edit, View, Project, Assignments, Processing, Tools, Window, and Help. The toolbar contains various icons for file operations and simulation. The Project Navigator on the left shows a tree structure with 'Files', 'Device Design Files', 'Software Files', and 'Other Files'. The 'Device Design Files' folder is expanded, showing files like 'd_ff_main_code.v' and '3_d_ff_one_chip.v'. The main editor window displays the Verilog code for the module 'd_ff_one_chip'. The code includes input/output declarations, wire declarations, assign statements, and module instantiations. A pink dashed arrow points from the '3_d_ff_one_chip.v' file in the Project Navigator to the code in the main editor. A pink oval highlights the module instantiation lines (15-19). The Status bar at the bottom shows the progress of the simulation, indicating 100% completion.

```
1 module d_ff_one_chip(clk_input, div_2, div_4, div_6);
2
3 input  clk_input;
4 output div_2;
5 output div_4;
6 output div_6;
7
8 wire  a;
9 wire  b;
10
11 assign div_2 = a;
12 assign div_4 = b;
13
14 // by name
15 d_ff_main_code d_ff_1(clk_input,a);
16
17 d_ff_main_code d_ff_2(a,b);
18
19 d_ff_main_code d_ff_3(b,div_6);
20
21
22 endmodule
```

Module	Progress %	Time
Simulator	100 %	00:00:02

One-Chip Design (22)

The screenshot displays the Quartus II IDE with the project '3_d_ff_one_chip_1' open. The Project Navigator on the left shows the design files, including '3_d_ff_one_chip_1.bdf'. The Flow Summary window on the right provides details about the compilation process. A 'Quartus II' dialog box is overlaid in the center, indicating that the full compilation was successful despite three warnings. The Status window at the bottom left shows the progress of various compilation steps, all of which are 100% complete.

Flow Summary

Flow Status	Successful - Mon Aug 28 16:45:01 2006
Quartus II Version	5.1 Build 216 03/06/2006 SP 2 SJ Full Version
Revision Name	3_d_ff_one_chip_1
Top-level Entity Name	3_d_ff_one_chip_1
Cyclone	EP1C6Q240C8
Final	Yes
	3 / 5,980 (< 1 %)
	4 / 185 (2 %)
	0
Total memory bits	0 / 92,160 (0 %)
Total PLLs	0 / 2 (0 %)

Quartus II

Full Compilation was successful (3 warnings)

確定

Status

Module	Progress %	Time
Full Compilation	100 %	00:00:10
Analysis & Synthesis	100 %	00:00:02
Fitter	100 %	00:00:04
Assembler	100 %	00:00:03
Timing Analyzer	100 %	00:00:01

One-Chip Design (23)

The screenshot displays the Quartus II software interface during a simulation. The top menu bar includes File, Edit, View, Project, Assignments, Processing, Tools, Window, and Help. The toolbar contains various icons for file operations and simulation control. The Project Navigator on the left shows the project structure, including Device Design Files, Software Files, and Other Files. The Simulation Waveforms window on the right shows the simulation mode set to Timing, with a Master Time Bar at 13.325 ns. A table lists the values at this time for various signals.

Name	Value at 13.33 ns
pin_...	B 0
pin_...	B 1
pin_...	B 1
pin_...	B 0

A dialog box titled "Quartus II" with the message "Simulator was successful" and a "確定" (OK) button is displayed in the center. The status bar at the bottom shows the Simulator progress at 100%.